Emergent Mathematics

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Contents

Part I

Symmetry

1 Spectral Theory

Definition 1.1 (Vector spaces and linear maps)**.** Let k be a field. A *vector space* over k is an abelian group V with scalar multiplication satisfying

$$a(v+w) = av + aw, \quad (a+b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map $T : V \to W$ is *linear* when $T(av + bw) = aT(v) + bT(w)$. Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For $T : V \to V$, a scalar λ is an *eigenvalue* when $\ker(T - \lambda I) \neq 0$; a nonzero vector in this kernel is an *eigenvector*.

Definition 1.2 (Dual spaces and tensor products)**.** The *dual space* of V is $V^\vee = \operatorname{Hom}_k(V, k)$. Its canonical evaluation pairing is

$$\mathrm{ev}_V : V^\vee \times V \longrightarrow k, \qquad (\varphi, v) \longmapsto \varphi(v).$$

The algebraic transpose of $T : V \to W$ is $T^\vee : W^\vee \to V^\vee$, defined by $T^\vee(\varphi) = \varphi \circ T$.

The *tensor product* $V \otimes W$ is characterized by a bilinear map $(v, w) \mapsto v \otimes w$ with the following universal property: every bilinear map $b : V \times W \to X$ factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\widetilde{b}} X.$$

The evaluation pairing induces a linear map $\mathrm{ev}_V : V^\vee \otimes V \to k$.

Definition 1.3 (Finite-dimensional vector spaces)**.** A vector space V is *finite-dimensional* when there is a coevaluation map

$$\mathrm{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the snake identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^\vee}) \circ (\mathrm{id}_{V^\vee} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^\vee}.$$

Exercise 1.1 (Coevaluation uniqueness)**.** Let $c, d : k \to V \otimes V^\vee$ both satisfy the snake identities with ev_V. Apply

$$\mathrm{id}_V \otimes \mathrm{ev}_V \otimes \mathrm{id}_{V^\vee}$$

to $d \otimes c : k \to V \otimes V^\vee \otimes V \otimes V^\vee$ and use the snake identity for d to identify the result with c. Use the snake identity for c to identify the same result with d, and deduce $c = d$.

Definition 1.4 (Trace and dimension)**.** For a finite-dimensional vector space V, let $s_{V,V^\vee} : V \otimes V^\vee \to V^\vee \otimes V$ exchange the factors. For $T \in \mathrm{End}(V)$, define

$$\mathrm{tr}(T) = \mathrm{ev}_V \circ s_{V,V^\vee} \circ (T \otimes \mathrm{id}_{V^\vee}) \circ \mathrm{coev}_V.$$

The *dimension* of V is the scalar

$$\dim_k(V) = \mathrm{tr}(\mathrm{id}_V) \in k.$$

Definition 1.5 (Exterior powers)**.** For $m \geq 0$, the mth *exterior power* of a vector space V is

$$\Lambda^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m : v_i = v_j \text{ for some } i \neq j \rangle,$$

and the image of $v_1 \otimes \cdots \otimes v_m$ is written $v_1 \wedge \cdots \wedge v_m$. An endomorphism T induces

$$\Lambda^m T(v_1 \wedge \cdots \wedge v_m) = Tv_1 \wedge \cdots \wedge Tv_m.$$

The products

$$\Lambda^r V \otimes \Lambda^s V \longrightarrow \Lambda^{r+s} V, \qquad (u_1 \wedge \cdots \wedge u_r) \otimes (v_1 \wedge \cdots \wedge v_s) \longmapsto u_1 \wedge \cdots \wedge u_r \wedge v_1 \wedge \cdots \wedge v_s$$

make $\Lambda^\bullet V = \bigoplus_{m \geq 0} \Lambda^m V$ the *exterior algebra* of V.

Exercise 1.2 (Exterior rank)**.** Let V be finite-dimensional. Prove that there is a unique $n \in \mathbb{Z}_{\geq 0}$ such that

$$\Lambda^n V \neq 0, \qquad \Lambda^{n+1} V = 0,$$

and that every endomorphism of $\Lambda^n V$ is a unique scalar multiple of the identity. Prove that every $\Lambda^m V$ is finite-dimensional and that

$$\dim_k(V) = n\, 1_k, \qquad \dim_k(\Lambda^m V) = \binom{n}{m} 1_k, \qquad \Lambda^m V = 0 \quad (m > n).$$

Prove that the exterior algebra is graded-commutative:

$$\xi \wedge \eta = (-1)^{rs} \eta \wedge \xi, \qquad \xi \in \Lambda^r V, \quad \eta \in \Lambda^s V.$$

Deduce that $\dim_k(V)$ determines n when $\operatorname{char} k = 0$, whereas it determines only the residue class of n modulo p when $\operatorname{char} k = p > 0$.

Definition 1.6 (Exterior rank and determinant)**.** The *exterior rank* of a finite-dimensional vector space V is the unique integer $\operatorname{rank}(V) = n$ such that

$$\Lambda^n V \neq 0, \qquad \Lambda^{n+1} V = 0.$$

For $T \in \operatorname{End}(V)$, its *determinant* is the unique scalar $\det(T)$ satisfying

$$\Lambda^n T = \det(T) \operatorname{id}_{\Lambda^n V}.$$

Exercise 1.3 (Determinant and exterior traces)**.** Let V have exterior rank n. Use functoriality of the highest exterior power to prove

$$\det(ST) = \det(S) \det(T), \qquad T \text{ is invertible} \iff \det(T) \neq 0.$$

For $T \in \operatorname{End}(V)$, extend scalars to $k[z]$ and prove

$$\det(I + zT) = \sum_{m=0}^{n} \operatorname{tr}(\Lambda^m T) z^m.$$

Recover $(1+z)^n = \sum_m \binom{n}{m} z^m$ by taking $T = I$.

Definition 1.7 (Rising factorial and binomial series)**.** For $a \in \mathbb{C}$ and $m \in \mathbb{Z}_{\geq 0}$, define

$$(a)_0 = 1, \qquad (a)_m = a(a+1)\cdots(a+m-1) \quad (m \geq 1).$$

For $|z| < 1$, define the *binomial series*

$$B_a(z) = \sum_{m=0}^{\infty} \frac{(a)_m}{m!} z^m.$$

Exercise 1.4 (Generalized binomial theorem)**.** Prove that the binomial series converges for $|z| < 1$ and satisfies

$$(1-z)B_a'(z) = aB_a(z), \qquad B_a(0) = 1.$$

For the branch normalized to equal 1 at $z = 0$, deduce

$$B_a(z) = (1-z)^{-a}.$$

For $a = -n$, $n \in \mathbb{Z}_{\geq 0}$, prove that the series terminates and, after replacing z by $-z$, recover

$$(1+z)^n = \sum_{m=0}^{n} \binom{n}{m} z^m.$$

Definition 1.8 (Hilbert spaces and orthogonality)**.** An *inner product* on a complex vector space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies $\langle x, y\rangle = \overline{\langle y, x\rangle}$. Its norm is $\|x\| = \sqrt{\langle x, x\rangle}$. A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace $M \subseteq H$, define

$$M^{\perp} = \{x : \langle m, x\rangle = 0 \text{ for every } m \in M\}.$$

Definition 1.9 (Bounded operators and adjoints)**.** A linear map $T : H \to K$ between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty.$$

Write $B(H, K)$ for the bounded operators and $B(H) = B(H, H)$. An *adjoint* of T is an operator $T^* : K \to H$ satisfying

$$\langle x, Ty\rangle = \langle T^* x, y\rangle.$$

An operator is self-adjoint when $T = T^*$, unitary when $T^*T = TT^* = I$, and normal when $T^*T = TT^*$. An *orthogonal projection* is an operator P satisfying $P = P^* = P^2$. A bounded operator is *compact* when it maps the closed unit ball to a subset with compact closure.

Exercise 1.5 (Hilbert-space foundations)**.** Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \qquad (\operatorname{im} T)^{\perp} = \ker T^*.$$

For every closed subspace $M \subseteq H$, prove $H = M \oplus M^{\perp}$ and construct the orthogonal projection $P_M : H \to M$.

Definition 1.10 (Measurable spaces)**.** A *σ-algebra* $\mathcal{F}$ on a set X contains $\varnothing$ and is closed under complements and countable unions. A *measure* is a countably additive map $\mu : \mathcal{F} \to [0, \infty]$ with $\mu(\varnothing) = 0$, and $(X, \mathcal{F}, \mu)$ is a *measure space*. A map is *measurable* when inverse images of measurable sets are measurable.

Definition 1.11 (L^p spaces)**.** For a measure space $(X, \mathcal{F}, \mu)$ and $1 \le p < \infty$, define

$$L^p(X,\mu) = \left\{ f : X \to \mathbb{C} \text{ measurable} : \int_X |f|^p \, d\mu < \infty \right\} /\text{equality a.e..}$$

Its norm is

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p}.$$

Let $L^\infty(X, \mu)$ consist of essentially bounded measurable functions modulo equality almost everywhere, with

$$\|f\|_\infty = \operatorname*{ess\,sup}_{x \in X} |f(x)|.$$

Exercise 1.6 (Lebesgue integration and convergence)**.** Construct the Lebesgue integral from nonnegative simple functions. Prove the monotone convergence theorem and Fatou's lemma.

Exercise 1.7 (Dominated convergence)**.** Prove the dominated convergence theorem by applying Fatou's lemma to an integrable dominating function. Deduce convergence in L^1 and justify interchanging a convergent series with an integral under an integrable majorant.

Exercise 1.8 (Product integration)**.** Construct product measure from measurable rectangles. Prove Tonelli's theorem for nonnegative functions and Fubini's theorem for integrable functions, stating the hypotheses that distinguish them. Derive the change-of-variables formula for a C^1 diffeomorphism between open subsets of finite-dimensional Euclidean spaces.

Exercise 1.9 (L^p geometry)**.** Prove Hölder's and Minkowski's inequalities and deduce that $L^p(X, \mu)$ is complete for $1 \le p \le \infty$. For $1 < p < \infty$ with $p^{-1} + q^{-1} = 1$, assume that $(X, \mathcal{F}, \mu)$ is σ-finite and prove that

$$L^q(X,\mu) \longrightarrow L^p(X,\mu)^*, \qquad g \longmapsto \left(f \mapsto \int_X \overline{g} f \, d\mu \right)$$

is an isometric isomorphism.

Definition 1.12 (Gamma and beta integrals)**.** For $s, t > 0$, define

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} \, dx, \qquad B(s,t) = \int_0^1 x^{s-1} (1-x)^{t-1} \, dx.$$

Exercise 1.10 (Beta–gamma identities)**.** Use Tonelli's theorem and the change of variables $x = ru$, $y = r(1-u)$ to prove

$$B(s,t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)}.$$

Use integration by parts to prove $\Gamma(s+1) = s\Gamma(s)$, and use the pushforward of Euclidean measure under the norm map to obtain

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \qquad \int_{\mathbb{R}^n} e^{-|x|^2/2} \, dx = (2\pi)^{n/2}.$$

State the uses of Tonelli, Fubini, monotone convergence, and dominated convergence in the derivation.

Definition 1.13 (C^*-algebras, spectrum, and positivity)**.** A *C^*-algebra* is a complete normed complex algebra A with a conjugate-linear involution satisfying

$$(ab)^* = b^* a^*, \qquad \|ab\| \le \|a\| \|b\|, \qquad \|a^* a\| = \|a\|^2.$$

For a unital C^*-algebra, define

$$\operatorname{spec}_A(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\}.$$

An element is *positive*, written $a \geq 0$, when $a = b^*b$ for some $b \in A$. It is *self-adjoint* when $a = a^*$ and *normal* when $a^*a = aa^*$.

Definition 1.14 (Gelfand spectrum)**.** For a commutative C^*-algebra A, let $\widehat{A}$ be the space of nonzero homomorphisms $\chi : A \to \mathbb{C}$ with its weak-$*$ topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\widehat{A}), \qquad \Gamma(a)(\chi) = \chi(a).$$

Definition 1.15 (Positive functionals and states)**.** A linear functional $\varphi : A \to \mathbb{C}$ on a unital C^*-algebra is *positive* when $\varphi(a^*a) \geq 0$ for every $a \in A$. A *state* is a positive functional satisfying $\varphi(1) = 1$.

Exercise 1.11 (Spectrum and spectral radius)**.** Prove that $\operatorname{spec}_A(a)$ is nonempty and compact and that

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}.$$

For normal a, prove $r(a) = \|a\|$.

Exercise 1.12 (Commutative Gelfand theory)**.** Prove that every character of a commutative C^*-algebra is continuous and that the Gelfand transform is an isometric $*$-isomorphism $A \cong C_0(\widehat{A})$.

Exercise 1.13 (Continuous functional calculus)**.** For every normal element a of a unital C^*-algebra, apply commutative Gelfand theory to $C^*(1, a)$ and prove that there is a unique unital isometric $*$-homomorphism

$$C(\operatorname{spec}(a)) \longrightarrow C^*(1, a), \qquad f \longmapsto f(a),$$

sending the coordinate function to a, and prove the continuous spectral mapping theorem. Deduce that positive elements have unique positive square roots and that $a \geq 0$ if and only if $a = a^*$ and $\operatorname{spec}_A(a) \subseteq [0, \infty)$.

Definition 1.16 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X is a map E from measurable subsets of X to orthogonal projections on H satisfying

$$E(\varnothing) = 0, \qquad E(X) = I, \qquad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral $\int_X f\,dE$ for bounded measurable f is defined by uniform approximation from simple functions. For a topological space X, the measure is *regular* when every scalar measure $B \mapsto \langle x, E(B)y\rangle$ is a regular Borel measure.

Exercise 1.14 (Positive Riesz–Markov representation)**.** Let X be locally compact Hausdorff. Prove that every positive linear functional $\varphi : C_c(X) \to \mathbb{C}$ that is bounded on functions supported in each compact set has a unique regular Borel measure μ_φ satisfying

$$\varphi(f) = \int_X f\,d\mu_\varphi.$$

Exercise 1.15 (Complex Riesz–Markov representation)**.** Prove that the real and imaginary parts of a bounded functional on $C_0(X)$ are differences of positive functionals. Apply the positive Riesz–Markov theorem to prove that every $\varphi \in C_0(X)^*$ has a unique finite complex regular measure μ_φ, and prove $\|\varphi\| = |\mu_\varphi|(X)$.

Exercise 1.16 (Spectral distributions of states)**.** Let A be a unital C^*-algebra, let $a = a^* \in A$, and let φ be a state. Apply the continuous functional calculus and Riesz–Markov theorem to prove that there is a unique Borel probability measure $\mu_{\varphi,a}$ on $\mathrm{spec}(a)$ such that

$$\varphi(f(a)) = \int_{\mathrm{spec}(a)} f(\lambda)\, d\mu_{\varphi,a}(\lambda), \qquad f \in C(\mathrm{spec}(a)).$$

Deduce

$$\varphi(a) = \int \lambda\, d\mu_{\varphi,a}(\lambda), \qquad \varphi(a^2) - \varphi(a)^2 = \int (\lambda - \varphi(a))^2\, d\mu_{\varphi,a}(\lambda).$$

Exercise 1.17 ($C_0(X)$ representations)**.** Let X be locally compact Hausdorff. A representation $\pi : C_0(X) \to B(H)$ is *nondegenerate* when $\overline{\pi(C_0(X))H} = H$. Prove that every nondegenerate $*$-representation determines scalar measures by applying the preceding exercise to $f \mapsto \langle x, \pi(f)y\rangle$. Use polarization and multiplicativity to assemble them into a unique regular projection-valued measure E_π satisfying

$$\pi(f) = \int_X f(x)\, dE_\pi(x), \qquad f \in C_0(X).$$

Exercise 1.18 (Bounded normal spectral theorem)**.** For a normal operator $T \in B(H)$, apply the preceding representation theorem to the continuous functional calculus $C(\mathrm{spec}(T)) \to B(H)$ and prove that there is a unique projection-valued measure E_T on $\mathrm{spec}(T)$ satisfying

$$T = \int_{\mathrm{spec}(T)} z\, dE_T(z), \qquad f(T) = \int_{\mathrm{spec}(T)} f(z)\, dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space has the intrinsic spectral decomposition

$$H = \bigoplus_{\lambda \in \mathrm{spec}(T)} E_T(\{\lambda\})H, \qquad T|_{E_T(\{\lambda\})H} = \lambda I.$$

Exercise 1.19 (Compact spectral theorem)**.** For a compact normal operator K, prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_{\lambda \in \mathrm{spec}(K) \setminus \{0\}} \lambda E_K(\{\lambda\})$$

in operator norm, and prove that the spectral subspaces in the displayed sum are mutually orthogonal.

Definition 1.17 (Unbounded self-adjoint operators)**.** An unbounded operator is a linear map $A : \mathcal{D}(A) \to H$ on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax\rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For $y \in \mathcal{D}(A^*)$, the vector A^*y is determined by $\langle A^*y, x\rangle = \langle y, Ax\rangle$ for $x \in \mathcal{D}(A)$. The operator is *self-adjoint* when $A = A^*$, including equality of domains, and *closed* when its graph is closed in $H \oplus H$.

Exercise 1.20 (Cayley transform)**.** For a self-adjoint operator A, prove that $A \pm iI$ are bijective from $\mathcal{D}(A)$ to H with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with $\ker(I - C_A) = 0$. Prove conversely that every unitary C with $\ker(I - C) = 0$ determines a self-adjoint operator through the inverse Cayley transform.

Exercise 1.21 (Unbounded spectral theorem)**.** Apply the bounded normal spectral theorem to C_A and transport its projection-valued measure by the inverse Cayley map. Prove that there is a unique projection-valued measure E_A on $\mathbb{R}$ satisfying

$$A = \int_{\mathbb{R}} \lambda \, dE_A(\lambda), \qquad \mathcal{D}(A) = \left\{ x : \int_{\mathbb{R}} \lambda^2 \, d\langle x, E_A(\lambda)x\rangle < \infty \right\}.$$

Exercise 1.22 (Stone's theorem)**.** Use the measurable functional calculus of E_A to prove that e^{-itA} is a strongly continuous unitary group with generator $-iA$. Conversely, let $(U_t)_{t\in\mathbb{R}}$ be a strongly continuous unitary group and define

$$\mathcal{D}(A) = \left\{ x : \lim_{t\to 0} i\frac{U_t x - x}{t} \text{ exists} \right\}, \qquad Ax = \lim_{t\to 0} i\frac{U_t x - x}{t}.$$

Prove that $\mathcal{D}(A)$ is dense and that A is symmetric. Use the strong integrals

$$i\int_0^{\infty} e^{-t} U_t \, dt, \qquad -i\int_{-\infty}^{0} e^{t} U_t \, dt$$

to prove that $A - iI$ and $A + iI$ are surjective. Deduce that A is self-adjoint, prove $U_t = e^{-itA}$, and prove uniqueness of A.

Exercise 1.23 (Spring and string frequencies)**.** Let V be a finite-dimensional real vector space with positive-definite mass form m and nonnegative stiffness form k. Use m to define the self-adjoint operator A by $k(x, y) = m(x, Ay)$. Prove that solutions of $\ddot{x} + Ax = 0$ decompose intrinsically through the spectral projections of A and that the observed angular frequencies are $\sqrt{\lambda}$ for $\lambda \in \operatorname{spec}(A) \setminus \{0\}$.

For a stretched string of length L, wave speed c, and fixed endpoints, let D be the self-adjoint operator on $L^2((0, L); \mathbb{C})$ given by

$$D\phi = -c^2\phi'',$$

on the domain of functions ϕ for which ϕ and ϕ' are absolutely continuous on $[0, L]$, $\phi'' \in L^2((0, L))$, and $\phi(0) = \phi(L) = 0$. For the wave equation

$$\partial_t^2 u + Du = 0,$$

prove that a nonzero time-harmonic solution $u(t, x) = \operatorname{Re}(e^{i\omega t}\phi(x))$, with $\omega > 0$, exists exactly when, for some $n \geq 1$,

$$\omega = \omega_n = \frac{n\pi c}{L}, \qquad \phi \in (\mathbb{C}\phi_n) \setminus \{0\}, \qquad \phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right),$$

Deduce the harmonic resonance ratios and the nodal locations measured by driving the string at each ω_n.

Exercise 1.24 (Spectral lines and time evolution)**.** Let A be self-adjoint, let $B \in B(H)$, and set $B(t) = e^{itA/\hbar} B e^{-itA/\hbar}$. For eigenvalues λ, μ of A, write $P_\lambda = E_A(\{\lambda\})$ and prove

$$P_\mu B(t) P_\lambda = e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

Use the functional calculus to prove the factorized weak identity

$$B(t) = \left(\int_{\mathbb{R}} e^{it\mu/\hbar} \, dE_A(\mu) \right) B \left(\int_{\mathbb{R}} e^{-it\lambda/\hbar} \, dE_A(\lambda) \right).$$

When A has pure-point spectrum, deduce in the weak operator topology that

$$B(t) = \sum_{\lambda,\mu} e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

For an initial state ψ, define the line strength

$$I_{\mu\lambda}(\psi) = \|P_\mu B P_\lambda \psi\|^2.$$

Prove that a line at $(\mu - \lambda)/\hbar$ occurs only when this quantity is nonzero, and identify vanishing spectral blocks as selection rules.

Definition 1.18 (Hermite polynomials and functions)**.** Define the physicists' Hermite polynomials by

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}.$$

The normalized *Hermite functions* are

$$h_n(x) = \frac{e^{-x^2/2} H_n(x)}{(2^n n! \sqrt{\pi})^{1/2}}.$$

Exercise 1.25 (Hermite orthogonality)**.** Use the generating function and the Gaussian integral to prove that $(h_n)_{n\geq 0}$ is an orthonormal family in $L^2(\mathbb{R})$. Derive the Hermite differential equation and recurrence relations.

Exercise 1.26 (Mehler kernel)**.** For $|r| < 1$, use the Hermite generating function and Gaussian integration to derive Mehler's identity

$$\sum_{n=0}^{\infty} r^n h_n(x) h_n(y) = \frac{1}{\sqrt{\pi(1-r^2)}} \exp\left(-\frac{(1+r^2)(x^2+y^2) - 4rxy}{2(1-r^2)} \right).$$

Exercise 1.27 (Hermite completeness)**.** Let K_r be the integral operator with the Mehler kernel for $0 < r < 1$. Use its explicit Gaussian form to prove

$$K_r f \longrightarrow f \qquad \text{in } L^2(\mathbb{R}) \quad (r \uparrow 1).$$

Prove that the image of K_r lies in the closed linear span of the Hermite functions and deduce that $(h_n)_{n\geq 0}$ is complete in $L^2(\mathbb{R})$.

Exercise 1.28 (Oscillator spectroscopy)**.** On Schwartz functions let

$$(Q\psi)(x) = x\psi(x), \qquad (P\psi)(x) = -i\hbar\psi'(x).$$

For

$$\widehat{H} = -\frac{\hbar^2}{2M}\frac{d^2}{dx^2} + \frac{M\omega^2 x^2}{2},$$

use the Hermite functions to prove that

$$\psi_n(x) = \left(\frac{M\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{M\omega}{\hbar}}x\right), \qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right)$$

give the complete spectral decomposition of $\widehat{H}$. Construct the creation and annihilation operators and compute

$$\langle \psi_k, Q\psi_n\rangle = \sqrt{\frac{\hbar}{2M\omega}}(\sqrt{n+1}\,\delta_{k,n+1} + \sqrt{n}\,\delta_{k,n-1}).$$

Deduce the spectral-line prediction $|E_{n\pm 1} - E_n|/\hbar = \omega$ and the relative line strengths for a linearly coupled oscillator. For $\alpha \in \mathbb{C}$, prove convergence and normalization of

$$\psi_\alpha = e^{-|\alpha|^2/2}\sum_{n\ge 0}\frac{\alpha^n}{\sqrt{n!}}\psi_n,$$

show that its number statistics are Poisson with mean $|\alpha|^2$, and derive its sinusoidal position expectation. Analytically continue Mehler's identity with the usual $t - i0$ prescription to obtain the oscillator propagator for $t \notin (\pi/\omega)\mathbb{Z}$,

$$K_t(x,y) = \sqrt{\frac{M\omega}{2\pi i\hbar\sin(\omega t)}}\exp\left[\frac{iM\omega}{2\hbar\sin(\omega t)}((x^2+y^2)\cos(\omega t) - 2xy)\right].$$

Determine the Maslov phase obtained by continuation across each caustic. Prove

$$e^{-2\pi i\widehat{H}/(\hbar\omega)} = -I, \qquad e^{-4\pi i\widehat{H}/(\hbar\omega)} = I,$$

and distinguish the projective and observable revival period $2\pi/\omega$ from the exact operator period $4\pi/\omega$. Use these formulas to predict the spectral spacing, coherent oscillation frequency, occupation-number distribution, and revival period measured in an oscillator experiment.

Definition 1.19 (Airy function)**.** The Airy function Ai is the unique real solution of

$$\mathrm{Ai}''(x) = x\,\mathrm{Ai}(x)$$

normalized by

$$\mathrm{Ai}(x) \sim \frac{x^{-1/4}}{2\sqrt{\pi}}e^{-2x^{3/2}/3} \qquad (x\to+\infty).$$

Let $a_n > 0$ be determined by $\mathrm{Ai}(-a_n) = 0$, ordered increasingly.

Exercise 1.29 (Airy spectrum of the quantum bouncer)**.** On $L^2((0,\infty), dz)$ consider the self-adjoint Dirichlet realization

$$\widehat{H} = -\frac{\hbar^2}{2M}\frac{d^2}{dz^2} + Mgz, \qquad \psi(0) = 0.$$

With

$$\ell_g = \left(\frac{\hbar^2}{2M^2 g}\right)^{1/3},$$

derive

$$E_n = Mg\ell_g a_n, \qquad \psi_n(z) = C_n \operatorname{Ai}\left(\frac{z}{\ell_g} - a_n\right).$$

Use the spectral projections of $\widehat{H}$ to prove that a periodic perturbation $B\cos(\Omega t)$ can produce a transition from the nth to the kth level only through the block $E_{\widehat{H}}(\{E_k\})BE_{\widehat{H}}(\{E_n\})$, at the resonant frequency

$$\Omega_{kn} = \frac{Mg\ell_g}{\hbar}|a_k - a_n|.$$

Deduce the classical turning heights $z_n = \ell_g a_n$. Model an absorber with lower face at height h, width w, and loss strength $\Gamma > 0$ by the potential $-i\Gamma\mathbf{1}_{[h,h+w]}$. In first-order decay theory, derive

$$\gamma_n(h) = \frac{2\Gamma}{\hbar}\int_h^{h+w} |\psi_n(z)|^2\, dz$$

and the transmission factor $e^{-T\gamma_n(h)}$ after flight time T. Use the Airy asymptotic to prove that the onset occurs across a height interval of order ℓ_g near z_n, rather than as a sharp step. State how measured onset heights and resonance frequencies determine g, and separate the predicted Airy scales from absorber roughness, finite width, and loss.

2 Fourier Analysis

Definition 2.1 (Smooth manifolds and tangent spaces)**.** A *smooth manifold* of dimension n is a Hausdorff, second-countable space with a maximal atlas of homeomorphisms to open subsets of $\mathbb{R}^n$ whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of a smooth map $F : M \to N$ is

$$dF_p : T_pM \longrightarrow T_{F(p)}N, \qquad dF_p(v)(f) = v(f \circ F).$$

Definition 2.2 (Tangent and cotangent bundles)**.** Define

$$TM = \coprod_{p\in M} T_pM, \qquad T^*M = \coprod_{p\in M} T_p^*M.$$

A smooth vector bundle is a smooth map $\pi : E \to M$ locally isomorphic to $U \times W \to U$ for a finite-dimensional real vector space W, by fiberwise linear maps. A section is a smooth map $s : M \to E$ satisfying $\pi s = \mathrm{id}_M$. The bundles TM and T^*M have the bundle structures induced by the differentials of chart maps and their duals.

Definition 2.3 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. Its local flow is a family of local diffeomorphisms Φ_t satisfying

$$\Phi_0 = \mathrm{id}_M, \qquad \Phi_{s+t} = \Phi_s \circ \Phi_t, \qquad \frac{d}{dt} f(\Phi_t(p)) = X_{\Phi_t(p)}(f)$$

where both sides are defined. For vector fields X, Y, define

$$[X,Y](f) = X(Yf) - Y(Xf).$$

Definition 2.4 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Definition 2.5 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Definition 2.6 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Definition 2.7 (Lie groups)**.** A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Definition 2.8 (Riemannian metrics and Laplacians)**.** A *Riemannian metric* on an n-manifold M is a smooth family of positive-definite inner products g_p on T_pM. Its Riemannian density is characterized by

$$|\operatorname{vol}_g|_p(v_1 \wedge \cdots \wedge v_n) = \det\big(g_p(v_i, v_j)\big)^{1/2},$$

and determines the Riemannian measure $d\operatorname{vol}_g$. The gradient is defined by

$$g(\operatorname{grad}_g f, X) = Xf.$$

The divergence is determined by

$$\int_M Xf \, d\operatorname{vol}_g = -\int_M f \, \operatorname{div}_g X \, d\operatorname{vol}_g$$

for compactly supported f, and the Laplace–Beltrami operator is $\Delta_g = \operatorname{div}_g \operatorname{grad}_g$.

Exercise 2.1 (Riemannian invariance)**.** Prove that an isometry preserves the Riemannian density, gradient, divergence, and Laplace–Beltrami operator. Prove that a left-invariant Riemannian metric on a Lie group has left-invariant Riemannian measure. For the Euclidean metric, recover Lebesgue measure and $\Delta = \sum_j \partial_j^2$.

Definition 2.9 (Haar normalization)**.** Every locally compact group below is equipped with a chosen nonzero left Haar measure μ. Haar measure on a compact group is normalized by $\mu(G) = 1$.

Definition 2.10 (Regular representation)**.** Let G be a locally compact group with left Haar measure μ. For $f, g \in C_c(G)$, define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) \, d\mu(y).$$

The *left regular representation* on $L^2(G)$ is

$$(\lambda_s \xi)(t) = \xi(s^{-1}t).$$

Exercise 2.2 (Haar regular representation)**.** Identify Haar measure on $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$. Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the L^1 convolution inequality, and $\|f * g\|_2 \le \|f\|_1 \|g\|_2$. Prove that $g \mapsto \lambda_g$ is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over G to obtain an invariant one.

Definition 2.11 (Pontryagin dual)**.** For a locally compact abelian group G, its *Pontryagin dual* $\widehat{G}$ is the group of continuous characters $\chi : G \to U(1)$, with pointwise multiplication and the compact-open topology. For $f \in L^1(G)$, define

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, d\mu(x).$$

The *double-dual map* sends $x \in G$ to the character $\chi \mapsto \chi(x)$ of $\widehat{G}$.

Exercise 2.3 (Abelian Fourier duality)**.** Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \qquad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \qquad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. For a linear subspace $W \subseteq \mathbb{R}^n$ and a full-rank lattice $\Gamma \subseteq \mathbb{R}^n$, compute their annihilators and identify the duals of $\mathbb{R}^n/W$ and $\mathbb{R}^n/\Gamma$.

Definition 2.12 (Fourier series)**.** On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, write

$$dm_{\mathbb{T}}(x) = \frac{dx}{2\pi}$$

for normalized Haar measure and define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x)e^{-ikx}\, dm_{\mathbb{T}}(x), \qquad k \in \mathbb{Z}.$$

The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}$.

Definition 2.13 (Schwartz space and convolution)**.** The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ consists of the smooth functions f satisfying

$$\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta f(x)| < \infty$$

for all multi-indices α, β. For integrable functions, define

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)\, dy.$$

Definition 2.14 (Fourier transform)**.** Let V be an n-dimensional real Euclidean space with Lebesgue measure dq. Equip V^* with its dual Euclidean structure and the resulting Lebesgue measure dp. For $f \in \mathcal{S}(V)$, the semiclassical Fourier transform is

$$(\mathcal{F}_\hbar f)(p) = (2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q)\, dq.$$

Write $\mathcal{F} = \mathcal{F}_1$.

Exercise 2.4 (Fourier series and Plancherel)**.** Prove that $(e^{ikx})_{k\in\mathbb{Z}}$ is a complete orthonormal family in $L^2(\mathbb{T})$ and derive

$$f = \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}, \qquad \|f\|_2^2 = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

Exercise 2.5 (Euclidean Fourier inversion)**.** Prove Fourier inversion on $\mathcal{S}(V)$ and prove that $\mathcal{F}_\hbar$ extends uniquely to a unitary operator on $L^2(V)$. For $v \in V$, prove

$$\mathcal{F}_\hbar(-i\hbar d_v)\mathcal{F}_\hbar^{-1}\widehat{f}(p) = p(v)\widehat{f}(p).$$

Exercise 2.6 (Free-particle dispersion)**.** Substitute the plane wave $\psi(q,t) = e^{i(k(q)-\omega t)}$, with $k \in V^*$, into the free Schrödinger equation and prove

$$\hbar\omega = \frac{\hbar^2\|k\|^2}{2m}.$$

Set $p = \hbar k$ and $E = \hbar\omega$. Derive

$$E(p) = \frac{\|p\|^2}{2m}, \qquad v_{\text{group}} = \frac{p^\sharp}{m},$$

where $p^\sharp \in V$ is the Euclidean-dual vector to p. Deduce the time-of-flight displacement of a wave packet concentrated near p_0 and the de Broglie relation between its measured wavelength and momentum.

Definition 2.15 (Bessel functions)**.** For $\alpha > -1$, the Bessel function of the first kind is

$$J_\alpha(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,\Gamma(m+\alpha+1)}\left(\frac{z}{2}\right)^{2m+\alpha},$$

with a branch of z^α fixed when $\alpha \notin \mathbb{Z}$.

Exercise 2.7 (Radial Bessel transform)**.** Prove

$$z^2 J_\alpha''(z) + zJ_\alpha'(z) + (z^2 - \alpha^2)J_\alpha(z) = 0$$

and the recurrence identities

$$J_{\alpha-1}(z) + J_{\alpha+1}(z) = \frac{2\alpha}{z}J_\alpha(z), \qquad J_{\alpha-1}(z) - J_{\alpha+1}(z) = 2J_\alpha'(z).$$

Average the Euclidean Fourier kernel over spheres and express the Fourier transform of a radial Schwartz function on $\mathbb{R}^n$ as a Hankel transform with kernel $J_{n/2-1}$. Derive the corresponding inversion identity.

Exercise 2.8 (Bessel diffraction)**.** Let $j_{\nu,k}$ be the kth positive zero of J_ν. Separate the wave equation on a circular membrane of radius R with Dirichlet boundary and prove that its normal-mode frequencies are

$$\omega_{m,k} = \frac{c}{R}j_{|m|,k},$$

with radial factor $J_{|m|}(j_{|m|,k}r/R)$. Deduce the nodal circles and angular nodal lines and state how measured resonance ratios determine the ratios of Bessel zeros without knowledge of c/R.

For a uniformly illuminated circular aperture of radius a, average the Fraunhofer kernel over the aperture, put $k = 2\pi/\lambda$, and derive

$$\frac{I(\theta)}{I(0)} = \left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2.$$

Prove that the dark-ring angles satisfy $ka\sin\theta_k = j_{1,k}$ and obtain the small-angle prediction $\theta_1 \sim j_{1,1}\lambda/(2\pi a)$. Identify which membrane resonances and diffraction minima measure zeros of the same Bessel functions.

Definition 2.16 (Lattices and reciprocal lattices)**.** A *lattice* in $\mathbb{R}^n$ is a subgroup $\Gamma = A\mathbb{Z}^n$ for some invertible linear map $A : \mathbb{R}^n \to \mathbb{R}^n$. Its covolume is $\mathrm{vol}(\Gamma) = |\det A|$, and a fundamental cell is $F = A[0,1)^n$. Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

Definition 2.17 (Periodic summation and structure factors)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, its Γ-*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma\in\Gamma} f(x+\gamma).$$

For a finite motif $B \subset \mathbb{R}^n$ with scattering weights c_b, its *structure factor* is

$$S_B(q) = \sum_{b\in B} c_b e^{-iq\cdot b}.$$

Exercise 2.9 (Poisson summation)**.** Prove that $P_\Gamma f$ converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\mathrm{vol}(\Gamma)} \sum_{\xi\in\Gamma^*} \mathcal{F}f(\xi) e^{i\xi\cdot x}.$$

Set $x = 0$ to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

Exercise 2.10 (Crystal diffraction and Bragg peaks)**.** Let Γ_N be a finite rectangular portion of Γ. In the single-scattering approximation, prove that the amplitude at momentum transfer q factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma\in\Gamma_N} e^{-iq\cdot\gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at $q \in \Gamma^*$ as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude $2\pi/\lambda$, derive the Laue condition and, for planes spaced by d, the Bragg relation

$$2d\sin\theta = m\lambda.$$

Show how zeros of S_B produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

3 Peter–Weyl Theory

Exercise 3.1 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 3.2 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 3.1 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h\rangle_G = \frac{1}{|G|}\sum_{g\in G} \overline{f(g)}h(g).$$

Exercise 3.3 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that $\langle \chi_U, \chi_V\rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Definition 3.2 (Matrix coefficients)**.** A *unitary representation* of a topological group G on a Hilbert space H_π is a homomorphism $\pi : G \to U(H_\pi)$ such that $g \mapsto \pi(g)x$ is continuous for every $x \in H_\pi$. Its *matrix coefficients* are

$$g \longmapsto \langle y, \pi(g)x\rangle, \qquad x, y \in H_\pi.$$

It is *irreducible* when it has no proper nonzero closed invariant subspace. The *unitary dual* $\widehat{G}$ is the set of unitary-equivalence classes of irreducible unitary representations.

Exercise 3.4 (Exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 3.3 (Infinitesimal representations)**.** For a finite-dimensional smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \left.\frac{d}{dt}\right|_0 \rho(\exp tX).$$

Exercise 3.5 (Compact convolution operators)**.** Let G be compact with normalized Haar measure and $k \in C(G)$. Prove that

$$(T_k f)(x) = \int_G k(y^{-1}x)f(y)\,dy$$

defines a compact operator on $L^2(G)$ by uniformly approximating its continuous kernel by finite sums of product functions. Prove that T_k commutes with left translations and that $k(g^{-1}) = \overline{k(g)}$ makes T_k self-adjoint. Deduce that every nonzero eigenspace of such a self-adjoint convolution operator is a finite-dimensional invariant subspace of the left regular representation.

Exercise 3.6 (Density of matrix coefficients)**.** Construct continuous convolution kernels that approximate the identity on $C(G)$. Apply the compact spectral theorem to their self-adjoint symmetrizations and prove that the linear span of matrix coefficients of finite-dimensional unitary representations is uniformly dense in $C(G)$ and dense in $L^2(G)$.

Exercise 3.7 (Schur orthogonality)**.** For irreducible finite-dimensional unitary representations π and σ, prove that

$$A \longmapsto \int_G \pi(g)A\sigma(g)^*\,dg$$

is zero when $\pi \not\simeq \sigma$ and is $A \mapsto \mathrm{tr}(A)I_{H_\pi}/\dim H_\pi$ when $\pi = \sigma$. Deduce that the coefficient map identifies $H_\pi \otimes H_\pi^\vee$, after multiplication by $\sqrt{\dim H_\pi}$, with an isometric subspace of $L^2(G)$, and that distinct irreducible coefficient spaces are orthogonal, where

$$C_\pi : H_\pi \otimes H_\pi^\vee \longrightarrow L^2(G), \qquad C_\pi(x\otimes\varphi)(g) = \varphi(\pi(g)x).$$

Exercise 3.8 (Peter–Weyl decomposition)**.** Use density and Schur orthogonality to prove

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}}} H_\pi \otimes H_\pi^\vee,$$

with the left regular representation acting as $\pi \otimes I_{H_\pi^\vee}$ on the π-summand. For $f \in L^2(G) \subseteq L^1(G)$, define $\widehat{f}(\pi) = \int_G f(g)\pi(g)^*\,dg$ and prove

$$f = \sum_{\pi \in \widehat{G}} (\dim H_\pi)\,\mathrm{tr}\big(\widehat{f}(\pi)\pi(\,\cdot\,)\big), \qquad \|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim H_\pi)\,\mathrm{tr}\big(\widehat{f}(\pi)^*\widehat{f}(\pi)\big),$$

where the first sum converges in $L^2(G)$. Prove uniform convergence on $C(G)$ for the summability means supplied by the approximate identities in the density exercise.

Exercise 3.9 (Character inversion)**.** Let G be compact with normalized Haar measure, and let f be a finite linear combination of irreducible characters. For an irreducible unitary representation π, set

$$\widehat{f}(\pi) = \int_G f(s)\pi(s^{-1})\,ds.$$

If f is central, use Schur's lemma to write $\widehat{f}(\pi) = c_\pi I_{d_\pi}$. Derive

$$f(e) = \sum_{\pi \in \widehat{G}} d_\pi\,\mathrm{Tr}\big(\widehat{f}(\pi)\big) = \sum_{\pi \in \widehat{G}} d_\pi^2 c_\pi$$

and prove that convolution by f acts by the scalar c_π on every copy of π in $L^2(G)$.

Definition 3.4 (Rotations and spin)**.** Define

$$\begin{aligned} SO(3) &= \{R \in \mathrm{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\}, \\ SU(2) &= \{U \in \mathrm{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}. \end{aligned}$$

An *angular-momentum representation* is a finite-dimensional smooth unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 3.10 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations and the 4π restoration observed by neutron interferometry.

Exercise 3.11 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \ldots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \ldots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$. For $j = \frac{1}{2}$, relate the two J_3 outcomes to the two Stern–Gerlach beams.

Exercise 3.12 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 3.5 (Gauss hypergeometric function)**.** Using the rising factorial defined in Spectral Theory, for $c \notin \mathbb{Z}_{\le 0}$ the *Gauss hypergeometric function* is

$$ {}_2F_1\left(\begin{matrix} a, b \\ c \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a)_m (b)_m}{(c)_m} \frac{z^m}{m!}. $$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer and otherwise converges for $|z| < 1$. The Gauss hypergeometric operator is

$$ \mathcal{L}_{a,b,c} = z(1-z)\frac{d^2}{dz^2} + \big(c - (a+b+1)z\big)\frac{d}{dz} - ab. $$

Exercise 3.13 (Gauss equation and integral)**.** Prove that ${}_2F_1(a, b; c; z)$ is the solution of $\mathcal{L}_{a,b,c} f = 0$ analytic at $z = 0$ with $f(0) = 1$. Derive Euler's integral

$$ {}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\, du, $$

and Gauss's summation

$$ {}_2F_1(a, b; c; 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}. $$

State the parameter conditions for the integral and the summation.

Definition 3.6 (Kummer and Whittaker functions)**.** For $b \notin \mathbb{Z}_{\le 0}$, define the Kummer function

$$ M(a, b, z) = {}_1F_1(a; b; z) = \sum_{m=0}^{\infty} \frac{(a)_m}{(b)_m} \frac{z^m}{m!}. $$

For $2\mu + 1 \notin \mathbb{Z}_{\le 0}$, define

$$ M_{\lambda,\mu}(z) = e^{-z/2} z^{\mu+1/2} M\left(\mu - \lambda + \tfrac{1}{2}, 2\mu + 1, z\right), $$

with a branch of $z^{\mu+1/2}$ fixed. The Whittaker function $W_{\lambda,\mu}$ is the solution of

$$ w'' + \left(-\frac{1}{4} + \frac{\lambda}{z} + \frac{1/4 - \mu^2}{z^2}\right) w = 0 $$

normalized by $W_{\lambda,\mu}(z) \sim e^{-z/2} z^{\lambda}$ as $z \to +\infty$.

Definition 3.7 (Generalized Laguerre polynomials)**.** For $\alpha > -1$ and $n \in \mathbb{Z}_{\ge 0}$, define

$$ L_n^{(\alpha)}(x) = \frac{x^{-\alpha} e^x}{n!} \frac{d^n}{dx^n}\left(e^{-x} x^{n+\alpha}\right) = \frac{(\alpha+1)_n}{n!} M(-n, \alpha+1, x). $$

Write $L_n = L_n^{(0)}$.

Exercise 3.14 (Confluence and Laguerre functions)**.** Prove locally uniformly in z that

$$ \lim_{B \to +\infty} {}_2F_1\left(\begin{matrix} a, B \\ c \end{matrix}; \frac{z}{B}\right) = M(a, c, z), $$

and derive Kummer's equation

$$ zy'' + (b-z)y' - ay = 0. $$

Apply the substitution $w(z) = e^{-z/2} z^{\mu+1/2} y(z)$ to obtain the Whittaker equation with $a = \mu - \lambda + \frac{1}{2}$ and $b = 2\mu + 1$.

Derive the equality of the two displayed definitions of $L_n^{(\alpha)}$. Use the Rodrigues formula and integration by parts to prove

$$\int_0^\infty L_m^{(\alpha)}(x) L_n^{(\alpha)}(x) x^\alpha e^{-x}\, dx = \frac{\Gamma(n+\alpha+1)}{n!}\delta_{mn},$$

and derive

$$(n+1)L_{n+1}^{(\alpha)}(x) = (2n+\alpha+1-x)L_n^{(\alpha)}(x) - (n+\alpha)L_{n-1}^{(\alpha)}(x).$$

Prove

$$\sum_{n\geq 0} L_n^{(\alpha)}(x) t^n = (1-t)^{-\alpha-1}\exp\left(-\frac{xt}{1-t}\right), \qquad |t| < 1,$$

and obtain the corresponding coefficient contour integral around $t = 0$.

Definition 3.8 (Jacobi polynomials)**.** For $\alpha, \beta > -1$, the *Jacobi polynomial* is

$$P_n^{(\alpha,\beta)}(x) = \frac{(\alpha+1)_n}{n!}\, {}_2F_1\left(\begin{matrix} -n, n+\alpha+\beta+1 \\ \alpha+1 \end{matrix}; \frac{1-x}{2}\right).$$

Exercise 3.15 (Jacobi spectral equation)**.** Prove

$$\left((1-x^2)\frac{d^2}{dx^2} + (\beta - \alpha - (\alpha+\beta+2)x)\frac{d}{dx}\right) P_n^{(\alpha,\beta)} = -n(n+\alpha+\beta+1)P_n^{(\alpha,\beta)}.$$

Derive Jacobi orthogonality on $[-1,1]$ with weight $(1-x)^\alpha(1+x)^\beta$ and derive the three-term recurrence.

Exercise 3.16 (Spherical harmonics and atomic orbitals)**.** Identify S^2 with $SO(3)/SO(2)$ and $L^2(S^2)$ with the right $SO(2)$-invariant subspace of $L^2(SO(3))$. Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell$$

For normalized J_3-eigenstates $|\ell m\rangle$, define the Wigner matrix coefficients

$$D^\ell_{mn}(g) = \langle \ell m | \pi_\ell(g) | \ell n\rangle.$$

Prove that the functions $Y_{\ell m} \propto D^\ell_{m0}$, $-\ell \leq m \leq \ell$, form a complete orthogonal family in V_ℓ. For the unit round sphere, set

$$\Delta_{S^2} = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{\sin^2\theta}\partial_\phi^2.$$

Derive $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$ and completeness of the spherical harmonics. Derive the Jacobi realization

$$Y_{\ell m}(\theta,\phi) = C_{\ell m} e^{im\phi}(\sin\theta)^{|m|} P_{\ell-|m|}^{(|m|,|m|)}(\cos\theta)$$

with $C_{\ell m}$ fixed by unit norm. On

$$L^2(\mathbb{R}^3, d^3x) \cong L^2((0,\infty), r^2\,dr) \otimes L^2(S^2, d\Omega),$$

consider the Coulomb Hamiltonian

$$\widehat{H}_C = -\frac{\hbar^2}{2M}\Delta_{\mathbb{R}^3} - \frac{\kappa}{r}, \qquad M, \kappa > 0.$$

Separate $\widehat{H}_C$ in spherical coordinates and prove that its radial operator on the ℓth angular sector is

$$-\frac{\hbar^2}{2M}\left(\frac{d^2}{dr^2} + \frac{2}{r}\frac{d}{dr} - \frac{\ell(\ell+1)}{r^2}\right) - \frac{\kappa}{r}$$

on $L^2((0,\infty), r^2\,dr)$. Recover $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi)$. Put

$$a_0 = \frac{\hbar^2}{M\kappa}, \qquad \beta = \frac{\sqrt{-2ME}}{\hbar}, \qquad \rho = 2\beta r$$

for $E < 0$, and set $u(r) = rR(r)$. Reduce the radial eigenvalue equation to the Whittaker equation with

$$\lambda = \frac{1}{a_0\beta}, \qquad \mu = \ell + \frac{1}{2}.$$

Use regularity at the origin and square-integrability at infinity to prove that

$$\ell + 1 - \lambda = -n_r, \qquad n_r \in \mathbb{Z}_{\geq 0}.$$

Writing $n = n_r + \ell + 1$, deduce

$$E_n = -\frac{M\kappa^2}{2\hbar^2 n^2}, \qquad R_{n\ell}(r) \propto e^{-r/(na_0)}\left(\frac{2r}{na_0}\right)^{\ell} L_{n-\ell-1}^{(2\ell+1)}\left(\frac{2r}{na_0}\right).$$

Prove that the spatial multiplicity of E_n is n^2 and derive the spectral-line frequencies

$$\omega_{n_i \to n_f} = \frac{M\kappa^2}{2\hbar^3}\left|\frac{1}{n_f^2} - \frac{1}{n_i^2}\right|.$$

Determine the angular nodes of the s, p, and d orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare the predicted momentum-space nodal suppression with Weiß et al., *Nature Communications* **6** (2015), and distinguish the orbital-symmetry prediction, plane-wave approximation, measured intensity, and final-state scattering corrections.

Definition 3.9 (Clebsch–Gordan and Wigner coefficients)**.** Choose the normalized J_3-eigenstates in each irreducible angular-momentum representation by the Condon–Shortley ladder convention

$$J_-|jm\rangle = \hbar\sqrt{(j+m)(j-m+1)}\,|j, m-1\rangle.$$

Define

$$|jm\rangle = \sum_{m_1+m_2=m} C^{jm}_{j_1m_1,j_2m_2}|j_1m_1\rangle \otimes |j_2m_2\rangle,$$

with all coefficients real and with the coefficient of $|j_1, j_1\rangle \otimes |j_2, j - j_1\rangle$ in $|jj\rangle$ positive, and set

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j+1}} C^{j,-m_3}_{j_1m_1,j_2m_2}.$$

Define the Wigner $6j$ symbol by

$$|(j_1j_2)j_{12}, j_3; jm\rangle = \sum_{j_{23}} (-1)^{j_1+j_2+j_3+j}\sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1, (j_2j_3)j_{23}; jm\rangle.$$

Exercise 3.17 (Wigner and Jacobi functions)**.** For a rotation through angle β about the second axis, set

$$d^j_{m'm}(\beta) = \langle jm'|e^{-i\beta J_2/\hbar}|jm\rangle.$$

Use the $SU(2)$ differential equation and the value at $\beta = 0$ to express $d^j_{m'm}(\beta)$ as a normalized product of powers of $\sin(\beta/2)$ and $\cos(\beta/2)$ with a Jacobi polynomial in $\cos\beta$. Deduce its orthogonality from Peter–Weyl. Use the Clebsch–Gordan decomposition to derive the linearization formula for a product of two Wigner coefficients and identify its coefficients as products of Clebsch–Gordan coefficients.

Exercise 3.18 (Angular amplitudes)**.** Derive the orthogonality relations for the Clebsch–Gordan coefficients and the triangle and magnetic selection rules for the $3j$ symbols. Prove the Gaunt identity

$$\int_{S^2} \overline{Y_{\ell_f m_f}} Y_{1q} Y_{\ell_i m_i}\, d\Omega = (-1)^{m_f} \sqrt{\frac{3(2\ell_i+1)(2\ell_f+1)}{4\pi}} \begin{pmatrix} \ell_i & 1 & \ell_f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_i & 1 & \ell_f \\ m_i & q & -m_f \end{pmatrix}.$$

Deduce the electric-dipole rules $\ell_f = \ell_i \pm 1$ and $m_f = m_i + q$, and prove that the relative Zeeman-resolved line intensities are the squared absolute values of the second $3j$ symbol after a common reduced matrix element is removed. Use the $6j$ identity to derive the corresponding relative intensities when orbital and spin angular momenta are coupled in different orders. For an oriented initial state, express the dipole radiation pattern as the squared absolute value of the resulting spherical-harmonic amplitude and identify its symmetry-enforced dark directions and absent spectral lines.

4 Harish–Chandra Theory

Definition 4.1 (The rank-one symmetric-space model)**.** Let

$$G = SL_2(\mathbb{R}), \qquad K = SO(2),$$

and let G act on the upper half-plane by fractional linear transformations. Define

$$a_r = \begin{pmatrix} e^{r/2} & 0 \\ 0 & e^{-r/2} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $r, x \in \mathbb{R}$. Give the upper half-plane the metric $ds^2 = (dx^2 + dy^2)/y^2$ and fix its curvature radius to be one.

Exercise 4.1 (Rank-one Iwasawa and Cartan decompositions)**.** Use the polar and QR decompositions of an element of $SL_2(\mathbb{R})$ to prove

$$K \times \mathbb{R} \times \mathbb{R} \longrightarrow G, \qquad (k, r, x) \longmapsto k a_r n_x$$

is a diffeomorphism and

$$G = K\{a_r : r \geq 0\}K.$$

Prove that K is the stabilizer of i and that the orbit map identifies G/K isometrically with the hyperbolic plane. Identify the two-point Weyl group and prove that a K-bi-invariant function is determined by its value on a_r, $r \geq 0$.

Definition 4.2 (Rank-one spherical transform)**.** For $\lambda \in \mathbb{R}$, let φ_λ be the normalized smooth K-invariant Laplace–Beltrami eigenfunction on $G/K = \mathbb{H}^2$ satisfying

$$\Delta_{\mathbb{H}^2}\varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \qquad \varphi_\lambda(eK) = 1.$$

Regarded as a K-bi-invariant function on G and written in the radial coordinate $a_r K$, it is determined by

$$\varphi_\lambda''(r) + \coth r\, \varphi_\lambda'(r) = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda(r), \qquad \varphi_\lambda(0) = 1, \qquad \varphi_\lambda'(0) = 0.$$

For a compactly supported smooth radial function, define

$$\widetilde{f}(\lambda) = 2\pi \int_0^\infty f(r)\varphi_\lambda(r)\sinh r\, dr.$$

Define $c(\lambda)$ by the asymptotic normalization

$$\varphi_\lambda(r) = c(\lambda)e^{(i\lambda-1/2)r} + c(-\lambda)e^{(-i\lambda-1/2)r} + o(e^{-r/2})$$

for real $\lambda \neq 0$.

Exercise 4.2 (Conical eigenfunctions on $\mathbb{H}^2$)**.** For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\mathrm{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta\varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \qquad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For real λ, μ, derive the Wronskian identity

$$(\mu^2 - \lambda^2)\int_0^R \varphi_\lambda(r)\varphi_\mu(r)\sinh r\, dr = \Big[\sinh r\,(\varphi_\mu\varphi_\lambda' - \varphi_\lambda\varphi_\mu')\Big]_0^R.$$

Fix one self-adjoint boundary condition $a\varphi(R) + b\varphi'(R) = 0$, with $(a, b) \neq (0, 0)$. Prove orthogonality only for the discrete parameters λ_j satisfying this condition. Compare the large-r equation with a constant-coefficient equation and show how the discrete parameters become continuous as $R \to \infty$.

Exercise 4.3 (Harish–Chandra normalization)**.** Use the large-r asymptotics of the conical Legendre function and the Wronskian identity to derive

$$c(\lambda) = \frac{\Gamma(i\lambda)}{\sqrt{\pi}\,\Gamma(\frac{1}{2} + i\lambda)}, \qquad |c(\lambda)|^{-2} = \pi\lambda\tanh(\pi\lambda).$$

Exercise 4.4 (Mehler–Fock inversion)**.** For $\lambda, \mu > 0$, prove in the distributional sense that

$$2\pi\int_0^\infty \varphi_\lambda(r)\varphi_\mu(r)\sinh r\, dr = \frac{2\pi}{\lambda\tanh(\pi\lambda)}\delta(\lambda - \mu).$$

Deduce the Mehler–Fock inversion and Plancherel formulas

$$f(r) = \frac{1}{2\pi^2}\int_0^\infty \widetilde{f}(\lambda)P_{-1/2+i\lambda}(\cosh r)|c(\lambda)|^{-2}\, d\lambda,$$

$$\|f\|_2^2 = \frac{1}{2\pi^2}\int_0^\infty |\widetilde{f}(\lambda)|^2|c(\lambda)|^{-2}\, d\lambda.$$

Exercise 4.5 (Hyperbolic wave propagation)**.** Let $u_{tt} = \Delta_{\mathbb{H}^2} u$ have radial initial data $u(0,r) = f(r)$ and $u_t(0,r) = g(r)$ in C_c^∞. Put $\omega_\lambda = (\lambda^2 + 1/4)^{1/2}$. Use Mehler–Fock inversion to prove

$$u(t,r) = \frac{1}{2\pi^2} \int_0^\infty \left(\cos(t\omega_\lambda)\widetilde{f}(\lambda) + \frac{\sin(t\omega_\lambda)}{\omega_\lambda}\widetilde{g}(\lambda) \right) P_{-1/2+i\lambda}(\cosh r)|c(\lambda)|^{-2}\, d\lambda.$$

Deduce the radial arrival profile of a localized pulse and its high-frequency spectral density from $|c(\lambda)|^{-2} = \pi\lambda \tanh(\pi\lambda)$.

Exercise 4.6 (Hyperbolic-drum resonances)**.** Let $D_R \subset \mathbb{H}^2$ be the geodesic disk of radius R in curvature -1, impose Dirichlet boundary conditions, and let the wave speed be v. Prove that the radial resonances are indexed by the positive solutions of

$$P_{-1/2+i\lambda_j}(\cosh R) = 0$$

and have angular frequencies

$$\omega_j = v\sqrt{\lambda_j^2 + \tfrac{1}{4}}.$$

Use the c-function asymptotics to derive the large-R quantization law

$$\lambda_j R + \arg c(\lambda_j) = \left(j + \tfrac{1}{2}\right)\pi + o(1),$$

with the indexing convention stated explicitly, and deduce the predicted radial frequency spacings.

For the $\{3,7\}$ tessellation used in Lenggenhager et al., *Nature Communications* **13** (2022), let Q_R be the graph Laplacian in their sign convention, put $L_R = -Q_R \geq 0$, and write $\Lambda_j(L_R)$ for its eigenvalues. The experiment uses curvature -4 and lattice parameter $h = \tanh(0.545275)$. Prove the long-wavelength calibration

$$\Lambda_j(L_R) = \frac{7}{4}h^2\nu_j^{(-4)} + O(h^3), \qquad \nu_j^{(-4)} = 4\nu_j^{(-1)},$$

where $\nu_j^{(-1)} = \lambda_j^2 + \frac{1}{4}$ is the curvature--1 continuum eigenvalue. Prove that the metric and geodesic-radius normalizations satisfy $g_{-1} = 4g_{-4}$ and $R_{-1} = 2R_{-4}$.

For circuit capacitance C and grounding inductance L, derive from the grounded circuit Laplacian

$$J(\omega) = -i\omega C Q_R + \frac{1}{i\omega L} I$$

that the measured circuit resonance satisfies

$$\omega_j^{\mathrm{circ}} = \frac{1}{\sqrt{LC\,\Lambda_j(L_R)}}, \qquad \nu_j^{(-1)} \sim \frac{\Lambda_j(L_R)}{7h^2} = \frac{1}{28\pi^2 h^2 LC (f_j^{\mathrm{circ}})^2}.$$

Use this calibration to compare the inferred conical parameters, measured voltage eigenmodes, and radial phase fronts with the conical-function zeros and Mehler–Fock wave kernel. State separately the curvature rescaling, continuum prediction, circuit resonance law, measured quantities, and the errors caused by boundary shape, loss, and discretization.

5 Automorphic Forms

Definition 5.1 (Integral lattices and theta series)**.** A *Euclidean lattice* $L \subset \mathbb{R}^d$ is a discrete subgroup for which $\mathbb{R}^d/L$ is compact. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y\rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if $L \subseteq L^*$, *unimodular* if $L = L^*$, and *even* if $\|x\|^2 \in 2\mathbb{Z}$ for every $x \in L$. For τ in the upper half-plane and $q = e^{2\pi i\tau}$, its theta series is

$$\Theta_L(\tau) = \sum_{x\in L} q^{\|x\|^2/2}.$$

Its coefficient of q^n counts lattice vectors of squared length $2n$. If the same lattice is used as a crystal lattice, its physical reciprocal lattice is

$$L^*_{\text{physical}} = 2\pi L^*.$$

Exercise 5.1 (Poisson duality for lattice theta series)**.** Apply Exercise 2.9 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\mathrm{vol}(L)}\Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives $\Theta_L(\tau+1) = \Theta_L(\tau)$. Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

Definition 5.2 (The modular group and modular forms)**.** The modular group $SL_2(\mathbb{Z})$ acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$. A *modular form of integral weight k* for $SL_2(\mathbb{Z})$ is a holomorphic function f on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion $f(\tau) = \sum_{n\geq 0} a_n q^n$ has no negative powers. It is a *cusp form* if $a_0 = 0$. Write M_k for the vector space of modular forms of weight k.

Definition 5.3 (Zeta and Eisenstein functions)**.** For $\mathrm{Re}\, s > 1$, define the Riemann zeta function by

$$\zeta(s) = \sum_{n\geq 1} n^{-s}.$$

For a d-dimensional Euclidean lattice L, with $d \geq 1$, and $\mathrm{Re}\, s > d/2$, define its Epstein zeta function by

$$Z_L(s) = \sum_{x\in L\setminus\{0\}} \|x\|^{-2s}.$$

For $m \geq 2$ and τ in the upper half-plane, define

$$G_{2m}(\tau) = \sum_{(c,d)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{(c\tau + d)^{2m}}, \qquad E_{2m}(\tau) = \frac{G_{2m}(\tau)}{2\zeta(2m)}.$$

Exercise 5.2 (Theta Mellin transform and zeta functional equation)**.** Put $\theta_L(t) = \Theta_L(it)$. For $\operatorname{Re} s > d/2$, prove

$$\pi^{-s}\Gamma(s)Z_L(s) = \int_0^\infty (\theta_L(t) - 1)t^{s-1}\,dt.$$

Split the integral at 1 and use Poisson duality to prove the meromorphic continuation

$$\begin{aligned}\pi^{-s}\Gamma(s)Z_L(s) = &\int_1^\infty (\theta_L(t) - 1)t^{s-1}\,dt \\ &+ \frac{1}{\operatorname{vol}(L)}\int_1^\infty (\theta_{L^*}(t) - 1)t^{d/2-s-1}\,dt + \frac{1}{\operatorname{vol}(L)(s - d/2)} - \frac{1}{s}.\end{aligned}$$

Deduce

$$\pi^{-s}\Gamma(s)Z_L(s) = \frac{1}{\operatorname{vol}(L)}\pi^{-(d/2-s)}\Gamma(d/2 - s)Z_{L^*}(d/2 - s),$$

$Z_L(0) = -1$, and

$$\operatorname*{res}_{s=d/2} Z_L(s) = \frac{\pi^{d/2}}{\operatorname{vol}(L)\Gamma(d/2)}.$$

For $L = \mathbb{Z}$, use $Z_{\mathbb{Z}}(s) = 2\zeta(2s)$ to derive the functional equation of ζ. Use Fourier series and Parseval to prove

$$\zeta(2) = \frac{\pi^2}{6}, \qquad \zeta(4) = \frac{\pi^4}{90},$$

and then deduce

$$\zeta(-1) = -\frac{1}{12}, \qquad \zeta(-3) = \frac{1}{120}.$$

Exercise 5.3 (Holomorphic Eisenstein series)**.** Prove absolute local uniform convergence of G_{2m} and prove

$$G_{2m}\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{2m}G_{2m}(\tau)$$

for every $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$. Using Poisson summation in one lattice direction, derive

$$G_{2m}(\tau) = 2\zeta(2m) + \frac{2(2\pi i)^{2m}}{(2m-1)!}\sum_{n\geq 1}\sigma_{2m-1}(n)q^n, \qquad \sigma_r(n) = \sum_{d|n} d^r.$$

Deduce that E_{2m} is a modular form of weight $2m$ and that

$$E_4(\tau) = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n, \qquad E_6(\tau) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n.$$

Definition 5.4 (Theta and eta functions)**.** Set

$$\begin{aligned}\vartheta_2(\tau) &= \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n\in\mathbb{Z}} q^{n^2/2}, \qquad\qquad \eta(\tau) = q^{1/24}\prod_{m\geq 1}(1 - q^m). \\ \vartheta_4(\tau) &= \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2},\end{aligned}$$

Exercise 5.4 (Theta transformations)**.** Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2}\vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2}\vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

Use the branch positive for τ on the positive imaginary axis. Prove directly from the series that

$$\vartheta_2(\tau+1) = e^{\pi i/4}\vartheta_2(\tau), \qquad \vartheta_3(\tau+1) = \vartheta_4(\tau), \qquad \vartheta_4(\tau+1) = \vartheta_3(\tau).$$

Write these S- and T-laws as a vector-valued transformation law of weight $\frac{1}{2}$ for $(\vartheta_2, \vartheta_3, \vartheta_4)^T$ and prove that it is not an instance of the preceding scalar integral-weight definition.

Exercise 5.5 (Jacobi triple product and eta transformations)**.** Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2/2} = \prod_{m\geq 1} (1-q^m)(1+zq^{m-1/2})(1+z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Deduce

$$2\eta(\tau)^3 = \vartheta_2(\tau)\vartheta_3(\tau)\vartheta_4(\tau).$$

Use the theta transformations and the branch positive on the positive imaginary axis to derive

$$\eta(\tau+1) = e^{\pi i/12}\eta(\tau), \qquad \eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau).$$

Prove that these laws give η weight $\frac{1}{2}$ with a nontrivial scalar multiplier, rather than the integral-weight trivial-multiplier law defined above. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

Definition 5.5 (Modular functions and Hauptmoduln)**.** Put $PSL_2(\mathbb{R}) = SL_2(\mathbb{R})/\{\pm I\}$ and define $PSL_2(\mathbb{Z})$ similarly. Let $\Gamma \subset PSL_2(\mathbb{R})$ be commensurable with $PSL_2(\mathbb{Z})$ and suppose that the cusp at infinity has width one. A *modular function* for Γ is a Γ-invariant function meromorphic on the upper half-plane and at the cusps. Write X_Γ for the compactified quotient. A *normalized Hauptmodul* is a modular function inducing an isomorphism $X_\Gamma \cong \mathbb{P}^1(\mathbb{C})$ and having Fourier expansion

$$q^{-1} + O(q)$$

at infinity.

Definition 5.6 (The modular discriminant and J-function)**.** Define

$$\Delta(\tau) = \eta(\tau)^{24}, \qquad j(\tau) = \frac{E_4(\tau)^3}{\Delta(\tau)}, \qquad J(\tau) = j(\tau) - 744 = \sum_{n\geq -1} c(n)q^n.$$

Exercise 5.6 (Valence formula and the J-function)**.** Let $\mathbb{H}$ be the upper half-plane, let $\rho = e^{2\pi i/3}$, and let $\mathcal{F}$ be the standard fundamental domain for $PSL_2(\mathbb{Z})$. Apply the argument principle to a truncated copy of $\mathcal{F}$, pair its boundary arcs by S and T, and prove that every nonzero modular form f of weight k satisfies

$$v_\infty(f) + \frac{1}{2}v_i(f) + \frac{1}{3}v_\rho(f) + \sum_{\substack{[z]\in PSL_2(\mathbb{Z})\backslash\mathbb{H} \\ [z]\neq[i],[\rho]}} v_z(f) = \frac{k}{12}.$$

Use the eta transformation laws and product to show that Δ is a weight-12 cusp form with a simple zero at infinity and no zero in the upper half-plane. Deduce

$$M_{12} = \mathbb{C}E_4^3 \oplus \mathbb{C}\Delta, \qquad \Delta = \frac{E_4^3 - E_6^2}{1728}.$$

Show that j has one simple pole on $X_{PSL_2(\mathbb{Z})}$ and hence induces an isomorphism with $\mathbb{P}^1(\mathbb{C})$. Expand the products and prove that

$$J(\tau) = q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + O(q^4),$$

so J is the normalized Hauptmodul for $PSL_2(\mathbb{Z})$.

Definition 5.7 (Leech lattice)**.** A *root* of an integral Euclidean lattice is a vector of squared length 2. A *Leech lattice* is an even unimodular Euclidean lattice of rank 24 having no roots.

Exercise 5.7 (Leech theta series and its first diffraction shell)**.** Take the rootless case of the Niemeier classification as theorem input: Leech lattices exist and any two are isometric. Fix one and denote it by Λ. Use Poisson duality to prove that Θ_Λ is a modular form of weight 12. Use the absence of roots and the preceding description of M_{12} to derive

$$\Theta_\Lambda = E_4^3 - 720\Delta = 1 + 196560q^2 + O(q^3).$$

For the idealized 24-dimensional crystal with lattice Λ and a one-point motif, deduce that the first nonzero reciprocal-lattice shell has radius 4π in the physical reciprocal convention and consists of 196560 Bragg peaks. State how a nontrivial motif changes their intensities without changing the shell radius.

Exercise 5.8 (Theta diffraction)**.** Let $T_L = \mathbb{R}^d/L$ carry its flat metric, and normalize L^* by $\langle x, \xi\rangle \in \mathbb{Z}$. Let $K_t(x, y)$ be the heat kernel of the flat Laplacian and prove

$$\int_{T_L} K_t(x,x)\,dx = \sum_{\xi\in L^*} e^{-4\pi^2 t\|\xi\|^2} = \Theta_{L^*}(4\pi i t).$$

Apply Poisson duality to derive the short-time image sum and its leading term $\mathrm{vol}(T_L)(4\pi t)^{-d/2}$. For a periodic array with motif B, prove that the elastic intensity at reciprocal vector $2\pi\xi$ is proportional to

$$|S_B(2\pi\xi)|^2.$$

Prove that the heat-trace exponents and diffraction-peak radii determine the same reciprocal squared lengths $\|\xi\|^2$. Deduce the predicted peak locations, systematic absences, and thermal decay rates from the theta coefficients of L^*.

Exercise 5.9 (Zeta-regularized Casimir pressure)**.** For a massless scalar field between parallel Dirichlet plates of area A and separation a, neglect edge effects and take the normal-mode frequencies to be

$$\omega_{n,k} = c\sqrt{|k|^2 + \left(\frac{\pi n}{a}\right)^2}, \qquad n \geq 1, \quad k \in \mathbb{R}^2.$$

For $\mu > 0$ and $\mathrm{Re}\, s$ sufficiently large, define the regulated vacuum energy density by

$$\frac{\mathcal{E}_D(s)}{A} = \frac{\hbar c}{2}\mu^{2s}\sum_{n\geq 1}\int_{\mathbb{R}^2}\frac{d^2k}{(2\pi)^2}\left(|k|^2 + \frac{\pi^2 n^2}{a^2}\right)^{1/2-s}.$$

Use the gamma integral to continue this expression as

$$\frac{\mathcal{E}_D(s)}{A} = \frac{\hbar c\, \mu^{2s}}{8\pi} \frac{\Gamma(s-3/2)}{\Gamma(s-1/2)} \left(\frac{\pi}{a}\right)^{3-2s} \zeta(2s-3).$$

After subtracting the infinite-separation and plate-local terms, evaluate at $s=0$ and prove

$$\frac{\mathcal{E}_D}{A} = -\frac{\pi^2 \hbar c}{1440 a^3}.$$

For the ideal electromagnetic field, account for the two polarizations and deduce

$$\frac{\mathcal{E}_{\mathrm{EM}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \qquad P = -\frac{\partial(\mathcal{E}_{\mathrm{EM}}/A)}{\partial a} = -\frac{\pi^2 \hbar c}{240 a^4}.$$

State how a force measurement tests the a^{-4} law and its coefficient, separating the ideal prediction from finite conductivity, temperature, roughness, alignment, and edge corrections.

Part II

Counting

6 Graded Characters

Definition 6.1 (Probability laws)**.** A probability space is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A random variable with values in a metric space (S, d) is a measurable map $X : \Omega \to S$, and its law is the pushforward $X_*\mathbb{P}$. For an integrable complex random variable define $\mathbb{E}X = \int_\Omega X\, d\mathbb{P}$. For square-integrable complex random variables define

$$\mathrm{Var}(X) = \mathbb{E}|X - \mathbb{E}X|^2, \quad \mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)\overline{(Y - \mathbb{E}Y)}].$$

Random variables $X_i : \Omega \to S_i$, $1 \le i \le r$, are independent when their joint law is the product of their marginal laws. For random variables $X_N, X : \Omega \to S$, define convergence in probability by

$$\mathbb{P}(d(X_N, X) > \varepsilon) \longrightarrow 0 \qquad (\varepsilon > 0),$$

and convergence in distribution by convergence of their laws against bounded continuous functions. The standard normal law $N(0,1)$ on $\mathbb{R}$ has density $(2\pi)^{-1/2} e^{-x^2/2}$ with respect to Lebesgue measure.

Definition 6.2 (Symmetric powers and graded traces)**.** For a finite-dimensional complex vector space V, define

$$\mathrm{Sym}^m V = V^{\otimes m} / \langle \xi - \sigma\xi : \xi \in V^{\otimes m},\ \sigma \in S_m \rangle.$$

An endomorphism T induces $\mathrm{Sym}^m T$. Using the exterior powers and trace defined in Spectral Theory, define

$$Z_B(T; z) = \sum_{m \ge 0} \mathrm{tr}(\mathrm{Sym}^m T) z^m, \qquad Z_F(T; z) = \sum_{m \ge 0} \mathrm{tr}(\Lambda^m T) z^m.$$

Definition 6.3 (Partitions)**.** An *integer partition* of $n \geq 0$ is a finite nonincreasing sequence $\lambda = (\lambda_1, \ldots, \lambda_r)$ of positive integers with $|\lambda| = \sum_i \lambda_i = n$. Let $p(n)$ be the number of partitions, with $p(0) = 1$. For partitions of equal size, write $\mu < \lambda$ in dominance order when

$$\sum_{i=1}^{k} \mu_i \leq \sum_{i=1}^{k} \lambda_i$$

for every $k \geq 1$, with missing parts interpreted as zero, and at least one inequality strict.

Definition 6.4 (Symmetric polynomials)**.** For $N \geq 1$, let

$$\Lambda_N = \mathbb{C}[x_1, \ldots, x_N]^{S_N}.$$

For a partition λ of length at most N, let m_λ be the sum of the distinct permutations of $x_1^{\lambda_1} \cdots x_N^{\lambda_N}$. Define

$$e_r = \sum_{1 \leq i_1 < \cdots < i_r \leq N} x_{i_1} \cdots x_{i_r}, \qquad h_r = \sum_{1 \leq i_1 \leq \cdots \leq i_r \leq N} x_{i_1} \cdots x_{i_r},$$

$$p_r = \sum_{i=1}^{N} x_i^r, \qquad e_0 = h_0 = 1.$$

Set $e_r = 0$ for $r > N$ and $h_r = 0$ for $r < 0$.

Exercise 6.1 (Fundamental symmetric identities)**.** Prove that every element of Λ_N is uniquely a polynomial in $e_1, \ldots, e_N$. Prove

$$\sum_{r=0}^{N} e_r t^r = \prod_{i=1}^{N} (1 + x_i t), \qquad \sum_{r \geq 0} h_r t^r = \prod_{i=1}^{N} \frac{1}{1 - x_i t},$$

and deduce

$$\left(\sum_{r \geq 0} h_r t^r \right) \left(\sum_{r=0}^{N} (-1)^r e_r t^r \right) = 1, \qquad r h_r = \sum_{j=1}^{r} p_j h_{r-j}.$$

Exercise 6.2 (Bosonic and fermionic trace generating functions)**.** Let V be a finite-dimensional complex vector space and $T \in \mathrm{End}(V)$. Using the intrinsic determinant defined by the action on the highest exterior power, prove

$$Z_B(T; z) = \frac{1}{\det(I - zT)}, \qquad Z_F(T; z) = \det(I + zT).$$

The first identity holds for $|z|$ below the reciprocal spectral radius; the second is a polynomial identity. Prove them first for a direct sum of generalized spectral subspaces and then by polynomial continuation, without choosing vectors in those subspaces. If $\lambda_1, \ldots, \lambda_N$ are the eigenvalues counted with algebraic multiplicity, prove

$$\mathrm{tr}(\mathrm{Sym}^m T) = h_m(\lambda_1, \ldots, \lambda_N), \qquad \mathrm{tr}(\Lambda^m T) = e_m(\lambda_1, \ldots, \lambda_N).$$

Let the one-particle Hamiltonian have one-dimensional spectral subspaces at energies $j\hbar\omega$, $j \geq 1$. Apply the identities to the finite spectral truncation and then pass coefficientwise to the limit to derive

$$\prod_{j \geq 1} \frac{1}{1 - q^j} = \sum_{n \geq 0} p(n) q^n, \qquad \prod_{j \geq 1} (1 + q^j) = \sum_{n \geq 0} p_{\mathrm{dist}}(n) q^n,$$

where $p_{\text{dist}}(n)$ counts partitions into distinct parts. Construct the multiplicity map separating powers of two and prove

$$\prod_{j\geq 1}(1+q^j) = \prod_{j\geq 1} \frac{1}{1-q^{2j-1}}.$$

Interpret the coefficient of q^n as the exact degeneracy at excitation energy $n\hbar\omega$ for the corresponding bosonic or fermionic system.

For the bosonic system put $q = e^{-\beta_{\text{th}}\hbar\omega}$, where $\beta_{\text{th}} = (k_B T)^{-1}$, and define

$$Z(\beta_{\text{th}}) = \prod_{j\geq 1} \frac{1}{1-e^{-\beta_{\text{th}} j\hbar\omega}}, \qquad \Psi(\beta_{\text{th}}) = \log Z(\beta_{\text{th}}).$$

Define

$$U = -\frac{d\Psi}{d\beta_{\text{th}}}.$$

For

$$\mathcal{S} = k_B(\Psi + \beta_{\text{th}} U), \qquad C = k_B \beta_{\text{th}}^2 \frac{d^2\Psi}{d\beta_{\text{th}}^2},$$

prove $d\mathcal{S} = k_B \beta_{\text{th}}\, dU$ as β_{th} varies and express $d^2\Psi/d\beta_{\text{th}}^2$ as the energy variance.

7 Modular and Moonshine Characters

Definition 7.1 (Virasoro characters)**.** The Virasoro algebra has generators L_n, $n \in \mathbb{Z}$, and a central generator C, with

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{1}{12}(m^3 - m)\delta_{m+n,0} C, \qquad [C, L_n] = 0.$$

A module has central charge c when C acts by cI. A highest-weight vector $|h\rangle$ satisfies $L_0|h\rangle = h|h\rangle$ and $L_n|h\rangle = 0$ for $n > 0$. For τ in the upper half-plane put $q = e^{2\pi i\tau}$. If H_a is a positive-energy module with finite-dimensional L_0 spectral subspaces, define

$$\chi_a(\tau) = \mathrm{Tr}_{H_a}\, q^{L_0 - c/24}$$

when the displayed operator is trace class; the same expression denotes the formal graded trace otherwise.

Exercise 7.1 (Ordered Virasoro descendants)**.** Order the negative Virasoro modes by their indices. Use the commutation relations to rewrite every word as an ordered word. Check every three-letter overlap and use the Jacobi identity to prove that the rewriting is confluent. Filter by word length and deduce that the associated graded algebra is the symmetric algebra on the negative modes. Conclude that the ordered monomials in the negative modes are linearly independent in the universal highest-weight module.

Exercise 7.2 (Verma and vacuum characters)**.** Use the ordered descendants to prove that the Verma-module character is

$$\chi_{c,h}^{\text{Verma}}(\tau) = q^{h-c/24} \prod_{m\geq 1}(1-q^m)^{-1}.$$

For the vacuum impose $L_{-1}|0\rangle = 0$ and derive the corresponding generic vacuum product.

Exercise 7.3 (Null-vector quotients)**.** Let a singular vector of degree r in a Verma module generate a submodule isomorphic to the Verma module of highest weight $h+r$. Prove that the quotient character is

$$\chi^{\mathrm{Verma}}_{c,h} - \chi^{\mathrm{Verma}}_{c,h+r}.$$

For several singular vectors, prove that their submodules must be combined by inclusion–exclusion, and show that independent subtraction is valid only when the intersections of their descendant submodules have been accounted for.

Definition 7.2 (Unitary modular character systems)**.** A finite character vector χ is a *unitary modular character system* when

$$\chi(-1/\tau) = S\chi(\tau), \qquad \chi(\tau+1) = T\chi(\tau)$$

for unitary matrices S, T. For a matrix M with nonnegative integral entries define

$$Z_M(\tau,\bar\tau) = \sum_{a,b} \overline{\chi_a(\tau)} M_{ab}\chi_b(\tau).$$

It is *modular invariant* when

$$S^* M S = M, \qquad T^* M T = M.$$

Exercise 7.4 (Ising characters and modular invariance)**.** For the Ising character system with $c = 1/2$ and weights $0, 1/2, 1/16$, choose the square roots whose q-expansions have the displayed positive leading coefficients and put

$$\begin{aligned}
\chi_0 &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}}\right), \\
\chi_{1/2} &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}}\right), \\
\chi_{1/16} &= \frac{1}{\sqrt{2}}\sqrt{\frac{\vartheta_2}{\eta}}.
\end{aligned}$$

Use the theta and eta transformations developed in Automorphic Forms to derive

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \qquad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Expand each character as $q^{h-c/24}$ times a power series, identify its coefficients with exact descendant counts, and verify the modular invariance of $\sum_a |\chi_a|^2$.

Exercise 7.5 (Eta asymptotics and bosonic thermodynamics)**.** Use Exercise 5.5 to prove

$$\prod_{j\ge 1}(1-e^{-tj})^{-1} \sim \sqrt{\frac{t}{2\pi}}\exp\left(\frac{\pi^2}{6t}\right) \qquad (t \downarrow 0).$$

For $t = \beta_{\mathrm{th}}\hbar\omega$, differentiate the transformed eta-product identity to strengthen this to

$$\Psi(\beta_{\mathrm{th}}) = \frac{\pi^2}{6t} + \frac{1}{2}\log\frac{t}{2\pi} - \frac{t}{24} + O(e^{-4\pi^2/t})$$

and derive

$$U = \hbar\omega\left(\frac{\pi^2}{6t^2} - \frac{1}{2t} + \frac{1}{24} + o(1)\right),$$

$$C = k_B\left(\frac{\pi^2}{3t} - \frac{1}{2} + o(1)\right), \qquad \mathcal{S} = k_B\left(\frac{\pi^2}{3t} + \frac{1}{2}\log\frac{t}{2\pi} - \frac{1}{2} + o(1)\right).$$

Exercise 7.6 (Partition saddle)**.** Insert this estimate into Cauchy's coefficient integral, locate its positive real saddle, prove that its equation is

$$n = \frac{U}{\hbar\omega},$$

and identify the leading saddle exponent with the Legendre transform $\mathcal{S}/k_B$. Expand the phase through quadratic order and derive the major-arc contribution

$$\frac{1}{4n\sqrt{3}}\exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Exercise 7.7 (Partition minor arcs)**.** On the coefficient circle through the positive saddle, use

$$|1 - re^{i\theta}|^2 = (1-r)^2 + 4r\sin^2(\theta/2)$$

together with the modular transformation at each rational cusp met by the circle dissection to bound the complementary arcs by an exponentially smaller quantity. Deduce

$$p(n) \sim \frac{1}{4n\sqrt{3}}\exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Exercise 7.8 (Ising state counts)**.** For the diagonal Ising system, define nonnegative integers $d_a(N)$ by

$$\chi_a(\tau) = q^{h_a - c/24}\sum_{N\geq 0} d_a(N)q^N$$

and define the resolved two-sided degeneracy

$$\rho_{\text{Ising}}(N,\bar{N}) = \sum_{a\in\{0,1/2,1/16\}} d_a(N)d_a(\bar{N}).$$

Use the explicit theta-product formulas and the S-transformation to control each character in a complex neighborhood of the positive real saddle. Bound the complementary arcs as in the partition calculation and prove, as $N,\bar{N}\to\infty$ with $N/\bar{N}$ in a fixed compact subset of $(0,\infty)$,

$$\log\rho_{\text{Ising}}(N,\bar{N}) \sim 2\pi\sqrt{\frac{cN}{6}} + 2\pi\sqrt{\frac{c\bar{N}}{6}}, \qquad c = \frac{1}{2}.$$

Derive the resulting asymptotic number of states below a specified high energy on a circle and the corresponding entropy. State the circumference, propagation speed, energy window, vacuum assumption, sector resolution, saddle-point regime, and finite-size limitations required when comparing these predictions with measured or transfer-matrix spectra.

Definition 7.3 (Positive-energy vertex operator algebras)**.** A *positive-energy vertex operator algebra* is a graded complex vector space

$$V = \bigoplus_{m\geq 0} V_m, \qquad \dim V_m < \infty, \qquad V_0 = \mathbb{C}\mathbf{1},$$

with a vacuum $\mathbf{1}$, a conformal vector $\omega \in V_2$, and a state–field map

$$Y(v,z) = \sum_{n\in\mathbb{Z}} v_n z^{-n-1} \in (\operatorname{End} V)[[z,z^{-1}]].$$

For homogeneous $v \in V_r$, put $\operatorname{wt}(v) = r$. The map satisfies $v_n w = 0$ for n sufficiently large, $v_n V_m \subseteq V_{m+\operatorname{wt}(v)-n-1}$,

$$Y(\mathbf{1},z) = \operatorname{id}_V, \qquad Y(v,z)\mathbf{1} \in v + zV[[z]],$$

and, for every $u, v \in V$, there is $N \geq 0$ such that

$$(z-w)^N [Y(u,z), Y(v,w)] = 0.$$

Writing

$$Y(\omega,z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2},$$

the modes satisfy the Virasoro relations with central charge c, $L_0|_{V_m} = m\,\operatorname{id}_{V_m}$, and $Y(L_{-1}v,z) = \partial_z Y(v,z)$. An automorphism preserves $\mathbf{1}$, ω, and the state–field map.

Definition 7.4 (Lattice oscillator state spaces)**.** For an even positive-definite lattice L of rank d, let $\varepsilon : L \times L \to \{\pm 1\}$ satisfy

$$\varepsilon(\lambda,0) = \varepsilon(0,\lambda) = 1, \qquad \varepsilon(\lambda,\mu)\varepsilon(\lambda+\mu,\nu) = \varepsilon(\mu,\nu)\varepsilon(\lambda,\mu+\nu),$$
$$\frac{\varepsilon(\lambda,\mu)}{\varepsilon(\mu,\lambda)} = (-1)^{\langle\lambda,\mu\rangle}.$$

Let $\mathbb{C}_\varepsilon[L]$ have symbols e^λ, $\lambda \in L$, with $e^\lambda e^\mu = \varepsilon(\lambda,\mu)e^{\lambda+\mu}$, and define

$$\mathcal{F}_L = \operatorname{Sym}\left(\bigoplus_{n\geq 1}(L\otimes_{\mathbb{Z}}\mathbb{C})t^{-n}\right) \otimes \mathbb{C}_\varepsilon[L],$$

where $\operatorname{Sym}(W) = \bigoplus_{r\geq 0}\operatorname{Sym}^r(W)$. Give $(L\otimes_{\mathbb{Z}}\mathbb{C})t^{-n}$ degree n, give the element e^λ of the twisted group algebra degree $\|\lambda\|^2/2$, and let L_0 act by total degree.

Exercise 7.9 (Leech lattice and oscillator character)**.** The standard lattice vertex-operator construction equips $\mathcal{F}_L$ with a central-charge-d vertex operator algebra structure. Use the symmetric-power trace formula and the definition of the lattice theta series to prove

$$\operatorname{Tr}_{\mathcal{F}_L} q^{L_0 - d/24} = \frac{\Theta_L(\tau)}{\eta(\tau)^d}.$$

For the Leech lattice, use Exercise 5.7 to derive

$$\operatorname{Tr}_{\mathcal{F}_\Lambda} q^{L_0 - 1} = \frac{\Theta_\Lambda}{\eta^{24}} = J + 24.$$

Show directly from the graded construction that the coefficient 24 counts the degree-one oscillator states and that

$$196884 = 196560 + 324, \qquad 324 = 24 + \dim\operatorname{Sym}^2(\Lambda\otimes_{\mathbb{Z}}\mathbb{C}),$$

splits the degree-two states into nonzero lattice momenta and oscillator states.

Definition 7.5 (McKay–Thompson series)**.** Let G act by automorphisms on a positive-energy vertex operator algebra V. For $g \in G$, define its *McKay–Thompson series* by

$$T_g(\tau) = \mathrm{Tr}_V\left(gq^{L_0 - c/24}\right).$$

Let $\mathbb{M}$ denote the Monster sporadic simple group.

Exercise 7.10 (Monstrous moonshine)**.** Take the Frenkel–Lepowsky–Meurman construction and Borcherds's monstrous moonshine theorem as theorem input. If θ lifts $\lambda \mapsto -\lambda$ to the Leech-lattice vertex operator algebra and $V_\Lambda^{T,+}$ is the $+1$ eigenspace in its θ-twisted module, the orbifold

$$V^\natural = V_\Lambda^+ \oplus V_\Lambda^{T,+}$$

is a central-charge-24 vertex operator algebra with

$$\mathrm{Aut}(V^\natural) = \mathbb{M}, \qquad T_1(\tau) = J(\tau),$$

and, for every $g \in \mathbb{M}$, the series T_g is the normalized Hauptmodul for the Conway–Norton genus-zero group Γ_g. Deduce

$$\dim V_0^\natural = 1, \qquad \dim V_1^\natural = 0, \qquad \dim V_2^\natural = 196884, \qquad \dim V_3^\natural = 21493760.$$

If $\mathbf{d}$ denotes the indicated irreducible Monster representation of dimension d, verify the first decompositions supplied by the Monster character table,

$$V_2^\natural \cong \mathbf{1} \oplus \mathbf{196883}, \qquad V_3^\natural \cong \mathbf{1} \oplus \mathbf{196883} \oplus \mathbf{21296876}.$$

With $p = e^{2\pi i\sigma}$, use Borcherds's denominator identity

$$p^{-1} \prod_{m>0,\ n\in\mathbb{Z}} (1 - p^m q^n)^{c(mn)} = J(\sigma) - J(\tau)$$

as a formal product in the region $|p| < |q| < 1$. Compare coefficients through order p and derive the first replication identity

$$\frac{1}{2}\left(J(\tau)^2 - \sum_{n\in\mathbb{Z}} c(n)q^{2n}\right) - \sum_{n\in\mathbb{Z}} c(2n)q^n = 196884.$$

Exercise 7.11 (Monster-resolved finite-size spectrum)**.** Let χ_ρ be the character of an irreducible Monster representation ρ. Use finite-group character orthogonality to prove that its multiplicity in $V_m^\natural$ is

$$\mu_\rho(m) = \frac{1}{|\mathbb{M}|} \sum_{g\in\mathbb{M}} \overline{\chi_\rho(g)}\,[q^{m-1}]T_g(\tau).$$

Deduce that the symmetry-resolved series

$$Z_\rho(\tau) = \frac{\dim(\rho)}{|\mathbb{M}|} \sum_{g\in\mathbb{M}} \overline{\chi_\rho(g)} T_g(\tau)$$

has coefficient $\dim(\rho)\mu_\rho(m)$ at q^{m-1}. Model an exactly Monster-symmetric chiral edge of circumference ℓ and propagation speed u by

$$H = \frac{2\pi\hbar u}{\ell}(L_0 - 1).$$

Derive the excitation energies

$$E_m - E_0 = \frac{2\pi\hbar u}{\ell} m$$

and predict a missing level at $2\pi\hbar u/\ell$, followed by levels of degeneracy 196884 and 21493760 at $4\pi\hbar u/\ell$ and $6\pi\hbar u/\ell$. Express the symmetry-resolved degeneracy at level m as $\dim(\rho)\mu_\rho(m)$. State the assumptions of exact Monster symmetry, linear chiral dispersion, periodic boundary conditions, access to the twined sectors, and the low-energy finite-size regime. Explain why a purely chiral central-charge-24 theory must be realized as an anomalous edge or accompanied by a compensating opposite-chirality sector, and distinguish the derived spectral prediction from evidence for a particular physical realization.

8 Transfer Matrices

Definition 8.1 (Transfer-matrix counts)**.** Let $\mathcal{R}_N$ be a finite set of allowed width-N configurations and let $T_N(z)$ be an operator on the configuration space whose matrix element between $a, b \in \mathcal{R}_N$ is the weight of adjoining b to a. For a height-M periodic lattice define

$$Z_{N,M}(z) = \operatorname{Tr}(T_N(z)^M).$$

When $T_N(z)$ has nonnegative entries, write $\lambda_N(z)$ for its spectral radius.

Definition 8.2 (Hard-hexagon rows)**.** Let

$$\mathcal{R}_N = \{a \in \{0,1\}^{\mathbb{Z}/N\mathbb{Z}} : a_i a_{i+1} = 0 \text{ for every } i\}.$$

Two rows $a, b \in \mathcal{R}_N$ are compatible when

$$a_i b_i = 0, \qquad a_i b_{i+1} = 0 \quad (i \in \mathbb{Z}/N\mathbb{Z}).$$

For $z \geq 0$, with $|a| = \sum_i a_i$, define

$$T_N(z)_{ab} = \mathbf{1}_{\{a,b \text{ compatible}\}} z^{(|a|+|b|)/2}.$$

Exercise 8.1 (Hard-hexagon trace counts)**.** Identify the horizontal, vertical, and diagonal nearest-neighbor edges encoded by the two compatibility conditions. Prove

$$\operatorname{Tr}(T_N(z)^M) = \sum_I z^{|I|},$$

where the sum is over independent vertex sets of the specified triangular $N \times M$ torus. Prove that every coefficient is an exact state count. Let 0 denote the empty row. For $z > 0$, prove

$$(T_N(z)^2)_{ab} \geq T_N(z)_{a0} T_N(z)_{0b} > 0.$$

Apply the Perron–Frobenius theorem to show that $\lambda_N(z)$ is simple, that every other spectral value has smaller modulus, and that

$$\lim_{M\to\infty} \frac{1}{NM} \log Z_{N,M}(z) = \frac{1}{N} \log \lambda_N(z).$$

Put $\eta = \log z = \beta_{\text{th}}\mu$ and

$$\Psi_{N,M}(\eta) = \log Z_{N,M}(e^\eta).$$

Set $A = \partial_\eta \Psi_{N,M}$ and prove

$$A = \mathbb{E}_\eta |I|, \qquad \partial_\eta A = \mathrm{Var}_\eta(|I|),$$

and derive the mean occupied fraction and isothermal finite-size susceptibility

$$\rho_{N,M} = \frac{A}{NM}, \qquad \chi_{N,M} = \frac{\partial \rho_{N,M}}{\partial \mu} = \frac{\beta_{\mathrm{th}}}{NM} \mathrm{Var}_\eta(|I|).$$

State the lattice geometry, boundary conditions, and finite N, M limitations required when these quantities are compared with adsorption or exclusion-process data.

Definition 8.3 (Planar Ising model)**.** Let $G = (V, E)$ be a finite connected graph with a fixed planar embedding. For couplings $K_e > 0$, define

$$Z_G^{\mathrm{f}}(K) = \sum_{s:V\to\{-1,1\}} \exp\left(\sum_{e=uv} K_e s_u s_v\right).$$

Let G^* be the planar dual, fix the spin of its outer-face vertex to $+1$, and denote the corresponding partition function by $Z_{G^*}^{+}(K^*)$. Let $\mathcal{E}(G)$ be the family of edge subsets in which every vertex has even degree. The dual couplings are determined by

$$e^{-2K^*_{e^*}} = \tanh K_e.$$

Exercise 8.2 (Ising duality and the square-lattice critical point)**.** Expand each edge factor as

$$e^{K_e s_u s_v} = \cosh K_e (1 + s_u s_v \tanh K_e)$$

and prove

$$Z_G^{\mathrm{f}}(K) = 2^{|V|} \prod_{e\in E} \cosh K_e \sum_{\gamma\in\mathcal{E}(G)} \prod_{e\in\gamma} \tanh K_e.$$

Identify $\mathcal{E}(G)$ with the domain walls of dual spin configurations whose outer-face spin is fixed, and deduce the exact planar duality

$$Z_G^{\mathrm{f}}(K) = 2^{|V|} \prod_{e\in E} (\cosh K_e \, e^{-K^*_{e^*}}) Z_{G^*}^{+}(K^*).$$

For horizontal and vertical square-lattice couplings K_h, K_v, show that the duality interchanges the directions and that its fixed locus is

$$\sinh(2K_h)\sinh(2K_v) = 1.$$

Assuming the infinite-volume ferromagnetic model has a single transition and that duality interchanges its high- and low-temperature phases, deduce in the isotropic case

$$K_c = \frac{1}{2}\log(1 + \sqrt{2}).$$

State separately the exact finite-graph identity, the infinite-volume assumptions, and the conclusion about the critical temperature.

Exercise 8.3 (Ising star–triangle transformation)**.** For $K_1, K_2, K_3 > 0$ and boundary spins s_1, s_2, s_3, prove that there are unique $A > 0$ and real L_{12}, L_{23}, L_{31} satisfying

$$\sum_{s_0=\pm 1} e^{K_1 s_0 s_1 + K_2 s_0 s_2 + K_3 s_0 s_3} = A e^{L_{12} s_1 s_2 + L_{23} s_2 s_3 + L_{31} s_3 s_1}.$$

Put

$$w_0 = 2\cosh(K_1 + K_2 + K_3), \quad w_1 = 2\cosh(-K_1 + K_2 + K_3),$$
$$w_2 = 2\cosh(K_1 - K_2 + K_3), \quad w_3 = 2\cosh(K_1 + K_2 - K_3).$$

Derive

$$A = (w_0 w_1 w_2 w_3)^{1/4},$$
$$e^{4L_{12}} = \frac{w_0 w_3}{w_1 w_2}, \qquad e^{4L_{23}} = \frac{w_0 w_1}{w_2 w_3}, \qquad e^{4L_{31}} = \frac{w_0 w_2}{w_1 w_3}.$$

Show that replacing every star of one sublattice of the honeycomb model by a triangle changes the partition function only by the product of the local factors A. In the isotropic case derive

$$e^{4L} = \frac{\cosh(3K)}{\cosh K}.$$

Combine this transformation with planar duality and, under the single-transition hypothesis of the preceding exercise, obtain

$$K_c^{\text{tri}} = \frac{1}{4}\log 3, \qquad K_c^{\text{hex}} = \frac{1}{2}\log(2 + \sqrt{3}).$$

Identify the coupling normalization and lattice geometry needed when these critical temperatures are compared with an Ising magnet or a programmable spin array.

Definition 8.4 (Ising row transfer and Clifford operators)**.** Let $\mathcal{S}_N = \{-1, 1\}^{\mathbb{Z}/N\mathbb{Z}}$ and $K_h, K_v > 0$. For row configurations $s, t \in \mathcal{S}_N$ define

$$T_N^{\text{Is}}(K_h, K_v)_{s,t} = \exp\left[\frac{K_h}{2}\sum_i (s_i s_{i+1} + t_i t_{i+1}) + K_v \sum_i s_i t_i\right].$$

Let R_N be cyclic row translation. On $\mathcal{H}_N = \ell^2(\mathcal{S}_N)$ let

$$(Z_j f)(s) = s_j f(s), \qquad (X_j f)(s) = f(s^{(j)}), \qquad Y_j = i X_j Z_j,$$

where $s^{(j)}$ is obtained by reversing s_j. Put

$$P = \prod_{j=1}^{N} X_j, \qquad c_{2j-1} = \left(\prod_{\ell<j} X_\ell\right) Z_j, \qquad c_{2j} = \left(\prod_{\ell<j} X_\ell\right) Y_j.$$

Let $K_v^* > 0$ satisfy $e^{-2K_v^*} = \tanh K_v$, and set

$$\mathcal{K}_+ = \left\{\frac{(2m+1)\pi}{N} : 0 \le m < N\right\}, \qquad \mathcal{K}_- = \left\{\frac{2m\pi}{N} : 0 \le m < N\right\}.$$

Write a_x and a_t for the spatial and transfer-step lattice spacings and $L = N a_x$ for the circumference.

Exercise 8.4 (Free-fermion Ising transfer spectrum)**.** Verify

$$R_N T_N^{\mathrm{Is}}(K_h, K_v) = T_N^{\mathrm{Is}}(K_h, K_v) R_N.$$

Prove

$$T_N^{\mathrm{Is}}(K_h, K_v) = (2\sinh(2K_v))^{N/2} e^{\frac{K_h}{2}\sum_j Z_j Z_{j+1}} e^{K_v^* \sum_j X_j} e^{\frac{K_h}{2}\sum_j Z_j Z_{j+1}}.$$

Verify the Clifford relations

$$c_r c_s + c_s c_r = 2\delta_{rs} I, \qquad X_j = i c_{2j-1} c_{2j}, \qquad Z_j Z_{j+1} = i c_{2j} c_{2j+1} \quad (j < N),$$

and

$$Z_N Z_1 = -iP c_{2N} c_1.$$

On the $P = +1$ and $P = -1$ sectors use, respectively, the distinguished Fourier modes with momenta in $\mathcal{K}_+$ and $\mathcal{K}_-$ to diagonalize the induced action on the Clifford generators. Derive

$$\cosh\gamma(k) = \cosh(2K_h)\cosh(2K_v^*) - \sinh(2K_h)\sinh(2K_v^*)\cos k, \qquad \gamma(k) \geq 0,$$

and prove that the transfer eigenvalues are

$$\Lambda = (2\sinh(2K_v))^{N/2} \exp\left(\frac{1}{2}\sum_{k\in\mathcal{K}_\pm} \epsilon_k \gamma(k)\right), \qquad \epsilon_k \in \{-1, 1\},$$

with the fermion-parity condition determined by the chosen P sector.

At $K_h = K_v = K_c$ prove

$$\gamma(k) = 2\,\mathrm{arsinh}\,|\sin(k/2)| = |k| + O(|k|^3).$$

After subtracting the sector ground energy and measuring gaps in units $2\pi(a_x/a_t)/L$, use the parity projections to obtain the chiral scaling products

$$\chi_0(q) = \frac{q^{-1/48}}{2}\left[\prod_{m\geq 0}(1 + q^{m+1/2}) + \prod_{m\geq 0}(1 - q^{m+1/2})\right],$$
$$\chi_{1/2}(q) = \frac{q^{-1/48}}{2}\left[\prod_{m\geq 0}(1 + q^{m+1/2}) - \prod_{m\geq 0}(1 - q^{m+1/2})\right],$$
$$\chi_{1/16}(q) = q^{1/24}\prod_{m\geq 1}(1 + q^m).$$

Use the Jacobi triple product to recover the theta–eta formulas in the Ising character exercise.

Exercise 8.5 (Theta-character transfer spectra)**.** Let $\Lambda_0(L) > 0$ be the Perron eigenvalue of $T_N^{\mathrm{Is}}(K_c, K_c)$ and let $\Lambda_{a,r,\bar r}(L)$ denote its low-lying symmetry-resolved eigenvalues. Define

$$E_{a,r,\bar r}(L) = -\frac{1}{a_t}\log|\Lambda_{a,r,\bar r}(L)|.$$

Use the critical Clifford spectrum and Euler–Maclaurin summation to prove the finite-size asymptotic

$$E_{a,r,\bar r}(L) = e_\infty L + \frac{2\pi v}{L}\left(h_a + \bar h_a + r + \bar r - \frac{c + \bar c}{24}\right) + o(L^{-1}).$$

If R_N acts on the same state by $e^{i\mathsf{p}a_x}$, prove

$$\mathsf{p} = \frac{2\pi}{L}(h_a - \bar{h}_a + r - \bar{r}) \pmod{2\pi/a_x}.$$

Prove that $v = a_x/a_t$ for the displayed isotropic transfer normalization. Use the Ising character system from the preceding section to prove that the coefficient of $q^{h_a+r-c/24}\bar{q}^{\bar{h}_a+\bar{r}-\bar{c}/24}$ gives the predicted multiplicity at these energy and momentum values. Express every excitation gap directly as

$$E_j(L) - E_0(L) = -\frac{1}{a_t}\log\left|\frac{\Lambda_j(L)}{\Lambda_0(L)}\right|.$$

Compute the first spectral gaps and degeneracies in the three Ising sectors. Deduce the universal gap ratios and the L^{-1} Casimir-energy correction. State how finite-size transfer eigenvalues determine c, the primary weights, and v, including the symmetry-sector resolution, anisotropy corrections, and irrelevant-operator corrections required for the comparison.

9 Determinantal Ensembles

Exercise 9.1 (Hermite Gaussian moments)**.** For $X \sim N(0,1)$, prove

$$\mathbb{E}[e^{tX}] = e^{t^2/2}.$$

Define the probabilists' Hermite polynomials by

$$e^{xt-t^2/2} = \sum_{n\geq 0} \mathrm{He}_n(x)\frac{t^n}{n!}$$

and prove

$$\mathbb{E}[\mathrm{He}_m(X)\,\mathrm{He}_n(X)] = n!\delta_{mn}, \qquad \mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n n!}.$$

Relate He_n to the oscillator Hermite functions developed in Spectral Theory.

Definition 9.1 (Dyson classes)**.** For $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, put $\beta_{\mathrm{D}} = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}$. The Gaussian Dyson ensemble on the self-adjoint operators on $\mathbb{F}^N$ has density

$$d\mathbb{P}_{N,\beta_{\mathrm{D}}}(H) = C_{N,\beta_{\mathrm{D}}} \exp\left[-\frac{\beta_{\mathrm{D}} N}{4\sigma^2}\,\mathrm{Tr}(H^2)\right] dH$$

for the invariant Hilbert–Schmidt volume. For $\lambda = (\lambda_1, \ldots, \lambda_N)$ define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^N.$$

Exercise 9.2 (Unitary orbit Jacobian)**.** Use the exterior-power determinant to prove

$$\Delta(\lambda) = \prod_{i<j}(\lambda_j - \lambda_i).$$

Compute the Jacobian of the unitary conjugation-orbit map on the regular spectral locus, including the stabilizer and permutation factors, and derive

$$p_{N,2}(\lambda) = \frac{1}{Z_{N,2}} e^{-N\sum_i \lambda_i^2/(2\sigma^2)} |\Delta(\lambda)|^2.$$

Exercise 9.3 (Dyson level repulsion)**.** For the real and quaternionic classes, show that each off-diagonal spectral direction has real multiplicity β_{D} and deduce

$$p_{N,\beta_{\mathrm{D}}}(\lambda) = \frac{1}{Z_{N,\beta_{\mathrm{D}}}} e^{-\beta_{\mathrm{D}} N \sum_i \lambda_i^2/(4\sigma^2)} |\Delta(\lambda)|^{\beta_{\mathrm{D}}}.$$

For 2×2 GOE and GUE, separate the trace from the traceless operator, rescale the gap to unit mean, and obtain

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \qquad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Identify the exponent β_{D} as the small-gap repulsion prediction and state the symmetry-sector separation required in spectral data.

Definition 9.2 (Fredholm determinants)**.** Let A be a positive compact operator on a Hilbert space. It is *trace class* when its nonzero spectral values, repeated with multiplicity, satisfy $\sum_j \lambda_j(A) < \infty$. Define

$$\det(I + zA) = \prod_j (1 + z\lambda_j(A)), \qquad \operatorname{Tr}(A^m) = \sum_j \lambda_j(A)^m.$$

Exercise 9.4 (Fredholm trace identity)**.** Prove absolute local uniform convergence of the defining product and

$$\det(I - zA) = \exp\left(- \sum_{m \geq 1} \frac{z^m}{m} \operatorname{Tr}(A^m) \right) \qquad (|z| < \|A\|^{-1}).$$

Prove that the determinant is entire, identify its zeros, and show that its finite-rank specialization agrees with the exterior-power identity from the opening section.

Definition 9.3 (Point processes and correlation densities)**.** Let X be a locally compact second-countable Hausdorff space and let μ be a Radon measure on X. A simple point process is a random locally finite counting measure $\Xi = \sum_i \delta_{x_i}$ with distinct points almost surely. When its kth factorial moment measure is absolutely continuous with respect to $\mu^{\otimes k}$, its density ρ_k is defined, for every nonnegative compactly supported measurable function f, by

$$\mathbb{E} \sum_{i_1,\ldots,i_k}^{\neq} f(x_{i_1}, \ldots, x_{i_k}) = \int_{X^k} f \rho_k \, d\mu^{\otimes k}.$$

A *determinantal point process* with measurable Hermitian kernel K is one for which

$$\rho_k(x_1, \ldots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^k.$$

For the existence and counting formulas below, the integral operator defined by K is a locally trace-class positive contraction on $L^2(X, \mu)$.

Exercise 9.5 (Determinantal counting statistics)**.** For a relatively compact measurable set B, put $N_B = \Xi(B)$ and $K_B = \mathbf{1}_B K \mathbf{1}_B$. Use factorial moments and the Fredholm series to prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z-1)K_B), \qquad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Prove that N_B is distributed as a sum of independent Bernoulli variables whose parameters are the eigenvalues of K_B.

Definition 9.4 (Unitary polynomial ensembles)**.** Let μ be a Radon measure on $\mathbb{R}$ such that $w(x)d\mu(x)$ has infinite support and finite moments, and let P_j be its monic orthogonal polynomials, normalized by

$$\int P_j(x)P_k(x)w(x)d\mu(x) = H_j\delta_{jk}, \qquad H_j > 0.$$

The associated unitary polynomial ensemble has density, with respect to $\mu^{\otimes N}$,

$$\mathfrak{p}_N(x_1,\dots,x_N) = \frac{1}{N!\prod_{j=0}^{N-1} H_j}\Delta(x)^2\prod_{i=1}^{N} w(x_i)$$

and kernel

$$K_N(x,y) = \sqrt{w(x)w(y)}\sum_{j=0}^{N-1}\frac{P_j(x)P_j(y)}{H_j}.$$

Exercise 9.6 (Andréief and Christoffel–Darboux identities)**.** Prove Andréief's identity

$$\int_{X^N}\det[f_i(x_j)]\det[g_i(x_j)]\prod_j d\mu(x_j) = N!\det\left[\int_X f_i g_j\,d\mu\right].$$

Deduce that the unitary polynomial ensemble is determinantal with kernel K_N. Derive the Christoffel–Darboux formula from the three-term recurrence and prove

$$\int K_N(x,z)K_N(z,y)d\mu(z) = K_N(x,y), \qquad \int K_N(x,x)d\mu(x) = N.$$

Definition 9.5 (GUE and its limiting kernels)**.** For the GUE weight $w_N(x) = e^{-Nx^2/(2\sigma^2)}$, write ψ_j^{H} for the normalized oscillator Hermite functions developed in Spectral Theory and define

$$\phi_{j,N}(x) = \left(\frac{N}{2\sigma^2}\right)^{1/4}\psi_j^{\mathrm{H}}\left(\sqrt{\frac{N}{2\sigma^2}}x\right), \qquad K_N^{\mathrm{GUE}}(x,y) = \sum_{j=0}^{N-1}\phi_{j,N}(x)\phi_{j,N}(y).$$

Define

$$\rho_{\mathrm{sc}}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2-x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x),$$

$$K_{\sin}(u,v) = \frac{\sin\pi(u-v)}{\pi(u-v)},$$

$$K_{\mathrm{Ai}}(u,v) = \frac{\mathrm{Ai}(u)\,\mathrm{Ai}'(v) - \mathrm{Ai}'(u)\,\mathrm{Ai}(v)}{u-v}, \qquad F_2(s) = \det(I - K_{\mathrm{Ai}})_{L^2(s,\infty)}.$$

Diagonal values are defined by continuity.

Definition 9.6 (Hastings–McLeod function)**.** The *Hastings–McLeod function* is the real pole-free solution q of the second Painlevé equation

$$q''(s) = sq(s) + 2q(s)^3$$

with

$$q(s) \sim \mathrm{Ai}(s) \quad (s \to +\infty).$$

Exercise 9.7 (Hermite steepest descent)**.** Derive a contour integral for the normalized Hermite functions from their generating function. For $E \in (-2\sigma, 2\sigma)$, perform steepest descent at the two simple saddles; at 2σ, rescale the coalescing saddles to the Airy integral. Establish uniform asymptotic formulas on compact bulk windows and uniform exponentially weighted bounds on edge half-lines.

Exercise 9.8 (GUE bulk statistics)**.** Insert the bulk Hermite asymptotics into the Christoffel–Darboux identity and prove

$$\frac{1}{N\rho_{\mathrm{sc}}(E)} K_N^{\mathrm{GUE}}\left(E + \frac{u}{N\rho_{\mathrm{sc}}(E)}, E + \frac{v}{N\rho_{\mathrm{sc}}(E)}\right) \longrightarrow K_{\sin}(u, v)$$

locally uniformly. For the limiting sine process prove

$$\operatorname{Var} N_{[0,L]} = \int_0^L K_{\sin}(u, u)\, du - \int_0^L \int_0^L |K_{\sin}(u, v)|^2\, du\, dv = \frac{1}{\pi^2} \log L + O(1).$$

State the unfolding, symmetry-sector separation, and finite-window assumptions required when testing the spacing and number-variance predictions with resonance or Hamiltonian spectra.

Exercise 9.9 (GUE edge statistics)**.** Insert the turning-point asymptotics into the Christoffel–Darboux identity and prove

$$\frac{\sigma}{N^{2/3}} K_N^{\mathrm{GUE}}\left(2\sigma + \frac{\sigma u}{N^{2/3}}, 2\sigma + \frac{\sigma v}{N^{2/3}}\right) \longrightarrow K_{\mathrm{Ai}}(u, v).$$

Use the exponentially weighted bounds to pass to the Fredholm series and deduce the GUE largest-eigenvalue law

$$\mathbb{P}\left(\frac{N^{2/3}}{\sigma}(\lambda_{\max} - 2\sigma) \le s\right) \longrightarrow F_2(s).$$

State the centering, scale calibration, symmetry class, and finite-size assumptions required when testing this prediction with extreme resonance or Hamiltonian levels.

Exercise 9.10 (Painlevé form of the Tracy–Widom law)**.** Let K_s be the Airy-kernel operator on $L^2((s, \infty))$ and define

$$Q_s = (I - K_s)^{-1} \operatorname{Ai}, \qquad P_s = (I - K_s)^{-1} \operatorname{Ai}'.$$

With the inner product on $L^2((s, \infty))$, put

$$q(s) = Q_s(s), \qquad p(s) = P_s(s), \qquad u(s) = \langle Q_s, \operatorname{Ai}\rangle, \qquad v(s) = \langle P_s, \operatorname{Ai}\rangle.$$

Use the integrable form of K_{Ai}, the Airy equation, and differentiation of the moving endpoint to prove

$$q' = p - qu, \qquad p' = sq + pu - 2qv, \qquad u' = -q^2, \qquad v' = -pq.$$

Prove that $u^2 - 2v - q^2$ is constant and vanishes as $s \to +\infty$. Deduce

$$q'' = sq + 2q^3, \qquad q(s) \sim \operatorname{Ai}(s) \quad (s \to +\infty).$$

Use the positive integral representation

$$K_{\mathrm{Ai}}(x, y) = \int_0^\infty \operatorname{Ai}(x + t) \operatorname{Ai}(y + t)\, dt$$

to prove that $I - K_s$ is invertible for every real s. Conclude that the resolvent construction gives the real pole-free Hastings–McLeod solution. For the resolvent $R_s = K_s(I - K_s)^{-1}$, prove

$$\frac{d}{ds} \log \det(I - K_s) = R_s(s, s) = u(s) = \int_s^\infty q(x)^2\, dx.$$

Conclude that the Fredholm and Painlevé definitions agree:

$$F_2(s) = \exp\left[-\int_s^\infty (x - s) q(x)^2\, dx\right].$$

Use the Airy asymptotic of the Hastings–McLeod solution to derive

$$1 - F_2(s) \sim \frac{e^{-4s^{3/2}/3}}{16\pi s^{3/2}} \quad (s \to +\infty).$$

State how exceedance counts for centered and edge-scaled extreme eigenvalues or resonance levels test this right-tail prediction, including the finite-size range over which the asymptotic is used.

10 Schur Measures

Definition 10.1 (Schur polynomials)**.** For a partition λ of length at most N, define

$$s_\lambda(x_1, \dots, x_N) = \det[h_{\lambda_i - i + j}]_{i,j=1}^N.$$

Exercise 10.1 (Schur identities)**.** Derive the alternant formula

$$s_\lambda(x) = \frac{\det[x_i^{\lambda_j + N - j}]_{i,j=1}^N}{\det[x_i^{N-j}]_{i,j=1}^N}$$

from the fundamental symmetric identities. Prove $s_{(r)} = h_r$ and $s_{(1^r)} = e_r$.

Definition 10.2 (Row insertion and RSK)**.** A semistandard Young tableau has weakly increasing rows and strictly increasing columns. To insert an integer x, replace the first entry strictly larger than x in the first row and insert the replaced entry in the next row; append when no such entry exists. For $W = (W_{ij}) \in M_{m,n}(\mathbb{Z}_{\geq 0})$, form the lexicographically ordered two-line word containing W_{ij} copies of (i, j), insert the lower letters, and record the upper letters in the created boxes. The resulting pair (P, Q) of tableaux of common shape λ is the RSK image of W.

Definition 10.3 (Last-passage values)**.** For nonnegative weights $W = (W_{ij})$, define

$$G(m, n) = \max_{\pi:(1,1)\nearrow(m,n)} \sum_{(i,j)\in\pi} W_{ij}.$$

For parameters $\alpha_i, \gamma_j \geq 0$ with $\alpha_i\gamma_j < 1$, the geometric model has independent weights

$$\mathbb{P}(W_{ij} = k) = (1 - \alpha_i\gamma_j)(\alpha_i\gamma_j)^k, \qquad k \in \mathbb{Z}_{\geq 0}.$$

Exercise 10.2 (RSK and the finite Schur measure)**.** Prove that row insertion is reversible and hence that RSK is a bijection. Show that the length of the first row of its shape is the maximal weight of an up-right path:

$$\lambda_1 = G(m, n).$$

For variables $\alpha = (\alpha_1, \dots, \alpha_m)$, prove the tableau formula

$$s_\lambda(\alpha) = \sum_T \prod_i \alpha_i^{m_i(T)}.$$

Use the contents of the two RSK tableaux and the geometric weight law to derive

$$\mathbb{P}(\lambda) = \prod_{i,j}(1 - \alpha_i \gamma_j) s_\lambda(\alpha) s_\lambda(\gamma)$$

and the Cauchy identity

$$\sum_\lambda s_\lambda(\alpha) s_\lambda(\gamma) = \prod_{i,j}(1 - \alpha_i \gamma_j)^{-1}.$$

Definition 10.4 (Interlacing partitions)**.** For partitions μ and λ, write $\mu \prec \lambda$ when

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots .$$

Exercise 10.3 (Nonintersecting Schur paths)**.** Prove the branching formula

$$s_\lambda(\alpha_1, \dots, \alpha_m) = \sum_{\varnothing = \lambda^{(0)} \prec \cdots \prec \lambda^{(m)} = \lambda} \prod_{k=1}^m \alpha_k^{|\lambda^{(k)}| - |\lambda^{(k-1)}|}.$$

Encode each interlacing chain by nonintersecting paths on the integer lattice. For a finite directed acyclic graph with compatible sources u_i and sinks v_i, let $Z(u_i, v_j)$ be the weighted sum of paths from u_i to v_j. Use the first intersection and tail exchange as a sign-reversing involution to prove

$$\sum_{\substack{(\pi_1, \dots, \pi_r) \\ \text{vertex-disjoint}}} \prod_i \mathrm{wt}(\pi_i) = \det[Z(u_i, v_j)]_{i,j=1}^r.$$

Apply this identity to the interlacing-chain encoding of s_λ.

Definition 10.5 (Finite Schur kernel)**.** For the preceding finite alphabets, set

$$\Phi(z) = \frac{\prod_j (1 - \gamma_j / z)}{\prod_i (1 - \alpha_i z)}.$$

Choose concentric contours with

$$\max_j \gamma_j < |w| < |z| < \min_i \alpha_i^{-1}.$$

For $r, s \in \mathbb{Z} + \frac{1}{2}$, define

$$K_{\alpha,\gamma}(r, s) = \frac{1}{(2\pi i)^2} \oint \oint \frac{\Phi(z)}{\Phi(w)} \frac{1}{z - w} z^{-r-1/2} w^{s-1/2} \, dz \, dw.$$

Exercise 10.4 (Schur correlation kernel)**.** Encode λ by

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \tfrac{1}{2} : i \geq 1\}.$$

Glue the two nonintersecting path ensembles associated with $s_\lambda(\alpha)$ and $s_\lambda(\gamma)$ along their common boundary λ. Apply the Andréief identity successively to prove that the occupation correlations of the glued ensemble are determinants. Evaluate the resulting one-particle propagators by coefficient extraction and derive the kernel $K_{\alpha,\gamma}$. Deduce the exact finite Fredholm formula

$$\mathbb{P}(G(m, n) \leq L) = \det(I - K_{\alpha,\gamma})_{\ell^2(\{L + \frac{1}{2}, L + \frac{3}{2}, \dots\})}.$$

State how repeated last-passage or growth experiments with the specified geometric weights test this distribution without invoking an asymptotic limit.

Definition 10.6 (Exponential last-passage model)**.** Let E_{ij} be independent rate-one exponential random variables and define

$$G_{\exp}(m,n) = \max_{\pi:(1,1)\nearrow(m,n)} \sum_{(i,j)\in\pi} E_{ij}.$$

Exercise 10.5 (Exponential last-passage kernel)**.** Set $\alpha_i = \gamma_j = e^{-\varepsilon/2}$ in the $n \times n$ geometric model and prove that εW_{ij} converges in distribution to a rate-one exponential variable as $\varepsilon \downarrow 0$. Take this limit in the finite Schur Fredholm formula and identify the resulting finite-n kernel with the Laguerre Christoffel–Darboux kernel.

Exercise 10.6 (Laguerre edge limit)**.** Use the Laguerre contour integral to locate the double saddle at the upper spectral edge. Perform the cubic rescaling, prove uniform exponential bounds on the deformed contours, and derive convergence of the rescaled Laguerre kernel to K_{Ai}. Use the bounds to pass to the Fredholm series.

Exercise 10.7 (Last-passage fluctuations)**.** Combine the exponential last-passage determinant with the Laguerre edge limit to prove

$$\frac{G_{\exp}(n,n) - 4n}{2^{4/3}n^{1/3}} \Longrightarrow \chi_2, \qquad \mathbb{P}(\chi_2 \le s) = F_2(s).$$

Deduce the $n^{1/3}$ height-fluctuation scale. Balance this scale against the quadratic curvature of the deterministic limit shape to predict the transverse $n^{2/3}$ scale. State the independent-weight, boundary, centering, scale-calibration, and finite-size assumptions required when the Tracy–Widom prediction is compared with interface-height or directed transport data.

Part III

Quantization

11 Geometric Quantization

Definition 11.1 (Differential forms)**.** A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$; write $\Omega^k(M)$ for the space of differential k-forms. The exterior products on the cotangent spaces define

$$\wedge : \Omega^r(M) \times \Omega^s(M) \longrightarrow \Omega^{r+s}(M).$$

For a smooth map $F : M \to N$, define

$$(F^*\alpha)_p(v_1, \ldots, v_k) = \alpha_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

For a vector field X, define contraction by

$$(\iota_X \alpha)_p(v_2, \ldots, v_k) = \alpha_p(X_p, v_2, \ldots, v_k).$$

The *exterior derivative* is the family $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and $F^*d = dF^*$. For a vector field X with local flow Φ_t, define

$$\mathcal{L}_X \alpha = \left.\frac{d}{dt}\right|_{t=0} \Phi_t^* \alpha.$$

Exercise 11.1 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

Exercise 11.2 (Local Poincaré lemma)**.** Let U be a star-shaped open neighborhood of 0 in a finite-dimensional real vector space. For $\alpha \in \Omega^k(U)$ with $k > 0$, define

$$(K\alpha)_x(v_1, \ldots, v_{k-1}) = \int_0^1 t^{k-1} \alpha_{tx}(x, v_1, \ldots, v_{k-1})\, dt.$$

Use Cartan's identity for the radial flow to prove

$$dK + Kd = \mathrm{id}$$

on positive-degree forms. Deduce that every closed form of positive degree on U is exact, with a primitive vanishing at the origin.

Definition 11.2 (Cotangent phase space)**.** For the projection $\pi : T^*Q \to Q$, define the tautological one-form by

$$\theta_\alpha(v) = \alpha(d\pi(v)), \qquad \alpha \in T^*Q, \quad v \in T_\alpha(T^*Q).$$

The canonical two-form on T^*Q is $\omega = d\theta$.

Exercise 11.3 (From linear to cotangent phase space)**.** For a finite-dimensional real vector space V, construct the canonical identifications

$$TV \cong V \times V, \qquad T^*V \cong V \times V^*.$$

Prove that a phase $S \in C^\infty(Q)$ determines the momentum $p = dS_q \in T_q^*Q$ and wave covector $p/\hbar$. For a diffeomorphism $F : Q \to Q$, construct its cotangent lift and prove that it preserves θ and $d\theta$. Under $T^*V \cong V \times V^*$, prove

$$\omega((q, p), (q', p')) = p(q') - p'(q).$$

Definition 11.3 (Symplectic manifolds)**.** A *symplectic form* on a smooth manifold M is a closed nondegenerate two-form ω. A *symplectic vector space* is a finite-dimensional real vector space with a nondegenerate alternating bilinear form, regarded as a constant symplectic form. For $f \in C^\infty(M)$, define

$$\iota_{X_f}\omega = -df, \qquad \{f, g\} = -\omega(X_f, X_g).$$

The local flow of X_f is the *Hamiltonian flow* of f. A diffeomorphism $F : (M, \omega_M) \to (N, \omega_N)$ is a *canonical transformation* when $F^*\omega_N = \omega_M$.

Exercise 11.4 (Hamiltonian identities)**.** Prove that the canonical two-form $d\theta$ on T^*Q is nondegenerate and hence symplectic. Prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. For the Hamiltonian flow Φ_t of H, prove

$$\frac{d}{dt} f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

Exercise 11.5 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the volume form $\omega^n/n!$, and every density depending only on the Hamiltonian. On a symplectic vector space, prove that the infinitesimal generator of a linear Hamiltonian flow has trace zero and that the flow has determinant one.

Definition 11.4 (Contact manifolds)**.** Let Y be a smooth manifold of dimension $2n+1$. A one-form α is a *contact form* when

$$\alpha \wedge (d\alpha)^n$$

is nowhere zero. Its kernel $\xi = \ker \alpha$ is the associated cooriented *contact structure*; contact forms defining the same cooriented contact structure differ by multiplication by a positive smooth function. The *Reeb vector field* R_α is defined by

$$\alpha(R_\alpha) = 1, \qquad \iota_{R_\alpha} d\alpha = 0.$$

An immersed n-manifold $\iota : L \to Y$ is *Legendrian* when $\iota^*\alpha = 0$. A diffeomorphism $F : (Y, \xi) \to (Y', \xi')$ is a *contact transformation* when $F^*\alpha' = h\alpha$ for a positive smooth function h.

Exercise 11.6 (Contactification and symplectization)**.** Let $(M, \omega = d\lambda)$ be an exact symplectic manifold. On $M \times \mathbb{R}$, with coordinate function u on the second factor, prove that

$$\alpha = du - \lambda$$

is a contact form and that R_α is the infinitesimal generator of translation in the $\mathbb{R}$ factor. For a manifold X, identify $J^1X = T^*X \times \mathbb{R}$ with this contactification and prove that

$$j^1 f : X \longrightarrow J^1X, \qquad j^1 f(x) = (df_x, f(x)),$$

is Legendrian for every $f \in C^\infty(X)$. Prove that every section of $J^1X \to X$ with Legendrian image is of this form.

For a contact manifold (Y, α) and the positive coordinate r on $\mathbb{R}_{>0}$, prove that

$$\Omega = d(r\alpha)$$

is symplectic on $\mathbb{R}_{>0} \times Y$. Determine the vector field Z satisfying $\iota_Z \Omega = r\alpha$ and prove that multiplication of α by a positive function gives a symplectomorphic symplectization.

Definition 11.5 (Integration of forms)**.** An *orientation* of an n-manifold is a continuous choice of component of the nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 11.7 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form B and electric one-form E, prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt}\int_S B \qquad \text{and} \qquad dE = -\partial_t B,$$

recovering the observed flux–emf law.

Definition 11.6 (de Rham cohomology)**.** A differential form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Since $d^2 = 0$, the *de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) = \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\operatorname{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes $H^*_{\mathrm{dR}}(M)$ a graded-commutative algebra.

Exercise 11.8 (de Rham classes and periods)**.** Use the local Poincaré lemma and Stokes' theorem to prove that an exact k-form has zero integral over every closed oriented k-manifold mapped smoothly into M. Prove $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, with the isomorphism given by integration.
On $\mathbb{R}^2 \setminus \{0\}$, show that

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed and has integral 2π around the unit circle, hence is not exact. Compute $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$. Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential $F = dA$ exactly when $[F] = 0$ in de Rham cohomology.

Definition 11.7 (Connections and parallel transport)**.** A *connection* on a vector bundle $E \to M$ is an $\mathbb{R}$-bilinear map $\nabla : \Gamma(TM) \times \Gamma(E) \to \Gamma(E)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fs) = X(f)s + f\nabla_X s$. Along a curve γ it induces a covariant derivative D/dt; its solutions $Ds/dt = 0$ define *parallel transport* between endpoint fibers.

Definition 11.8 (Curvature and holonomy)**.** The *curvature* of ∇ is the $\mathrm{End}(E)$-valued two-form

$$R^{\nabla}(X,Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s.$$

The *holonomy group* at p consists of the automorphisms of E_p obtained by parallel transport around loops based at p.

Exercise 11.9 (Parallel transport and curvature)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

Definition 11.9 (Principal connections)**.** For a Lie group G, a *principal G-bundle* is a smooth map $P \to M$ with a free right G-action, transitive on each fiber, and local G-equivariant trivializations. For a representation $G \to \mathrm{GL}(W)$, its associated vector bundle is $P \times_G W = (P \times W)/((pg, w) \sim (p, gw))$. A *principal connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M.

Exercise 11.10 (Gauge covariance and Bianchi identity)**.** Prove $d_A F_A = 0$. For a gauge transformation Φ, prove $F_{\Phi^* A} = \Phi^* F_A$ and that holonomy changes by conjugacy. Prove that a flat connection has trivial holonomy on contractible loops and is gauge-equivalent to a product connection on a connected base exactly when its holonomy representation is trivial. For $G = U(1)$, compute the holonomy around a generator of the fundamental group and exhibit the obstruction to removing a flat connection globally.

Definition 11.10 (Prequantum line bundles)**.** A *prequantum line bundle* over (M, ω) is a Hermitian line bundle $L \to M$ with a unitary connection ∇ satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega.$$

For $f \in C^\infty(M)$, its *Kostant–Souriau operator* is

$$\mathcal{Q}_{\mathrm{pre}}(f) = -i\hbar \nabla_{X_f} + f.$$

Definition 11.11 (Metaplectic correction)**.** A *complex polarization* is an involutive Lagrangian subbundle $P \subset T_{\mathbb{C}}M$. Its annihilator and canonical line are

$$P^\circ = \{\alpha \in T^*_{\mathbb{C}}M : \alpha|_P = 0\}, \qquad K_P = \Lambda^{\mathrm{top}} P^\circ.$$

A *half-form bundle* is a line bundle δ_P with $\delta_P^{\otimes 2} \cong K_P$, equipped with the partial connection along P induced by Lie derivatives. A corrected polarized section of $L \otimes \delta_P$ is covariantly constant along P and square-integrable for the density determined by the half-form pairing. The corrected quantum space is its Hilbert completion when this pairing is positive.

Exercise 11.11 (Prequantum integrality)**.** Prove that a prequantum line bundle exists if and only if the period condition

$$\frac{1}{2\pi\hbar}\int_\Sigma f^*\omega \in \mathbb{Z}$$

holds for every smooth map from a closed oriented surface $f : \Sigma \to M$. Construct the line-bundle transition functions from local symplectic potentials and prove that the period condition is exactly their cocycle condition.

Exercise 11.12 (Kostant–Souriau commutators)**.** Prove

$$[\mathcal{Q}_{\mathrm{pre}}(f), \mathcal{Q}_{\mathrm{pre}}(g)] = i\hbar\mathcal{Q}_{\mathrm{pre}}(\{f, g\}).$$

Exercise 11.13 (Contactification)**.** Let $\pi : S(L) \to M$ be the unit-circle bundle of L, let $A \in \Omega^1(S(L); i\mathbb{R})$ be its connection form normalized by $A(\xi) = i$ on the circle generator ξ, and put $\alpha = -iA$. Prove

$$d\alpha = -\frac{1}{\hbar}\pi^*\omega,$$

that α is a contact form with Reeb vector field ξ, and that sections of L correspond to functions $F : S(L) \to \mathbb{C}$ satisfying $F(ue^{i\theta}) = e^{-i\theta}F(u)$.

Exercise 11.14 (Polarized quantum spaces)**.** For T^*V with the vertical polarization, recover $L^2(V)$ and the operators

$$\psi(x) \longmapsto \ell(x)\psi(x), \qquad \psi(x) \longmapsto -i\hbar\, d\psi_x(v).$$

For a compact Kähler manifold with $P = T^{0,1}M$, prove $P^\circ = T^{*1,0}M$ and $K_P = K_M$. For a positive prequantum line bundle, identify the corrected quantum space with $H^0(M, L \otimes K_M^{1/2})$.

Definition 11.12 (Theta functions with characteristics)**.** For $a, b \in \mathbb{R}$, $z \in \mathbb{C}$, and τ in the upper half-plane, define

$$\vartheta\begin{bmatrix} a \\ b \end{bmatrix}(z \mid \tau) = \sum_{n\in\mathbb{Z}} \exp\Big(\pi i(n+a)^2\tau + 2\pi i(n+a)(z+b)\Big).$$

Exercise 11.15 (Landau theta functions)**.** Let $Q = \mathbb{R}^2/(L_x\mathbb{Z} \oplus L_y\mathbb{Z})$ have area $A = L_xL_y$, let a particle of charge e experience the constant magnetic field $B\,dx \wedge dy$, and assume $eB > 0$. Put

$$N = \frac{eBA}{2\pi\hbar}, \qquad \tau = i\frac{L_y}{L_x}, \qquad z = \frac{x}{L_x} + \tau\frac{y}{L_y}.$$

Use the prequantization condition to prove that a magnetic line bundle $L \to Q$ exists exactly when $N \in \mathbb{Z}$. For $N > 0$, use the holomorphic-gauge automorphy laws

$$f(z+1) = f(z), \qquad f(z+\tau) = e^{-\pi i N\tau - 2\pi i N z} f(z)$$

to prove the magnetic-translation decomposition

$$H^0(Q,L) = \bigoplus_{r\in\mathbb{Z}/N\mathbb{Z}} \mathbb{C}\,\vartheta\begin{bmatrix} r/N \\ 0 \end{bmatrix}(Nz \mid N\tau).$$

Derive the automorphy and magnetic-translation eigenvalue laws directly from the defining series. Relate the holomorphic-gauge sections to physical wavefunctions by their common unitary-gauge factor.

For magnetic translations by L_x/N and L_y/N, use holonomy around a fundamental magnetic cell to prove

$$UV = e^{2\pi i/N} VU.$$

For the covariant Landau Hamiltonian, construct the raising and lowering operators and prove

$$E_n = \hbar\frac{eB}{m}\left(n+\frac{1}{2}\right), \qquad \dim E_{\widehat{H}}(\{E_n\})H = N.$$

Deduce the cyclotron resonance spacing, the degeneracy density $eB/(2\pi\hbar)$, and the magnetic-translation interference phase. State how measurements of these three quantities separately test the energy, flux-integrality, and theta-function predictions.

12 Deformation Quantization

Definition 12.1 (Coadjoint action and stabilizers)**.** For a Lie group G with Lie algebra $\mathfrak{g}$, define

$$\mathrm{Ad}_g = d(h\mapsto ghg^{-1})_e, \qquad (\mathrm{Ad}_g^*\,\ell)(X) = \ell(\mathrm{Ad}_{g^{-1}} X).$$

For $\ell\in\mathfrak{g}^*$, define

$$\mathcal{O}_\ell = \mathrm{Ad}^*(G)\ell, \qquad G_\ell = \{g\in G : \mathrm{Ad}_g^*\,\ell = \ell\},$$

and

$$\mathfrak{g}_\ell = \{X\in\mathfrak{g} : \ell([X,Y]) = 0 \text{ for every } Y\in\mathfrak{g}\}.$$

For $X\in\mathfrak{g}$, set

$$(\mathrm{ad}_X^*\,\ell)(Y) = -\ell([X,Y]).$$

Definition 12.2 (Kirillov form)**.** On $T_\ell\mathcal{O}_\ell$, define

$$\omega_\ell(\mathrm{ad}_X^*\,\ell, \mathrm{ad}_Y^*\,\ell) = \ell([X,Y]).$$

For $\eta = \mathrm{Ad}_g^*\,\ell$, transport ω_ℓ by the coadjoint action to define ω_η.

Exercise 12.1 (Geometry of coadjoint orbits)**.** Prove

$$T_\ell\mathcal{O}_\ell = \{\mathrm{ad}_X^*\,\ell : X\in\mathfrak{g}\} \cong \mathfrak{g}/\mathfrak{g}_\ell.$$

Prove that the Kirillov form is well-defined, alternating, nondegenerate, and G-invariant. Prove the cyclic identity

$$\ell([X,[Y,Z]]) + \ell([Y,[Z,X]]) + \ell([Z,[X,Y]]) = 0$$

and identify it with the closedness identity for the invariant two-form ω on $\mathcal{O}_\ell$.

Definition 12.3 (Projective representations)**.** A *normalized multiplier* on a locally compact group G is a continuous map $\sigma : G \times G \to \mathbb{T}$ satisfying

$$\sigma(x, e) = \sigma(e, x) = 1, \qquad \sigma(x, y)\sigma(xy, z) = \sigma(y, z)\sigma(x, yz).$$

A *σ-projective unitary representation* is a strongly continuous map $U : G \to U(H)$ satisfying

$$U_x U_y = \sigma(x, y)U_{xy}, \qquad U_e = I.$$

Multipliers σ and σ' are *cohomologous* when there is a continuous map $b : G \to \mathbb{T}$, $b(e) = 1$, such that

$$\sigma'(x, y) = b(x)b(y)\overline{b(xy)}\,\sigma(x, y).$$

Definition 12.4 (Weyl multipliers)**.** Let (E, ω) be a symplectic vector space. For $\lambda \in \mathbb{R}$, define

$$\sigma_{\lambda\omega}(z, z') = e^{i\lambda\omega(z,z')/2}.$$

For physical momentum variables and $\hbar > 0$, the *Weyl multiplier* is $\sigma_{\omega/\hbar}$.

Exercise 12.2 (The Weyl multiplier)**.** Prove that $\sigma_{\lambda\omega}$ is a normalized multiplier for every $\lambda \in \mathbb{R}$.

Definition 12.5 (Commutator bicharacter)**.** For a multiplier σ on an abelian group A, define

$$c_\sigma(x, y) = \sigma(x, y)\overline{\sigma(y, x)}.$$

Exercise 12.3 (Commutator bicharacter)**.** Prove that c_σ is an alternating bicharacter and depends only on the cohomology class of σ. For the Weyl multiplier on $V \oplus V^*$, prove

$$c_{\sigma_{\omega/\hbar}}\big((q, p), (q', p')\big) = e^{i(p(q')-p'(q))/\hbar}.$$

Prove

$$c_{\sigma_{\lambda\omega}}(z, z') = e^{i\lambda\omega(z,z')}.$$

Definition 12.6 (Heisenberg group)**.** For a symplectic vector space (E, ω), its *Heisenberg group* is

$$\mathbb{H}(E, \omega) = E \times \mathbb{R}$$

with multiplication

$$(z, s)(z', s') = (z + z', s + s' + \tfrac{1}{2}\omega(z, z')).$$

Write $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$ for the canonical symplectic form on $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$. Its Lie algebra is $\mathfrak{h}(E, \omega) = E \oplus \mathbb{R}Z$ with

$$[(z, sZ), (z', s'Z)] = \omega(z, z')Z.$$

For $E = V \oplus V^*$ and $\lambda \neq 0$, the *Schrödinger representation* on $L^2(V)$ is

$$\big(\pi_\lambda(q, p, s)\xi\big)(x) = e^{i\lambda(s+p(x-q/2))}\xi(x - q).$$

For $\hbar > 0$, $\ell \in V^*$, and $v \in V$, define on $\mathcal{S}(V)$

$$Q_\ell\psi(x) = \ell(x)\psi(x), \qquad P_v\psi(x) = -i\hbar\, d\psi_x(v).$$

Definition 12.7 (Polarization)**.** For a connected simply connected nilpotent group G and $\ell \in \mathfrak{g}^*$, a *polarization* is a Lie subalgebra $\mathfrak{p} \subseteq \mathfrak{g}$ satisfying

$$\ell([\mathfrak{p},\mathfrak{p}]) = 0, \qquad \dim \mathfrak{p} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

For $P = \exp(\mathfrak{p})$, set $\chi_\ell(\exp X) = e^{i\ell(X)}$. The induced representation $\operatorname{Ind}_P^G \chi_\ell$ acts by left translation on the L^2-sections over G/P satisfying $F(gp) = \chi_\ell(p)^{-1}F(g)$.

Exercise 12.4 (Central extensions)**.** Prove that $\mathbb{H}(E,\omega)$ is a locally compact group with center $\{0\} \times \mathbb{R}$ and that

$$\pi_\lambda(0,s) = e^{i\lambda s}I.$$

Prove that projective representations of E with multiplier $e^{i\lambda\omega/2}$ correspond to representations of $\mathbb{H}(E,\omega)$ with central character $e^{i\lambda s}$. Let $p : \mathbb{H}(E,\omega) \to E$ be the quotient map and define

$$\alpha_{(z,s)}(v,r) = r - \frac{1}{2}\omega(z,v).$$

Prove that α is left-invariant,

$$d\alpha = -p^*\omega,$$

and that α is a contact form whose Reeb flow is translation in the central direction.

Exercise 12.5 (Canonical commutation and Weyl relations)**.** Prove

$$[Q_\ell, P_v] = i\hbar\ell(v)I.$$

For

$$(T_q\psi)(x) = \psi(x-q), \qquad (M_p^\hbar\psi)(x) = e^{ip(x)/\hbar}\psi(x),$$

prove

$$M_p^\hbar T_q = e^{ip(q)/\hbar}T_q M_p^\hbar.$$

For

$$W_\hbar(q,p) = e^{-ip(q)/(2\hbar)}M_p^\hbar T_q,$$

prove

$$W_\hbar(z)W_\hbar(z') = e^{i\omega(z,z')/(2\hbar)}W_\hbar(z+z')$$

and identify $W_\hbar(z)$ with $\pi_{1/\hbar}(z,0)$ under the canonical set-theoretic section $z \mapsto (z,0)$. Prove that this section is not a homomorphism and that its failure to preserve multiplication is exactly the displayed multiplier.

Exercise 12.6 (Schrödinger model)**.** Identify

$$\mathfrak{h}(E,\omega)^* = E^* \oplus \mathbb{R}Z^*.$$

For $(\xi,\lambda) \in E^* \oplus \mathbb{R}Z^*$, prove

$$\operatorname{Ad}^*_{(z,s)}(\xi,\lambda) = (\xi - \lambda\omega(z,\cdot),\lambda).$$

Deduce that the orbits with $\lambda = 0$ are points and that, for $\lambda \neq 0$,

$$\mathcal{O}_\lambda = \{(\xi,\lambda) : \xi \in E^*\}.$$

Prove that

$$E \longrightarrow \mathcal{O}_\lambda, \qquad z \longmapsto (-\lambda\omega(z,\cdot), \lambda),$$

pulls the Kirillov form back to $\lambda\omega$.

For $E = V \oplus V^*$ and $\mathfrak{p} = V^* \oplus \mathbb{R}Z$, prove that $\mathfrak{p}$ is a polarization at $(0,\lambda)$. Identify $\mathbb{H}(E,\omega)/\exp(\mathfrak{p})$ with V and compute the induced action to obtain

$$(\pi_\lambda(q,p,s)\psi)(x) = e^{i\lambda(s+p(x-q/2))}\psi(x-q).$$

For $\lambda = 0$, recover every character of E from a point orbit.

Exercise 12.7 (Irreducibility of the Schrödinger representation)**.** Prove that an operator on $L^2(V)$ commuting with every translation and every modulation is scalar: first use the Fourier transform to identify the commutant of translations, and then impose commutation with modulations. Deduce that π_λ is irreducible for $\lambda \neq 0$.

Exercise 12.8 (Stone–von Neumann theorem)**.** Let U be a strongly continuous irreducible unitary representation of $\mathbb{H}(V \oplus V^*, \omega)$ with central character $U(0,s) = e^{i\lambda s}I$, where $\lambda \neq 0$. Integrate Schwartz functions on $V \oplus V^*$ against the canonical section of U. Use Fourier inversion and a Gaussian approximate identity to show that this integrated representation contains the same nonzero finite-rank operators as the Schrödinger model. Construct a nonzero intertwiner and use irreducibility to prove $U \simeq \pi_\lambda$.

Definition 12.8 (Trace and Hilbert–Schmidt classes)**.** For $x, y \in H$, let

$$\theta_{x,y}(\xi) = \langle y, \xi\rangle x.$$

The *trace class* $S_1(H)$ consists of operators admitting a representation

$$A = \sum_{j=1}^{\infty} \theta_{x_j,y_j}, \qquad \sum_{j=1}^{\infty} \|x_j\|\,\|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\mathrm{Tr}(A) = \sum_{j=1}^{\infty} \langle y_j, x_j\rangle.$$

The *Hilbert–Schmidt class* is

$$S_2(H) = \{A \in B(H) : A^*A \in S_1(H)\}, \qquad \|A\|_2^2 = \mathrm{Tr}(A^*A).$$

Exercise 12.9 (Trace ideals)**.** Prove that $S_1(H)$, Tr, and $S_2(H)$ are well-defined. Prove that $\langle A, B\rangle_{S_2} = \mathrm{Tr}(A^*B)$ makes $S_2(H)$ a Hilbert space and that the pairing $(A,T) \mapsto \mathrm{Tr}(AT)$ identifies $S_1(H)^*$ isometrically with $B(H)$.

Definition 12.9 (Coefficient transforms)**.** Let (X,μ) be a measure space and let $U : X \to U(H)$ be a weakly measurable map. The *coefficient transform* of the unitary system U is

$$\mathcal{V}_U : S_1(H) \longrightarrow L^\infty(X), \qquad \mathcal{V}_U(A)(x) = \mathrm{Tr}(AU_x).$$

Definition 12.10 (Fourier–Wigner transform)**.** Let V be an n-dimensional real vector space with Lebesgue measure dq, let $\hbar > 0$, and equip V^* with the dual measure dp for which $(2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q)\, dq$ is unitary on L^2. For $q \in V$ and $p \in V^*$, define

$$(T_q\xi)(x) = \xi(x-q), \qquad (M_p^\hbar \xi)(x) = e^{ip(x)/\hbar}\xi(x).$$

For $z = (q,p) \in V \oplus V^*$, define

$$W_\hbar(z) = e^{-ip(q)/(2\hbar)} M_p^\hbar T_q.$$

The *Fourier–Wigner transform* is

$$\mathcal{W}_\hbar : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \qquad \mathcal{W}_\hbar(A)(z) = (2\pi\hbar)^{-n/2}\operatorname{Tr}(AW_\hbar(z)).$$

Exercise 12.10 (Fourier–Wigner Plancherel)**.** Prove

$$W_\hbar(z)^* = W_\hbar(-z).$$

For $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$, prove the Fourier–Wigner orthogonality identity

$$\int_{V\oplus V^*} \overline{\mathcal{W}_\hbar(A)(z)}\mathcal{W}_\hbar(B)(z)\, dz = \operatorname{Tr}(A^*B).$$

Deduce that $\mathcal{W}_\hbar$ extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever $\mathcal{W}_\hbar(A) \in L^1(V \oplus V^*)$, prove

$$A = (2\pi\hbar)^{-n/2} \int_{V\oplus V^*} \mathcal{W}_\hbar(A)(z) W_\hbar(z)^*\, dz$$

in the weak operator topology.

Exercise 12.11 (Schrödinger character)**.** For $V = \mathbb{R}^n$, let $f \in C^\infty(\mathbb{H}_n)$ satisfy

$$(q,p,s) \longmapsto f\big(\exp(q,p,s)\big) \in \mathcal{S}(\mathbb{R}^{2n+1})$$

under the canonical identification $\mathfrak{h}_n = \mathbb{R}^n \oplus (\mathbb{R}^n)^* \oplus \mathbb{R}$. With Haar measure $dq\, dp\, ds$, put

$$\pi_\lambda(f) = \int_{\mathbb{H}_n} f(g)\pi_\lambda(g)\, dg.$$

Prove

$$\operatorname{Tr}\pi_\lambda(f) = (2\pi)^n |\lambda|^{-n} \int_{\mathbb{R}} f(0,0,s) e^{i\lambda s}\, ds.$$

Derive this identity directly from the Schrödinger kernel and identify the factor $|\lambda|^{-n}$ with the symplectic-volume scaling of $\mathcal{O}_\lambda$.

Exercise 12.12 (Heisenberg Plancherel and inversion)**.** For $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$, use Haar measure $dq\, dp\, ds$ and the scalar Fourier transform $\widehat{h}(\lambda) = \int_{\mathbb{R}} h(s)e^{-i\lambda s}\, ds$, and define

$$\widehat{f}(\lambda) = \int_{\mathbb{H}_n} f(g)\pi_\lambda(g)^*\, dg.$$

Compute the integral kernel of $\widehat{f}(\lambda)$ and apply Euclidean Plancherel in the central and phase-space variables. Deduce the Plancherel measure $(2\pi)^{-(n+1)}|\lambda|^n\, d\lambda$ and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R}\setminus\{0\}} \|\widehat{f}(\lambda)\|_{\mathrm{HS}}^2 |\lambda|^n\, d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

Definition 12.11 (Weyl quantization)**.** For $a \in \mathcal{S}(V \oplus V^*)$, define $\mathrm{Op}_\hbar^W(a) : \mathcal{S}(V) \to \mathcal{S}(V)$ by

$$(\mathrm{Op}_\hbar^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n} \int_{V \times V^*} e^{ip(x-y)/\hbar} a\left(\frac{x+y}{2}, p\right) \psi(y)\, dy\, dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on $\mathcal{S}(V)$.

Definition 12.12 (Wigner distributions)**.** For $\psi, \varphi \in \mathcal{S}(V)$, define

$$W_\hbar(\psi, \varphi)(q, p) = \frac{1}{(2\pi\hbar)^n} \int_V e^{-ip(y)/\hbar} \overline{\psi(q - y/2)}\, \varphi(q + y/2)\, dy.$$

Write $W_\hbar[\psi] = W_\hbar(\psi, \psi)$.

Exercise 12.13 (Weyl calculus and the Moyal product)**.** For $\ell \in V^*$ and $v \in V$, prove

$$\mathrm{Op}_\hbar^W((q,p) \mapsto \ell(q)) = Q_\ell, \qquad \mathrm{Op}_\hbar^W((q,p) \mapsto p(v)) = P_v.$$

Prove

$$\mathrm{Op}_\hbar^W(a)^* = \mathrm{Op}_\hbar^W(\overline{a}), \qquad \| \mathrm{Op}_\hbar^W(a)\|_{\mathrm{HS}} = (2\pi\hbar)^{-n/2}\|a\|_2,$$

and

$$\langle \psi, \mathrm{Op}_\hbar^W(a)\varphi \rangle = \int_{V \oplus V^*} a(q,p) W_\hbar(\psi, \varphi)(q,p)\, dq\, dp.$$

Prove that for $a, b \in \mathcal{S}(V \oplus V^*)$ there is a unique $a \star_\hbar b \in \mathcal{S}(V \oplus V^*)$ satisfying

$$\mathrm{Op}_\hbar^W(a \star_\hbar b) = \mathrm{Op}_\hbar^W(a)\, \mathrm{Op}_\hbar^W(b).$$

Derive the oscillatory-integral formula for $\star_\hbar$ and prove

$$a \star_\hbar b = ab + \frac{i\hbar}{2}\{a, b\} + O(\hbar^2), \qquad \frac{a \star_\hbar b - b \star_\hbar a}{i\hbar} = \{a, b\} + O(\hbar^2)$$

in the Schwartz-symbol topology.

Exercise 12.14 (Groenewold–van Hove obstruction)**.** Specialize to $V = \mathbb{R}$ and write $Q = Q_{\mathrm{id}_\mathbb{R}}$ and $P = P_1$. Prove that the polynomials of degree at most two form a Lie subalgebra for the Poisson bracket and that Weyl quantization is an exact Lie-algebra quantization on this subalgebra. Quantize q^2p^2 using

$$q^2p^2 = \frac{1}{9}\{q^3, p^3\} = \frac{1}{3}\{q^2p, qp^2\}$$

and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \qquad \mathcal{Q}(q) = Q, \qquad \mathcal{Q}(p) = P, \qquad \mathcal{Q}(q^m) = Q^m, \qquad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f, g\}) = (i\hbar)^{-1}[\mathcal{Q}(f), \mathcal{Q}(g)]$$

on a common invariant domain.

Definition 12.13 (Classical and quantum evolution)**.** Let $H \in C^\infty(V \oplus V^*, \mathbb{R})$ have bounded derivatives of order at least two, let Φ_t be its Hamiltonian flow, and let $\mathrm{Op}_\hbar^W(H)$ denote its self-adjoint realization. Define

$$U_\hbar(t) = e^{-it\, \mathrm{Op}_\hbar^W(H)/\hbar}.$$

For $a \in \mathcal{S}(V \oplus V^*)$, define

$$a_t = a \circ \Phi_t, \qquad A_\hbar(t) = U_\hbar(t)^*\, \mathrm{Op}_\hbar^W(a) U_\hbar(t).$$

Exercise 12.15 (Heisenberg equation and Egorov's theorem)**.** Prove

$$\frac{d}{dt}a_t = \{a_t, H\}, \qquad \frac{d}{dt}A_\hbar(t) = \frac{i}{\hbar}[\mathrm{Op}_\hbar^W(H), A_\hbar(t)].$$

For every $T > 0$, prove Egorov's theorem

$$\sup_{|t|\le T} \left\| A_\hbar(t) - \mathrm{Op}_\hbar^W(a \circ \Phi_t) \right\| = O_T(\hbar).$$

Prove exact equality for quadratic H. Derive Ehrenfest's equations for the expectations of Q_ℓ and P_v for every $\ell \in V^*$ and $v \in V$.

Exercise 12.16 (Linear symplectic covariance)**.** Let S be a linear symplectic automorphism of $V \oplus V^*$. Construct unitary implementers for cotangent lifts of linear automorphisms of V, for symplectic shears defined by quadratic forms, and for the exchange of position and momentum implemented by the Fourier transform. Prove for these generators that

$$\mu(S)\,\mathrm{Op}_\hbar^W(a)\mu(S)^* = \mathrm{Op}_\hbar^W(a \circ S^{-1}), \qquad W_\hbar[\mu(S)\psi] = W_\hbar[\psi] \circ S^{-1}.$$

Deduce the covariance for their products and prove that each implementer is determined up to phase. Show that coherent choices form the metaplectic double cover, whose two lifts of a fixed symplectic map differ by sign. Use the Groenewold–van Hove obstruction to show why exact covariance cannot extend to every nonlinear canonical transformation.

Exercise 12.17 (Wigner–Laguerre identity)**.** Use the generalized Laguerre polynomials and their orthogonality from Peter–Weyl Theory. For

$$H_{\mathrm{cl}}(q,p) = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}, \qquad \psi_n(q) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\!\left(\sqrt{\frac{m\omega}{\hbar}}q\right),$$

prove

$$W_\hbar[\psi_n](q,p) = \frac{(-1)^n}{\pi\hbar} e^{-2H_{\mathrm{cl}}(q,p)/(\hbar\omega)} L_n\!\left(\frac{4H_{\mathrm{cl}}(q,p)}{\hbar\omega}\right).$$

Verify the position and momentum marginals and identify the zeros and sign changes of the Wigner distribution from those of L_n.

Exercise 12.18 (Laguerre tomography)**.** For the dimensionless quadrature

$$X_\theta = \sqrt{\frac{m\omega}{\hbar}}Q\cos\theta + \frac{P}{\sqrt{m\hbar\omega}}\sin\theta,$$

prove that its probability density in a state ψ is the Radon transform of $W_\hbar[\psi]$ along the corresponding phase-space lines. Derive the Fourier-slice inversion formula reconstructing the Wigner distribution from the family of quadrature densities. For the oscillator state ψ_n, use the Wigner–Laguerre identity to prove that the reconstructed radial profile is

$$\frac{(-1)^n}{\pi\hbar} e^{-2H_{\mathrm{cl}}/(\hbar\omega)} L_n\!\left(\frac{4H_{\mathrm{cl}}}{\hbar\omega}\right),$$

while every quadrature density is a scaled $e^{-x^2}H_n(x)^2$. Deduce the predicted number of quadrature nodes, the radial zero circles determined by L_n, and the sign at the phase-space origin. State how homodyne quadrature data distinguish adjacent number states and detect Wigner negativity after inverse Radon reconstruction.

Definition 12.14 (Poisson manifolds)**.** A *Poisson manifold* is a smooth manifold M with a bracket on $C^\infty(M)$ that is a Lie bracket and a derivation in each argument. Equivalently, there is a bivector $\Pi \in \Gamma(\Lambda^2 TM)$ satisfying

$$\{f, g\} = \Pi(df, dg), \qquad [\Pi, \Pi]_{\mathrm{Sch}} = 0.$$

Definition 12.15 (Star products)**.** A *differential star product* on a Poisson manifold is an associative $\mathbb{C}[[\hbar]]$-bilinear product on $C^\infty(M)[[\hbar]]$ of the form

$$f \star g = fg + \sum_{r \geq 1} \hbar^r C_r(f, g),$$

where the C_r are bidifferential operators, 1 is a unit, and

$$C_1(f, g) - C_1(g, f) = i\{f, g\}.$$

Two star products are *equivalent* when they are intertwined by an operator $T = \mathrm{id} + \sum_{r \geq 1} \hbar^r T_r$ with differential T_r.

Exercise 12.19 (Affine Moyal product)**.** Let (E, ω) be a symplectic vector space and let $\Pi : E^* \to E$ be the inverse of $v \mapsto \omega(v, -)$, and write $\Pi(k, \ell) = \ell(\Pi(k))$. Let D_Π be the constant bidifferential operator characterized on decomposable functions by

$$D_\Pi(f \otimes g)(x, y) = d_y g(\Pi(d_x f)).$$

On $C^\infty(E)[[\hbar]]$, define

$$f \star g = m \circ \exp\!\left(\frac{i\hbar}{2} D_\Pi\right)(f \otimes g),$$

where m restricts a function on $E \times E$ to the diagonal. Prove associativity from the commutation of the induced constant bidifferential operators on E^3, and prove

$$f \star g = fg + \frac{i\hbar}{2}\{f, g\} + O(\hbar^2).$$

For $k, \ell \in E^*$ put $e_k(x) = e^{ik(x)}$ and derive

$$e_k \star e_\ell = e^{-i\hbar \Pi(k,\ell)/2} e_{k+\ell}.$$

Prove associativity directly by checking the multiplier identity for the phase. Differentiate in k and ℓ to recover the canonical commutation relations. Prove that every affine symplectic transformation of E preserves $\star$. For $E = V \oplus V^*$, prove that the restriction to Schwartz functions is the formal $\hbar$-expansion of the Weyl product defined by operator composition above.

Definition 12.16 (Magnetic lattice Hamiltonian)**.** Let U, V be unitary translations satisfying

$$UV = e^{2\pi i \alpha} VU,$$

where α is the magnetic flux through one lattice cell in units of the flux quantum. The magnetic nearest-neighbor Hamiltonian is

$$H_\alpha = U + U^* + V + V^*.$$

In the position realization on $\ell^2(\mathbb{Z})$, with transverse phase θ, it is the Harper operator

$$(H_{\alpha,\theta}\psi)_m = \psi_{m+1} + \psi_{m-1} + 2\cos(2\pi\alpha m + \theta)\psi_m.$$

Exercise 12.20 (Chebyshev magnetic bands)**.** Derive the magnetic-translation relation from the Moyal plane-wave identity. For rational flux $\alpha = p/q$, prove that the Harper equation has q-periodic coefficients and reduce it by the Floquet condition $\psi_{m+q} = e^{iqk}\psi_m$ to a q-dimensional spectral problem. At zero flux, use the Chebyshev identity

$$T_q(\cos t) = \cos(qt)$$

to derive the q-step discriminant

$$\Delta_{0,q}(E, \theta) = 2T_q\left(\frac{E - 2\cos\theta}{2}\right).$$

At half a flux quantum, compute the two bands

$$E_\pm(k, \theta) = \pm 2\sqrt{\cos^2 k + \cos^2 \theta}.$$

Deduce the flux-dependent band edges and the touching energy, and state how microwave, photonic, or electronic lattice spectroscopy tests the Chebyshev discriminant and the half-flux splitting. Prove that the commutator phase $e^{2\pi i\alpha}$ is the phase accumulated around one lattice cell. Show that nonintegral total flux on a periodic sample obstructs consistent torus boundary phases, while integral total flux makes their corner cocycle trivial. Prove that nonzero integral flux still obstructs a single globally periodic vector potential.

13 Quantum Groups

Definition 13.1 (Poisson–Lie groups)**.** A *Poisson–Lie group* is a Lie group G with a Poisson structure for which multiplication $G \times G \to G$ is Poisson. A *Lie bialgebra* is a Lie algebra $\mathfrak{g}$ with a linear map

$$\delta : \mathfrak{g} \to \mathfrak{g} \wedge \mathfrak{g}$$

that is a 1-cocycle for the adjoint action and whose transpose defines a Lie bracket on $\mathfrak{g}^*$. It is *coboundary* when

$$\delta(x) = [x \otimes 1 + 1 \otimes x, r]$$

for some $r \in \mathfrak{g} \otimes \mathfrak{g}$.

Exercise 13.1 (Infinitesimal Poisson–Lie structure)**.** Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. For a coboundary Lie bialgebra, define

$$\mathrm{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Prove that the co-Jacobi identity is equivalent to $\mathrm{CYB}(r)$ being invariant under the diagonal adjoint action. Under the additional triangular hypothesis $\mathrm{CYB}(r) = 0$, recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as $\mathrm{CYB}(r) = \Omega$ for an invariant tensor Ω and distinguish the case $\Omega \neq 0$.

Definition 13.2 (Bialgebras and Hopf algebras)**.** A *bialgebra* is a unital associative algebra H with unital algebra homomorphisms

$$\Delta : H \to H \otimes H, \qquad \varepsilon : H \to \mathbb{C}$$

satisfying coassociativity and the counit identities. A *Hopf algebra* is a bialgebra with a linear antipode $S : H \to H$ satisfying

$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta.$$

Definition 13.3 (Deformation parameters)**.** The generic quantum-group parameter is $0 < v < 1$, the basic-hypergeometric base is Q, the root-of-unity parameter is ζ, the DAHA parameters are q, t, the elliptic nome is p, and $\kappa \neq 0$ is the independent qKZ shift.

Definition 13.4 (Compact real form of $U_v(\mathfrak{sl}_2)$)**.** For $0 < v < 1$, $U_v(\mathfrak{sl}_2)$ is generated by E, F, K, K^{-1} with

$$KEK^{-1} = v^2 E, \qquad KFK^{-1} = v^{-2} F, \qquad [E, F] = \frac{K - K^{-1}}{v - v^{-1}},$$

and

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

Its counit, antipode, and compact real-form involution are

$$\varepsilon(E) = \varepsilon(F) = 0, \quad \varepsilon(K) = 1, \quad S(E) = -K^{-1}E, \quad S(F) = -FK, \quad S(K) = K^{-1},$$

$$K^* = K, \qquad E^* = KF, \qquad F^* = EK^{-1}.$$

Exercise 13.2 (The $U_v(\mathfrak{sl}_2)$ Hopf deformation)**.** Verify that the coproduct preserves the defining relations and that the displayed counit, antipode, and involution make $U_v(\mathfrak{sl}_2)$ a Hopf $*$-algebra. In the $\hbar$-adic completion put

$$v = e^{-\hbar}, \qquad K = v^H = e^{-\hbar H}.$$

Prove that

$$[H, E] = 2E, \qquad [H, F] = -2F, \qquad \frac{K - K^{-1}}{v - v^{-1}} = H + O(\hbar^2),$$

and that the relations and coproduct tend to those of $U(\mathfrak{sl}_2)$ as $\hbar \to 0$. Expand the coproduct through first order and compute the resulting coboundary Lie bialgebra.

Definition 13.5 (Gaussian binomial coefficients)**.** For an indeterminate u, $n \geq 0$, and $0 \leq r \leq n$, define

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \prod_{j=1}^{r} \frac{1 - u^{n-r+j}}{1 - u^j},$$

with value 1 for $r = 0$ and value 0 outside this range.

Exercise 13.3 (Gaussian and quantum-group binomial identities)**.** Derive

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \begin{bmatrix} n-1 \\ r \end{bmatrix}_u + u^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_u$$

and prove

$$\prod_{j=0}^{n-1} (1 + tu^j) = \sum_{r=0}^{n} u^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_u t^r.$$

For $YX = uXY$, prove the noncommutative identity

$$(X + Y)^n = \sum_{r=0}^{n} \begin{bmatrix} n \\ r \end{bmatrix}_u X^r Y^{n-r}.$$

Set $u = v^2$ and apply it to $X = E \otimes 1$ and $Y = K \otimes E$ in $U_v(\mathfrak{sl}_2)$ to obtain an explicit formula for $\Delta(E^n)$. Show that the limit $v \to 1$ recovers the ordinary binomial theorem and the undeformed coproduct.

Exercise 13.4 (Gaussian-binomial state counts)**.** Apply the Gaussian binomial theorem from the preceding exercise. Prove that the coefficient of u^m in $\begin{bmatrix} n \\ r \end{bmatrix}_u$ counts partitions of m whose diagrams fit inside an $r \times (n-r)$ rectangle. For n fermionic levels with energies $j\hbar\omega$, $0 \le j < n$, prove that the coefficient of $t^r u^{m+r(r-1)/2}$ in

$$\prod_{j=0}^{n-1} (1 + tu^j)$$

is the exact number of r-particle states at excitation energy $m\hbar\omega$ above the filled lowest configuration, and identify it with the coefficient of u^m in $\begin{bmatrix} n \\ r \end{bmatrix}_u$.

Definition 13.6 (Compact quantum groups and Haar states)**.** A *compact quantum group* $\mathbb{G}$ is a unital C^*-algebra $C(\mathbb{G})$ with a unital $*$-homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where $\otimes_{\min}$ is the closure of the algebraic tensor product in $B(H \otimes K)$ for faithful representations on H and K. The coproduct satisfies

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta$$

and

$$\overline{\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))} = C(\mathbb{G}) \otimes_{\min} C(\mathbb{G}) = \overline{\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)}.$$

A *Haar state* is a state h satisfying

$$(h \otimes \mathrm{id})\Delta(a) = h(a)1 = (\mathrm{id} \otimes h)\Delta(a).$$

Definition 13.7 (Representative functions)**.** A finite-dimensional *unitary corepresentation* on H_U is a unitary $U \in B(H_U) \otimes C(\mathbb{G})$ satisfying

$$(\mathrm{id} \otimes \Delta)(U) = U_{12} U_{13}.$$

For $\omega \in B(H_U)^*$, the corresponding *coefficient* is

$$u_\omega = (\omega \otimes \mathrm{id})(U) \in C(\mathbb{G}).$$

The representative-function algebra $\mathrm{Pol}(\mathbb{G})$ is the unital $*$-subalgebra generated by these coefficients for all finite-dimensional unitary corepresentations.

Definition 13.8 ($SU_v(2)$)**.** Let $\mathrm{Pol}(SU_v(2))$ be the universal unital $*$-algebra generated by α, γ so that

$$u = \begin{pmatrix} \alpha & -v\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary. Equivalently, its relations are

$$\alpha^*\alpha + \gamma^*\gamma = 1, \qquad \alpha\alpha^* + v^2\gamma\gamma^* = 1, \qquad \gamma^*\gamma = \gamma\gamma^*,$$
$$\alpha\gamma = v\gamma\alpha, \qquad \alpha\gamma^* = v\gamma^*\alpha,$$

Define $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$, and let $C(SU_v(2))$ be the universal C^*-completion.

Exercise 13.5 ($SU_v(2)$ as a compact quantum group)**.** Prove that Δ is a coassociative $*$-homomorphism satisfying the cancellation conditions. On $\ell^2(\mathbb{Z}_{\geq 0})$, define for $\xi \in \mathbb{T}$

$$\pi_\xi(\gamma)e_n = \xi v^n e_n, \qquad \pi_\xi(\alpha)e_n = (1-v^{2n})^{1/2}e_{n-1}, \qquad e_{-1} = 0.$$

Verify the relations and prove that

$$h(a) = (1-v^2)\sum_{n\geq 0} v^{2n}\int_{\mathbb{T}}\langle e_n, \pi_\xi(a)e_n\rangle\, dm_{\mathbb{T}}(\xi)$$

is the Haar state.

Definition 13.9 (Quantum-group coefficient pairing)**.** For a finite-dimensional type-1 $U_v(\mathfrak{sl}_2)$-module V, $\xi \in V$, and $\varphi \in V^\vee$, define

$$c_{\varphi,\xi}(x) = \varphi(x\xi) \qquad (x \in U_v(\mathfrak{sl}_2)).$$

Let $\mathcal{O}_v$ be the Hopf algebra generated by these coefficient functionals. Its pairing with $U_v(\mathfrak{sl}_2)$ is

$$\langle x, c_{\varphi,\xi}\rangle = c_{\varphi,\xi}(x),$$

with multiplication and coproduct characterized by

$$\langle x, ab\rangle = \langle \Delta x, a\otimes b\rangle, \qquad \langle xy, a\rangle = \langle x\otimes y, \Delta a\rangle.$$

Exercise 13.6 (Coefficient corepresentations)**.** Prove that the coefficient map of a tensor product is the product of the coefficient maps, that the coefficient map of a dual module is obtained from the antipode, and that the displayed pairing is a nondegenerate Hopf pairing. Apply it to the fundamental module and prove that sending its coefficient corepresentation to the displayed matrix u extends to a Hopf $*$-isomorphism

$$\mathcal{O}_v \longrightarrow \mathrm{Pol}(SU_v(2)).$$

Exercise 13.7 ($SU_v(2)$ Peter–Weyl decomposition)**.** Construct the irreducible type-1 unitary $U_v(\mathfrak{sl}_2)$-modules V_n from their highest-weight lines and form their coefficient corepresentations of $\mathrm{Pol}(SU_v(2))$ using the Hopf pairing. Use the explicit Haar state to derive quantum Schur orthogonality and prove

$$\mathrm{Pol}(SU_v(2)) = \bigoplus_{n\geq 0} C(V_n)$$

algebraically and as a Hilbert direct sum in the Haar GNS space.

Definition 13.10 ($\mathbb{C}$-linear and C^*-tensor categories)**.** A $\mathbb{C}$-*linear category* $\mathcal{C}$ consists of objects, complex vector spaces $\mathrm{Hom}_{\mathcal{C}}(X,Y)$, bilinear associative compositions

$$\mathrm{Hom}_{\mathcal{C}}(Y,Z)\otimes \mathrm{Hom}_{\mathcal{C}}(X,Y) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X,Z), \qquad (g,f)\longmapsto gf,$$

and identity morphisms id_X. A $\mathbb{C}$-*linear tensor category* also has a bilinear tensor product on objects and morphisms, a tensor unit $\mathbf{1}$, and isomorphisms

$$a_{X,Y,Z} : (X\otimes Y)\otimes Z \longrightarrow X\otimes(Y\otimes Z), \quad l_X : \mathbf{1}\otimes X \longrightarrow X, \quad r_X : X\otimes \mathbf{1}\longrightarrow X.$$

The tensor product satisfies

$$(f'f)\otimes(g'g) = (f'\otimes g')(f\otimes g), \qquad \mathrm{id}_X\otimes \mathrm{id}_Y = \mathrm{id}_{X\otimes Y},$$

and the structure maps commute with morphisms; in particular,

$$a_{X',Y',Z'}((f\otimes g)\otimes h)=(f\otimes(g\otimes h))a_{X,Y,Z}.$$

They satisfy the pentagon and triangle identities

$$a_{W,X,Y\otimes Z}a_{W\otimes X,Y,Z}=(\mathrm{id}_W\otimes a_{X,Y,Z})a_{W,X\otimes Y,Z}(a_{W,X,Y}\otimes\mathrm{id}_Z),$$

$$(\mathrm{id}_X\otimes l_Y)a_{X,\mathbf{1},Y}=r_X\otimes\mathrm{id}_Y\,.$$

A C^**-category* has Banach morphism spaces and conjugate-linear maps $f\mapsto f^*$ with

$$f^*:Y\longrightarrow X,\qquad (f^*)^*=f,\qquad (gf)^*=f^*g^*,$$

$$\|gf\|\le\|g\|\|f\|,\qquad \|f^*f\|=\|f\|^2.$$

A C^**-tensor category* is a $\mathbb{C}$-linear tensor category with this structure, with

$$(f\otimes g)^*=f^*\otimes g^*,\qquad a^*=a^{-1},\quad l^*=l^{-1},\quad r^*=r^{-1}.$$

A nonzero object X is *simple* when $\mathrm{End}(X)=\mathbb{C}\,\mathrm{id}_X$. An *orthogonal direct sum* $X=\bigoplus_{i=1}^r X_i$ is specified by morphisms $u_i:X_i\to X$ satisfying

$$u_i^*u_j=\delta_{ij}\,\mathrm{id}_{X_i},\qquad \sum_{i=1}^r u_iu_i^*=\mathrm{id}_X\,.$$

All C^*-tensor categories below have simple tensor unit $\mathrm{End}(\mathbf{1})=\mathbb{C}\,\mathrm{id}_{\mathbf{1}}$. Associators and unit maps are suppressed in subsequent tensor composites.

Definition 13.11 (Corepresentation tensor categories)**.** For a compact quantum group $\mathbb{G}$, let $\mathrm{Rep}(\mathbb{G})$ have finite-dimensional unitary corepresentations as objects and

$$\mathrm{Hom}_{\mathbb{G}}(U,V)=\{A:H_U\to H_V:(A\otimes 1)U=V(A\otimes 1)\}$$

as morphism spaces. Composition and adjoints are those of Hilbert-space operators. Define

$$U\boxtimes V=U_{13}V_{23}\quad\text{on } H_U\otimes H_V,$$

and take the trivial corepresentation on $\mathbb{C}$ as tensor unit, with the Hilbert-space associator and unit maps.

Exercise 13.8 (Corepresentations as a tensor category)**.** Prove that the adjoint of an intertwiner is an intertwiner, that $U\boxtimes V$ is a unitary corepresentation, and that tensor products of intertwiners are intertwiners. Verify the pentagon, triangle, and C^*-identities and deduce that $\mathrm{Rep}(\mathbb{G})$ is a C^*-tensor category. For $C(\mathbb{G})=\mathbb{C}$, identify it with $\mathrm{Hilb}_{\mathrm{fd}}$, the category of finite-dimensional Hilbert spaces and all linear maps.

Definition 13.12 (Duals and rigidity)**.** For a finite-dimensional H-module V, its *left dual* $V^\vee$ has action

$$(x\varphi)(v)=\varphi(S(x)v),$$

evaluation $\mathrm{ev}_V(\varphi\otimes v)=\varphi(v)$, and coevaluation coev_V.

In a $\mathbb{C}$-linear tensor category, a *left dual* of V is an object $V^\vee$ with morphisms

$$\mathrm{ev}_V:V^\vee\otimes V\longrightarrow\mathbf{1},\qquad \mathrm{coev}_V:\mathbf{1}\longrightarrow V\otimes V^\vee$$

satisfying
$$(\mathrm{id}_V \otimes \mathrm{ev}_V)(\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$
$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^\vee})(\mathrm{id}_{V^\vee} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^\vee}.$$

A *right dual* ${}^\vee V$ has maps
$$\widetilde{\mathrm{ev}}_V : V \otimes {}^\vee V \longrightarrow \mathbf{1}, \qquad \widetilde{\mathrm{coev}}_V : \mathbf{1} \longrightarrow {}^\vee V \otimes V$$
satisfying
$$(\widetilde{\mathrm{ev}}_V \otimes \mathrm{id}_V)(\mathrm{id}_V \otimes \widetilde{\mathrm{coev}}_V) = \mathrm{id}_V,$$
$$(\mathrm{id}_{{}^\vee V} \otimes \widetilde{\mathrm{ev}}_V)(\widetilde{\mathrm{coev}}_V \otimes \mathrm{id}_{{}^\vee V}) = \mathrm{id}_{{}^\vee V}.$$

The category is *rigid* when every object has left and right duals. For $f : V \to W$, the left-dual morphism is
$$f^\vee = (\mathrm{ev}_W \otimes \mathrm{id}_{V^\vee})(\mathrm{id}_{W^\vee} \otimes f \otimes \mathrm{id}_{V^\vee})(\mathrm{id}_{W^\vee} \otimes \mathrm{coev}_V) : W^\vee \longrightarrow V^\vee.$$

Definition 13.13 (Conjugates and quantum dimension)**.** In a C^*-tensor category, a *conjugate* of an object V is an object $\overline{V}$ with morphisms
$$R_V : \mathbf{1} \longrightarrow \overline{V} \otimes V, \qquad \overline{R}_V : \mathbf{1} \longrightarrow V \otimes \overline{V}$$
satisfying
$$(\overline{R}_V^* \otimes \mathrm{id}_V)(\mathrm{id}_V \otimes R_V) = \mathrm{id}_V, \qquad (R_V^* \otimes \mathrm{id}_{\overline{V}})(\mathrm{id}_{\overline{V}} \otimes \overline{R}_V) = \mathrm{id}_{\overline{V}}.$$
Its dual maps are
$$\mathrm{ev}_V = R_V^*, \qquad \mathrm{coev}_V = \overline{R}_V, \qquad \widetilde{\mathrm{ev}}_V = \overline{R}_V^*, \qquad \widetilde{\mathrm{coev}}_V = R_V.$$

A C^*-tensor category is rigid when every object has a conjugate. A solution of the conjugate equations is *standard* when
$$R_V^*(\mathrm{id}_{\overline{V}} \otimes T)R_V = \overline{R}_V^*(T \otimes \mathrm{id}_{\overline{V}})\overline{R}_V \qquad (T \in \mathrm{End}(V)).$$

For a standard solution, define
$$\mathrm{tr}_{\mathcal{C}}(T) = R_V^*(\mathrm{id}_{\overline{V}} \otimes T)R_V, \qquad \dim_{\mathcal{C}}(V) = \mathrm{tr}_{\mathcal{C}}(\mathrm{id}_V).$$

For $\mathcal{C} = \mathrm{Rep}(SU_v(2))$, write tr_v and $\dim_v$.

Exercise 13.9 (Standard categorical trace)**.** Let two standard solutions of the conjugate equations be given for V. Use the snake identities and standardness to construct the unitary isomorphism between their conjugate objects and prove that they define the same trace. For $S : V \to W$ and $T : W \to V$, prove
$$\mathrm{tr}_{\mathcal{C}}(TS) = \mathrm{tr}_{\mathcal{C}}(ST), \qquad \mathrm{tr}_{\mathcal{C}}(T^*T) \geq 0.$$

Construct standard solutions for tensor products, conjugates, and existing orthogonal direct sums and prove
$$\dim_{\mathcal{C}}(V \otimes W) = \dim_{\mathcal{C}}(V)\dim_{\mathcal{C}}(W), \qquad \dim_{\mathcal{C}}(\overline{V}) = \dim_{\mathcal{C}}(V),$$
$$\dim_{\mathcal{C}}(V \oplus W) = \dim_{\mathcal{C}}(V) + \dim_{\mathcal{C}}(W).$$

Exercise 13.10 (Finite Hilbert-space trace)**.** For a finite-dimensional Hilbert space V, let $j_V : \overline{V} \to V^\vee$ be the Riesz isomorphism $j_V(\overline{v}) = \langle v, - \rangle$. From the coevaluation and symmetry defined in Spectral Theory, put

$$\overline{R}_V = (\mathrm{id}_V \otimes j_V^{-1})\,\mathrm{coev}_V : \mathbb{C} \longrightarrow V \otimes \overline{V}, \qquad R_V = s_{V,\overline{V}}\overline{R}_V : \mathbb{C} \longrightarrow \overline{V} \otimes V.$$

Prove the conjugate equations and standardness. Prove, without choosing a basis, that

$$\mathrm{tr}_{\mathrm{Hilb_{fd}}}(T) = \mathrm{tr}(T), \qquad \dim_{\mathrm{Hilb_{fd}}}(V) = \dim_{\mathbb{C}}(V),$$

where the right-hand sides are the trace and dimension defined in Spectral Theory.

Exercise 13.11 ($SU_v(2)$ fusion and quantum dimensions)**.** Let V_n be the irreducible type-1 $U_v(\mathfrak{sl}_2)$-module of highest weight n. Construct its conjugate equations and prove

$$V_m \otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Normalize the conjugate solutions by the compact involution and compute

$$\dim_v(V_n) = [n+1]_v = \frac{v^{n+1} - v^{-(n+1)}}{v - v^{-1}}.$$

Prove that finite orthogonal direct sums of the V_n are closed under tensor products and conjugates and form a rigid C^*-tensor category. Prove

$$\begin{aligned} &\dim_{\mathbb{C}}(V_1) = 2, && \dim_v(V_1) = v + v^{-1}, \\ &\dim_{\mathbb{C}}(V_n) = n+1, && \lim_{v\to 1}\dim_v(V_n) = n+1, \\ & && \dim_v(V_n) > n+1 \quad (0 < v < 1,\ n \geq 1). \end{aligned}$$

Definition 13.14 (Basic hypergeometric series)**.** For $m \in \mathbb{N}$, define

$$(a;Q)_m = \prod_{j=0}^{m-1}(1 - aQ^j), \qquad (a;Q)_\infty = \prod_{j=0}^{\infty}(1 - aQ^j),$$

$$(a_1,\ldots,a_r;Q)_m = \prod_{i=1}^{r}(a_i;Q)_m, \qquad (a_1,\ldots,a_r;Q)_\infty = \prod_{i=1}^{r}(a_i;Q)_\infty.$$

The *basic hypergeometric series* is

$${}_r\phi_s\left(\begin{matrix} a_1,\ldots,a_r \\ b_1,\ldots,b_s \end{matrix}; Q, z\right) = \sum_{m=0}^{\infty} \frac{(a_1,\ldots,a_r;Q)_m}{(Q,b_1,\ldots,b_s;Q)_m}\left((-1)^m Q^{m(m-1)/2}\right)^{1+s-r} z^m.$$

It is terminating when one numerator parameter is Q^{-n}.

Exercise 13.12 (Basic hypergeometric summations)**.** Prove the Q-Gauss identity

$${}_2\phi_1\left(\begin{matrix} a, b \\ c \end{matrix}; Q, \frac{c}{ab}\right) = \frac{(c/a, c/b;Q)_\infty}{(c, c/(ab);Q)_\infty}, \qquad \left|\frac{c}{ab}\right| < 1,$$

and the terminating Q-Pfaff–Saalschütz identity

$${}_3\phi_2\left(\begin{matrix} a, b, Q^{-n} \\ c, abQ^{1-n}/c \end{matrix}; Q, Q\right) = \frac{(c/a, c/b;Q)_n}{(c, c/(ab);Q)_n}$$

by comparing coefficients after multiplication by the denominator Q-products in the first identity, and by showing that the two sides of the second identity satisfy the same recurrence in n and have the same initial value.

Definition 13.15 (Quantum coupling coefficients)**.** For irreducible type-1 modules V_a, V_b, V_c, let

$$C^c_{ab} : V_c \longrightarrow V_a \otimes V_b$$

be the intertwining isometry normalized by its positive highest-weight coefficient. Its matrix coefficients between the distinguished weight lines are the *v-Clebsch–Gordan coefficients*.

Exercise 13.13 (XXZ quantum-group symmetry)**.** Use the distinguished weight-line realization $V_1 = \mathbb{C}e_+ \oplus \mathbb{C}e_-$ of the fundamental type-1 module. Define the Pauli endomorphisms, relative to the ordered weight lines, by

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and let σ^a_j act as σ^a on the jth tensor factor and as the identity on the others. On $V_1^{\otimes L}$ let

$$\begin{aligned} H_{\mathrm{XXZ}} = J \sum_{j=1}^{L-1} \Big(\sigma^x_j \sigma^x_{j+1} + \sigma^y_j \sigma^y_{j+1} + \frac{v+v^{-1}}{2} (\sigma^z_j \sigma^z_{j+1} - I) \\ + \frac{v-v^{-1}}{2} (\sigma^z_j - \sigma^z_{j+1}) \Big). \end{aligned}$$

For $J \in \mathbb{R}$ and $0 < v < 1$, prove that this open-chain Hamiltonian is self-adjoint and commutes with the iterated coproduct action of $U_v(\mathfrak{sl}_2)$.

Exercise 13.14 (Dual Q-Hahn coupling coefficients)**.** Decompose two coupled irreducible spin sectors by the projections $C^c_{ab}(C^c_{ab})^*$ and derive the orthogonality relations for their v-Clebsch–Gordan coefficients. Prove that their weight dependence satisfies the dual Q-Hahn recurrence with $Q = v^2$ and normalized amplitudes with a terminating ${}_3\phi_2$ of base Q.

Exercise 13.15 (XXZ transition strengths)**.** Let the weak drive be an irreducible tensor operator $A^{(s)}_\mu$ for the $U_v(\mathfrak{sl}_2)$ action. Prove the v-Wigner–Eckart factorization of its matrix elements into a reduced matrix element and a v-Clebsch–Gordan coefficient. Derive the relative transition strengths at fixed reduced matrix element as squared v-Clebsch–Gordan coefficients. State the chain length, open boundary terms, resolved weights, energy scale J, and nondegeneracy assumptions under which the measured line positions and strengths determine v.

Definition 13.16 (Universal R-matrices)**.** A *universal R-matrix* for a Hopf algebra H is an invertible element $\mathcal{R}$ in a completion of $H \otimes H$ on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\mathrm{op}}(x), \quad (\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

For finite-dimensional modules V, W, define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}) : V \otimes W \longrightarrow W \otimes V.$$

The pair $(H, \mathcal{R})$ is *quasitriangular*.

Definition 13.17 (Braided tensor categories)**.** A *braiding* on a $\mathbb{C}$-linear tensor category $\mathcal{C}$ is a family of isomorphisms

$$c_{X,Y} : X \otimes Y \longrightarrow Y \otimes X$$

satisfying, for $f : X \to X'$ and $g : Y \to Y'$,

$$(g \otimes f)c_{X,Y} = c_{X',Y'}(f \otimes g),$$

and the two hexagon identities

$$c_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes c_{X,Z})(c_{X,Y} \otimes \mathrm{id}_Z),$$
$$c_{X\otimes Y,Z} = (c_{X,Z} \otimes \mathrm{id}_Y)(\mathrm{id}_X \otimes c_{Y,Z}).$$

The category is *braided* when it has a braiding. A braiding on a C^*-tensor category is *unitary* when $c^*_{X,Y} = c^{-1}_{X,Y}$.

Definition 13.18 (Spectral exchange operators)**.** Let V be finite-dimensional. A meromorphic family $R(u) \in \mathrm{End}(V \otimes V)$ satisfies the spectral Yang–Baxter equation when

$$R_{12}(u/v)R_{13}(u/w)R_{23}(v/w) = R_{23}(v/w)R_{13}(u/w)R_{12}(u/v).$$

It is regular when $R(1)$ is a nonzero scalar multiple of the flip and unitary up to scalar when $R_{12}(u)R_{21}(u^{-1})$ is scalar.

Exercise 13.16 (Hexagon identities)**.** Prove that $c_{V,W}$ is a module intertwiner satisfying naturality and that the two coproduct identities for $\mathcal{R}$ give the two hexagon identities, so that the finite-dimensional module category is braided. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

and, on $V^{\otimes 3}$,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Prove that the operators $c_{j,j+1}$ define braid-group representations on tensor powers of every finite-dimensional type-1 module.

Definition 13.19 (Ribbon categories)**.** Let $\mathcal{C}$ be a braided rigid $\mathbb{C}$-linear tensor category. A *twist* is a family of isomorphisms $\theta_X : X \to X$ satisfying

$$\theta_Y f = f\theta_X \qquad (f : X \to Y)$$

and

$$\theta_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}, \qquad \theta_{X\otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object X.

Exercise 13.17 (Ribbon elements)**.** For a quasitriangular Hopf algebra $(H, \mathcal{R})$, set

$$u = m(S \otimes \mathrm{id})(\mathcal{R}_{21}).$$

Let $\nu_R \in H$ be central and invertible and satisfy

$$\nu_R^2 = uS(u), \qquad S(\nu_R) = \nu_R, \qquad \varepsilon(\nu_R) = 1, \qquad \Delta(\nu_R) = (\mathcal{R}_{21}\mathcal{R})^{-1}(\nu_R \otimes \nu_R).$$

Prove that

$$\theta_V = \pi_V(\nu_R^{-1})$$

defines a ribbon twist on the category of finite-dimensional H-modules. Derive invariance of the quantum trace under framed ribbon isotopy.

Exercise 13.18 (Chebyshev quantum dimensions)**.** For the irreducible $SU_v(2)$ objects V_n, use $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$ to prove

$$d_{n+1} = (v + v^{-1})d_n - d_{n-1}, \qquad d_0 = 1, \quad d_1 = v + v^{-1},$$

where $d_n = \dim_v(V_n)$. If U_n is the Chebyshev polynomial determined by $U_0(x) = 1$, $U_1(x) = 2x$, and $U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)$, deduce

$$\dim_v(V_n) = U_n\left(\frac{v + v^{-1}}{2}\right) = \frac{v^{n+1} - v^{-(n+1)}}{v - v^{-1}}.$$

Definition 13.20 (Quantum recoupling coefficients)**.** The matrix coefficients of the recoupling map

$$(V_a \otimes V_b) \otimes V_c \longrightarrow V_a \otimes (V_b \otimes V_c)$$

between fixed intermediate irreducible summands are the v-$6j$ coefficients.

Exercise 13.19 (Quantum Racah recoupling)**.** Compute the change between the two coupling orders of $V_a \otimes V_b \otimes V_c$ and express its entries as terminating ${}_4\phi_3$ series with base $Q = v^2$. Identify them with Q-Racah polynomials and derive their orthogonality from unitarity of the recoupling map for the compact real form. Prove the pentagon identity by comparing the five coupling orders of four factors. For three resolved XXZ spin sectors, derive the recoupling probabilities as squared v-$6j$ coefficients and state the unitarity and spectral-resolution assumptions required for this probability interpretation.

Definition 13.21 (Type-A Hecke actions)**.** For $v \neq 0$, the Hecke algebra $\mathcal{H}_v(S_r)$ is generated by $T_1, \ldots, T_{r-1}$ with

$$T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \qquad T_iT_j = T_jT_i \quad (|i-j| > 1),$$
$$(T_i - v)(T_i + v^{-1}) = 0.$$

Let V_1 be the fundamental $U_v(\mathfrak{sl}_2)$-module. On $V_1 \otimes V_1 \cong V_2 \oplus V_0$, let P_2, P_0 be the invariant orthogonal projections, and define

$$T = vP_2 - v^{-1}P_0, \qquad \rho_r(T_i) = \mathrm{id}^{\otimes(i-1)} \otimes T \otimes \mathrm{id}^{\otimes(r-i-1)}.$$

Exercise 13.20 (Fundamental Hecke symmetry)**.** Prove that the coproduct action of $U_v(\mathfrak{sl}_2)$ commutes with $\rho_r(T_i)$. Derive the braid relation from the Q-Racah recoupling identity and the Hecke relation from the two displayed spectral projections. Put $e_i = (v - T_i)/(v + v^{-1})$ and prove that the action factors through the Temperley–Lieb relations

$$e_i^2 = e_i, \qquad e_ie_{i\pm1}e_i = (v + v^{-1})^{-2}e_i.$$

Exercise 13.21 (Hecke Poincaré identity)**.** Put $\tau = v^2$. For $w \in S_r$, let $\ell(w)$ be the least number of adjacent transpositions in an expression for w, and let $T_w = T_{i_1} \cdots T_{i_{\ell(w)}}$ for any reduced expression $w = s_{i_1} \cdots s_{i_{\ell(w)}}$. Prove from the braid relations that T_w is well-defined and that

$$\sum_{w \in S_r} \tau^{\ell(w)} = \prod_{j=1}^{r} \frac{1 - \tau^j}{1 - \tau}.$$

Prove that

$$e_+ = \frac{\sum_{w \in S_r} v^{\ell(w)} T_w}{\sum_{w \in S_r} v^{2\ell(w)}}, \qquad e_- = \frac{\sum_{w \in S_r} (-v^{-1})^{\ell(w)} T_w}{\sum_{w \in S_r} v^{-2\ell(w)}}$$

are idempotents satisfying $T_ie_+ = ve_+$ and $T_ie_- = -v^{-1}e_-$. Under the fundamental Hecke action from the preceding exercise, identify their images with the quantum symmetric and exterior state spaces and interpret the Poincaré polynomial as their graded state count.

Definition 13.22 (Affine Hecke algebras)**.** The extended affine Hecke algebra $\mathcal{H}_v^{\mathrm{aff}}(GL_r)$ is generated by $T_1, \dots, T_{r-1}$ and commuting invertible elements $X_1, \dots, X_r$, subject to the finite Hecke relations and

$$T_i X_i T_i = X_{i+1}, \qquad T_i X_j = X_j T_i \quad (j \neq i, i+1).$$

Equivalently, it is the Hecke quotient of the extended affine braid group of $\mathbb{Z}^r \rtimes S_r$.

Definition 13.23 (Baxterized Hecke operators)**.** For an invertible spectral parameter u, define

$$\check{R}_i(u) = uT_i - u^{-1}T_i^{-1}.$$

Its eigenvalues on the v and $-v^{-1}$ eigenspaces of T_i are

$$uv - u^{-1}v^{-1}, \qquad -uv^{-1} + u^{-1}v.$$

Exercise 13.22 (Baxterization)**.** Use the braid and Hecke relations for T_i to prove

$$\check{R}_i(u)\check{R}_{i+1}(uv)\check{R}_i(v) = \check{R}_{i+1}(v)\check{R}_i(uv)\check{R}_{i+1}(u).$$

Prove that $\check{R}_i(1)$ is a scalar multiple of the identity and compute $\check{R}_i(u)\check{R}_i(u^{-1})$. For $|u| = 1$, prove that $\check{R}_i(u)^* = \check{R}_i(u^{-1})$ and normalize the operators to be unitary wherever the scalar product is nonzero. Prove that their two eigenvalues depend on u and hence that the spectral family itself does not satisfy one parameter-independent Hecke polynomial.

Definition 13.24 (Six-vertex row transfer)**.** On the fundamental module V_1, put

$$\check{R}(u) = uT - u^{-1}T^{-1}, \qquad R(u) = \tau \circ \check{R}(u),$$

where $T = vP_2 - v^{-1}P_0$ and τ is the flip. For an auxiliary copy $V_a \cong V_1$ and a periodic row of length N, define

$$\mathsf{T}_N(u) = \mathrm{Tr}_{V_a}\left(R_{aN}(u) \cdots R_{a1}(u)\right) \quad \text{on } V_1^{\otimes N},$$

and for an $N \times M$ periodic lattice define

$$Z^{6\mathrm{V}}_{N,M}(u) = \mathrm{Tr}_{V_1^{\otimes N}} \mathsf{T}_N(u)^M.$$

Exercise 13.23 (Six-vertex integrability and the XXZ spectrum)**.** Use invariance of P_2 and P_0 under the Cartan action to prove

$$[R(u), K \otimes K] = 0.$$

Relative to the distinguished weight lines of V_1, deduce that exactly the six arrow configurations preserving total weight can have nonzero local weights, and compute those weights from P_2 and P_0. Prove the spectral Yang–Baxter equation for R from the Baxterization identity. Apply it to an auxiliary pair and take their intrinsic traces to prove

$$[\mathsf{T}_N(u), \mathsf{T}_N(w)] = 0.$$

Show that $Z^{6\mathrm{V}}_{N,M}(u)$ is the sum of products of the six local weights over arrow configurations on the periodic square lattice.

Use $\check{R}(1) = (v - v^{-1})I$ and compute the logarithmic derivative at the regular point. Prove that

$$\widehat{H}_N = J\left[(v - v^{-1})\mathsf{T}'_N(1)\mathsf{T}_N(1)^{-1} - N(v + v^{-1})I\right]$$

is the periodic version of the XXZ Hamiltonian in the preceding exercise, with $\Delta = (v + v^{-1})/2$ and with the telescoping boundary term equal to zero. On the sector with one reversed spin, apply the distinguished Fourier modes and derive

$$E(k) = 4J(\cos k - \Delta), \qquad k = \frac{2\pi m}{N}, \quad 0 \leq m < N.$$

Deduce the finite-size line spacings and the band edges. State the energy calibration, periodic-boundary, polarization-sector, and spectral-resolution assumptions required when these values are compared with an XXZ spin-chain simulator, and determine the scalar gauge and real spectral-parameter regime under which the same six weights are nonnegative Boltzmann weights.

Definition 13.25 (qKZ transport)**.** Let v be the quantum-group parameter and let $\kappa \neq 0$ be an independent shift parameter. Let $D \in \mathrm{End}(V)$ be invertible and satisfy $[D \otimes D, \check{R}(u)] = 0$. Let D_1 act on the transported tensor factor and define

$$\mathcal{K}_1(z_1, \ldots, z_r) = D_1 \check{R}_{1r}(z_1/z_r) \cdots \check{R}_{12}(z_1/z_2).$$

The first qKZ equation is

$$\Psi(z_2, \ldots, z_r, \kappa z_1) = \mathcal{K}_1(z_1, \ldots, z_r)\Psi(z_1, \ldots, z_r),$$

and the remaining equations are obtained by cyclic conjugation.

Exercise 13.24 (qKZ exchange transport)**.** Use Baxterization to prove consistency of the adjacent exchange rules and the qKZ difference equations. Identify the constant exchanges and winding operators with the generators of $\mathcal{H}_v^{\mathrm{aff}}(GL_r)$, keeping v and κ independent. For the unitary normalization, prepare and detect three resolved XXZ spin sectors and prove that the two factorizations of a three-spin exchange give the same output probabilities. Express this equality as the Q-Racah recoupling identity. Derive the phase accumulated in one κ-shift cycle and state the twist, unitarity, preparation, and detection assumptions under which it is an interference observable.

Definition 13.26 (Temperley–Lieb category and algebras)**.** For $d \in \mathbb{C}$, the *Temperley–Lieb category* TL_d has objects $n \in \mathbb{Z}_{\geq 0}$. The morphism space $\mathrm{Hom}_{\mathsf{TL}_d}(m, n)$ is the complex span of planar nonintersecting pairings of m lower and n upper boundary points. Composition is vertical stacking, with every resulting closed loop removed and replaced by the scalar d; tensor product is horizontal juxtaposition and has tensor unit 0. For $d \in \mathbb{R}$, reflection across a horizontal line defines f^*. The cup and cap diagrams give coevaluation and evaluation morphisms.

The *Temperley–Lieb algebra* is $TL_n(d) = \mathrm{End}_{\mathsf{TL}_d}(n)$. It is generated by $U_1, \ldots, U_{n-1}$ with

$$U_i^2 = dU_i, \qquad U_i U_{i\pm 1} U_i = U_i, \qquad U_i U_j = U_j U_i \quad (|i - j| > 1).$$

Its diagram trace is normalized by

$$\mathrm{tr}_0(1) = 1, \qquad \mathrm{tr}_{n+1}(x) = d\,\mathrm{tr}_n(x), \qquad \mathrm{tr}_{n+1}(xU_n) = \mathrm{tr}_n(x),$$

where $x \in TL_n(d)$ is embedded in $TL_{n+1}(d)$.

Definition 13.27 (Jones–Wenzl recursion)**.** Put

$$\zeta = e^{\pi i/(k+2)}, \qquad d = \zeta + \zeta^{-1}, \qquad [n]_\zeta = \frac{\zeta^n - \zeta^{-n}}{\zeta - \zeta^{-1}}.$$

With $f_0 = f_1 = 1$, define recursively in $TL_{n+1}(d)$

$$f_{n+1} = f_n - \frac{[n]_\zeta}{[n+1]_\zeta} f_n U_n f_n \qquad (1 \leq n \leq k).$$

Definition 13.28 (Tensor ideals and unitary fusion categories)**.** A *tensor ideal* $\mathcal{I}$ in a $\mathbb{C}$-linear tensor category $\mathcal{C}$ consists of subspaces $\mathcal{I}(X,Y) \subseteq \mathrm{Hom}_{\mathcal{C}}(X,Y)$ such that, for $f \in \mathcal{I}(X,Y)$,

$$gfh \in \mathcal{I}(X',Y'), \qquad f \otimes u \in \mathcal{I}(X \otimes Z, Y \otimes Z'), \qquad u \otimes f \in \mathcal{I}(Z \otimes X, Z' \otimes Y)$$

whenever the displayed composites are defined. The quotient $\mathcal{C}/\mathcal{I}$ has the same objects and morphism spaces

$$\mathrm{Hom}_{\mathcal{C}/\mathcal{I}}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(X,Y)/\mathcal{I}(X,Y).$$

The *idempotent completion* of $\mathcal{C}$ has objects (X,p) with $p \in \mathrm{End}_{\mathcal{C}}(X)$ and $p^2 = p$, and

$$\mathrm{Hom}((X,p),(Y,q)) = q\,\mathrm{Hom}_{\mathcal{C}}(X,Y)p.$$

The C^*-idempotent completion requires $p = p^*$.

A C^*-category is *semisimple* when every object is a finite orthogonal direct sum of simple objects. A *unitary fusion category* is a rigid semisimple C^*-tensor category with finite-dimensional morphism spaces and finitely many simple objects up to unitary isomorphism.

Definition 13.29 (Negligible morphisms)**.** At $d = \zeta + \zeta^{-1}$, a morphism $f : m \to n$ in TL_d is *negligible* when

$$\mathrm{tr}_m(gf) = 0 \qquad \text{for every } g : n \to m.$$

Let $\mathcal{N}(m,n)$ be the space of negligible morphisms. Let $\mathcal{C}_k$ be the idempotent completion of $\mathsf{TL}_d/\mathcal{N}$, and write V_a for the image of f_a, $0 \le a \le k$.

Exercise 13.25 (Jones–Wenzl trace)**.** Use the Temperley–Lieb relations to prove inductively

$$f_n^2 = f_n, \qquad f_n^* = f_n, \qquad U_i f_n = f_n U_i = 0 \quad (1 \le i < n), \qquad \mathrm{tr}(f_n) = [n+1]_\zeta.$$

Deduce $\mathrm{tr}(f_{k+1}) = [k+2]_\zeta = 0$.

Exercise 13.26 (Temperley–Lieb semisimplification)**.** Prove that $\mathcal{N}$ is the $*$-closed tensor ideal generated by f_{k+1}. Prove that

$$\langle f, g \rangle = \mathrm{tr}_m(f^* g), \qquad f, g : m \to n,$$

is positive semidefinite with radical $\mathcal{N}(m,n)$, and that the quotient removes every trace-zero summand.

Exercise 13.27 (Truncated fusion)**.** Derive

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b| \le c \le \min(a+b, 2k-a-b) \\ a+b+c \equiv 0 \pmod 2}} V_c$$

and

$$d_a = [a+1]_\zeta = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

Exercise 13.28 (Unitary fusion structure)**.** Use left composition on the positive quotient spaces and diagram reflection to equip $\mathcal{C}_k$ with C^*-norms. Use the cup and cap diagrams to construct standard conjugates. Prove from truncated fusion that every simple object is unitarily isomorphic to exactly one of

$$V_0, V_1, \ldots, V_k,$$

and deduce that $\mathcal{C}_k$ is a unitary fusion category. Prove that its standard categorical trace agrees with the quotient diagram trace and that the recoupling maps between simple-summand decompositions are unitary and satisfy the pentagon identity. With $\zeta^{1/2} = e^{\pi i/(2(k+2))}$, prove that

$$c_i = \zeta^{1/2}\,\mathrm{id} - \zeta^{-1/2} U_i$$

is unitary, satisfies the braid relations, preserves $\mathcal{N}$, and descends to a unitary braiding on $\mathcal{C}_k$.

Definition 13.30 ($SU(2)_k$ modular data)**.** For $0 \le a, b \le k$, define

$$S_{ab} = \sqrt{\frac{2}{k+2}}\sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \qquad \theta_a = \exp\left(\frac{\pi i\, a(a+2)}{2(k+2)}\right).$$

Let ${N_{ab}}^c$ be the multiplicity of V_c in $V_a \otimes V_b$.

Exercise 13.29 (Verlinde diagonalization)**.** Use sine orthogonality to prove that S is real, symmetric, and unitary. For each a, let $(N_a)_{bc} = {N_{ab}}^c$. Use the truncated Chebyshev recursion to prove

$$N_a = S\ \mathrm{diag}\left(\frac{S_{ax}}{S_{0x}}\right)_{x=0}^{k} S^{-1}$$

and hence

$${N_{ab}}^c = \sum_{x=0}^{k} \frac{S_{ax}S_{bx}\overline{S_{cx}}}{S_{0x}}.$$

Deduce that the number of fusion channels from $a_1, \dots, a_n$ to c is $(N_{a_1}\cdots N_{a_n})_{0c}$ and express it as a finite sum of products of sine functions. Apply the same formula to the Ising modular matrix derived in Counting and recover the Ising fusion multiplicities.

Definition 13.31 (Levin–Wen string-net Hamiltonian)**.** Let Γ be a trivalent cellulation of a closed oriented surface and let $\mathcal{C}_k$ be the preceding unitary fusion category with simple objects $V_0, \dots, V_k$, fusion multiplicities ${N_{ab}}^c$, standard categorical trace, and quantum dimensions d_a. Write $a^\vee$ for the unique label satisfying $\overline{V_a} \cong V_{a^\vee}$. Put

$$\mathcal{H}_\Gamma = \ell^2\left(\{0,\dots,k\}^{E(\Gamma)}\right), \qquad \mathcal{D}^2 = \sum_{a=0}^{k} d_a^2.$$

For each vertex v, let Q_v be the projection onto edge labelings admissible under the fusion rules. For a face p, let B_p^a insert a coevaluation–evaluation loop labelled V_a inside p and resolve it into the boundary edges by the associator and the quantum recoupling maps; set it equal to zero off the subspace on which the vertices incident to p are admissible. Define

$$B_p = \frac{1}{\mathcal{D}^2}\sum_{a=0}^{k} d_a B_p^a$$

and, for $\Delta_v, \Delta_p > 0$,

$$\widehat{H}_\Gamma = \Delta_v \sum_v (I - Q_v) + \Delta_p \sum_p (I - B_p).$$

Exercise 13.30 (String-net commuting projectors)**.** Use the snake identities, the standard categorical trace, and the quantum recoupling pentagon to prove

$$(B_p^a)^* = B_p^{a^\vee}, \qquad B_p^a B_p^b = \sum_c {N_{ab}}^c B_p^c.$$

Use

$$d_a d_b = \sum_c N_{ab}{}^c d_c$$

to prove

$$Q_v^2 = Q_v = Q_v^*, \qquad B_p^2 = B_p = B_p^*, \qquad [Q_v, B_p] = [B_p, B_{p'}] = 0.$$

Deduce that $\widehat{H}_\Gamma$ is positive and frustration-free and that its ground space is the simultaneous $+1$ space of all Q_v and B_p. Prove that recoupling across an edge gives a unitary identification of the ground spaces before and after the corresponding local change of cellulation.

For $k = 1$, identify the labels with $\mathbb{Z}/2\mathbb{Z}$, let Z_e act on a label a_e by $(-1)^{a_e}$, and let X_e reverse that label. Show that

$$Q_v = \frac{1}{2}\left(I + \prod_{e \ni v} Z_e\right), \qquad B_p = \frac{1}{2}\left(I + \prod_{e \subset \partial p} X_e\right).$$

On a torus, count the two independent relations among the vertex and face constraints and prove that the ground-state degeneracy is 4. Identify the two pairs of noncontractible string operators and their mutual sign, and state the boundary conditions and gap-preserving perturbation regime under which the fourfold low-energy multiplicity is a measurable prediction of a qubit realization.

Definition 13.32 (String-net loop operators)**.** On a torus let $W_a^{\alpha,+}$ and $W_a^{\alpha,-}$ insert a noncontractible V_a loop along a cycle α, using respectively the braiding and inverse braiding of $\mathcal{C}_k$ when the loop is resolved through the string net. Define $W_a^{\beta,+}$ and $W_a^{\beta,-}$ for a cycle β with intersection number one with α.

Exercise 13.31 (String-net loop spectroscopy)**.** Use local recoupling to reduce a torus cellulation to a trivalent spine and prove that the ground space of the $\mathcal{C}_k$ string-net Hamiltonian has sectors indexed by pairs (x, y) with $0 \le x, y \le k$. Deduce

$$\dim \ker \widehat{H}_\Gamma = (k+1)^2.$$

Prove the loop fusion relations

$$W_a^{\alpha,\pm} W_b^{\alpha,\pm} = \sum_c N_{ab}{}^c W_c^{\alpha,\pm}$$

and use Verlinde diagonalization to show that their eigenvalues in the sector (x, y) are

$$W_a^{\alpha,+} : \frac{S_{ax}}{S_{0x}}, \qquad W_a^{\alpha,-} : \overline{\frac{S_{ay}}{S_{0y}}}.$$

For a weak calibrated loop perturbation

$$\widehat{V} = -\sum_a \left(\epsilon_a^+ W_a^{\alpha,+} + \epsilon_a^- W_a^{\alpha,-} + \text{adjoints}\right),$$

derive the first-order splitting

$$\delta E_{x,y} = -2\,\mathrm{Re} \sum_a \left(\epsilon_a^+ \frac{S_{ax}}{S_{0x}} + \epsilon_a^- \overline{\frac{S_{ay}}{S_{0y}}}\right).$$

Insert the sine formula for S and list the distinct splittings for a chosen k. State the topological-gap, system-size, loop-support, calibration, and sector-resolution assumptions under which these spectral lines test the doubled $SU(2)_k$ string-net model.

Exercise 13.32 (Anyon interference and fusion counting)**.** Prove that the phases θ_a define a ribbon twist on $\mathcal{C}_k$. On every simple fusion channel $V_c \subseteq V_a \otimes V_b$, prove the balancing identity

$$c_{V_b,V_a} c_{V_a,V_b}|_{V_c} = \frac{\theta_c}{\theta_a \theta_b} \operatorname{id}_{V_c}.$$

For simple charges a, b, prove that the normalized monodromy amplitude is

$$M_{ab} = \frac{S_{ab} S_{00}}{S_{0a} S_{0b}}.$$

In a two-path anyon interferometer with path intensities I_1, I_2, derive

$$I_b(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2}\, \Re(e^{i\phi} M_{ab}).$$

Prove that a calibrated full framing twist of the probe path replaces M_{ab} by $\theta_a M_{ab}$ and shifts the interference phase by $\arg \theta_a$. Insert the sine formula and predict the visibility and phase for every enclosed charge. For a collection of prepared charges, use the Verlinde formula to predict the exact number of fusion outcomes in each total-charge sector. Identify indistinguishable charges for one probe and prove that the full family of probes distinguishes them. State the coherence, topological-gap, charge-preparation, and readout assumptions separately from the predicted fringes and integer state counts.

14 Integrable Systems

Exercise 14.1 (Elliptic pendulum period)**.** On T^*S^1 consider

$$H(\theta, p) = \frac{p^2}{2m\ell^2} + mg\ell(1 - \cos\theta).$$

For an oscillation of amplitude $0 < \theta_0 < \pi$, set $k = \sin(\theta_0/2)$ and define

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

Use energy conservation to prove

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}}\, K(k).$$

Apply the generalized binomial theorem with $a = \frac{1}{2}$. For

$$I_m = \int_0^{\pi/2} \sin^{2m} \phi \, d\phi,$$

use integration by parts to prove

$$I_0 = \frac{\pi}{2}, \qquad I_m = \frac{2m-1}{2m} I_{m-1} \quad (m \geq 1),$$

and obtain

$$K(k) = \frac{\pi}{2} \sum_{m=0}^{\infty} \left(\frac{(1/2)_m}{m!} \right)^2 k^{2m}, \qquad |k| < 1.$$

Deduce

$$T(\theta_0) = 2\pi\sqrt{\frac{\ell}{g}}\left(1 + \frac{\theta_0^2}{16} + O(\theta_0^4)\right).$$

Deduce the experimentally testable increase of period with amplitude and the logarithmic divergence as $\theta_0 \uparrow \pi$.

Definition 14.1 (Sutherland model)**.** For N identical particles of mass m on a ring of radius R and $g \geq 1$, let

$$\mathrm{Conf}_N(S^1) = \{(\theta_1, \ldots, \theta_N) : \theta_i \neq \theta_j \text{ for } i \neq j\}.$$

On $C_c^\infty(\mathrm{Conf}_N(S^1))^{S_N}$ define

$$\widehat{H}_g = \frac{\hbar^2}{2mR^2}\left[-\sum_{j=1}^N \partial_{\theta_j}^2 + \frac{g(g-1)}{2}\sum_{i<j}\frac{1}{\sin^2((\theta_i - \theta_j)/2)}\right]$$

and let the same symbol denote its Friedrichs realization in $L^2((S^1)^N)^{S_N}$. Define

$$\Psi_0(\theta) = \prod_{i<j}|e^{i\theta_i} - e^{i\theta_j}|^g.$$

For a partition λ of length at most N, use the monomial symmetric polynomial m_λ and dominance order defined in Counting, evaluated at $z_j = e^{i\theta_j}$.

Definition 14.2 (Jack polynomials)**.** The Jack polynomial $P_\lambda^{(1/g)}$ is the symmetric polynomial normalized by

$$P_\lambda^{(1/g)} = m_\lambda + \sum_{\mu<\lambda} c_{\lambda\mu}(g) m_\mu$$

and by satisfying

$$\left[\sum_i (z_i\partial_{z_i})^2 + g\sum_{i<j}\frac{z_i + z_j}{z_i - z_j}(z_i\partial_{z_i} - z_j\partial_{z_j})\right] P_\lambda^{(1/g)} = \left[\sum_i (\lambda_i^2 + g(N+1-2i)\lambda_i)\right] P_\lambda^{(1/g)};$$

here $<$ is dominance order.

Exercise 14.2 (Sutherland spectrum)**.** Conjugate the quantum Hamiltonian by Ψ_0, identify its commuting quantum integrals, and prove that

$$\Psi_\lambda = \Psi_0 P_\lambda^{(1/g)}(e^{i\theta_1}, \ldots, e^{i\theta_N})$$

is a joint eigenfunction with

$$E_\lambda = \frac{\hbar^2}{2mR^2}\left[\frac{g^2N(N^2-1)}{12} + \sum_{j=1}^N (\lambda_j^2 + g(N+1-2j)\lambda_j)\right].$$

At $g = 1$, prove that $P_\lambda^{(1)} = s_\lambda$ for the Schur polynomial defined in Counting.

Exercise 14.3 (Jack transition spectrum)**.** Let $\rho_r = \sum_{j=1}^N e^{ir\theta_j}$ be a density mode. Derive the Jack Pieri identity in orthonormal normalization for $\rho_1 P_\lambda^{(1/g)}$ and show that only nonincreasing tuples obtained from λ by increasing one entry by one occur. For a weak periodic drive proportional to $\rho_1 e^{-i\Omega t} + \rho_{-1} e^{i\Omega t}$, deduce the allowed resonance frequencies

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and express their relative intensities as squared normalized Jack Pieri coefficients. State how spectroscopy of particles confined to a ring determines g from the line shifts and tests the Jack selection rule from the missing lines.

Definition 14.3 (Double-affine braid group)**.** For $r \geq 2$, the *double-affine braid group* $\widetilde{B}_r$ of type GL_r is generated by T_i for $1 \leq i < r$, commuting invertible elements X_j, commuting invertible elements Y_j, and a central invertible element q, subject to the braid relations and

$$T_i X_i T_i = X_{i+1}, \qquad T_i X_j = X_j T_i \quad (j \neq i, i+1),$$
$$T_i^{-1} Y_i T_i^{-1} = Y_{i+1}, \qquad T_i Y_j = Y_j T_i \quad (j \neq i, i+1),$$
$$Y_1 X_1 \cdots X_r = \mathsf{q}\, X_1 \cdots X_r Y_1, \qquad X_1^{-1} Y_2 = Y_2 X_1^{-1} T_1^{-2}.$$

Exercise 14.4 (From affine to double-affine braids)**.** Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Interpret T_i as an adjacent braid and X_j, Y_j as motions of the jth point around the two cycles of T^2. Derive the displayed relations from these motions and identify $\widetilde{B}_r$ with the central extension of

$$\pi_1(\mathrm{Conf}_r(T^2)/S_r)$$

whose total cycle commutator is q. Prove that deleting either the X-family or the Y-family leaves an affine braid group.

Definition 14.4 (Type-A double affine Hecke algebra)**.** For nonzero parameters q, t, the *double affine Hecke algebra* $\mathsf{H}_{q,t}(GL_r)$ is the quotient of $\mathbb{C}(q,t)[\widetilde{B}_r]$ by

$$\mathsf{q} = q, \qquad (T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

Definition 14.5 (Rank-one DAHA)**.** For nonzero k_0, k_1, u_0, u_1 and a choice of $q^{1/2}$, the rank-one DAHA $\mathsf{H}_q(C_1^\vee, C_1)$ is generated by $T_0^{\pm 1}, T_1^{\pm 1}, (T_0^\vee)^{\pm 1}, (T_1^\vee)^{\pm 1}$ subject to

$$(T_i - k_i)(T_i + k_i^{-1}) = 0, \qquad (T_i^\vee - u_i)(T_i^\vee + u_i^{-1}) = 0 \quad (i = 0, 1),$$
$$T_1^\vee T_1 T_0 T_0^\vee = q^{-1/2}.$$

For the finite Hecke generator T_1, define

$$e = \frac{T_1 + k_1^{-1}}{k_1 + k_1^{-1}}.$$

Definition 14.6 (Rank-one polynomial representation)**.** Fix nonzero k_0, k_1, u_0, u_1 and choose $q^{1/2}$. On $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$ put

$$(s_1 f)(z) = f(z^{-1}), \qquad (s_0 f)(z) = f(q z^{-1}),$$
$$(T_1 f)(z) = k_1 f(z^{-1}) + \frac{(k_1 - k_1^{-1}) + (u_1 - u_1^{-1}) z}{1 - z^2} (f(z) - f(z^{-1})),$$
$$(T_0 f)(z) = k_0 f(q z^{-1}) + \frac{(k_0 - k_0^{-1}) + (u_0 - u_0^{-1}) q^{1/2} z^{-1}}{1 - q z^{-2}} (f(z) - f(q z^{-1})).$$

Let X act by multiplication by z. The Askey–Wilson parameters in this convention are

$$a = k_1 u_1, \qquad b = -k_1 u_1^{-1}, \qquad c = q^{1/2} k_0 u_0, \qquad d = -q^{1/2} k_0 u_0^{-1}.$$

Put

$$T_1^\vee = X^{-1} T_1^{-1}, \qquad T_0^\vee = q^{-1/2} T_0^{-1} X.$$

Definition 14.7 (Generalized hypergeometric series)**.** If no denominator parameter b_j is a nonpositive integer, define

$${}_rF_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_r)_m}{(b_1)_m \cdots (b_s)_m} \frac{z^m}{m!}.$$

The series is finite when a numerator parameter is a nonpositive integer. Otherwise it converges for every z when $r \le s$, for $|z| < 1$ when $r = s+1$, and only at $z = 0$ when $r > s+1$.

Definition 14.8 (Askey–Wilson operator and polynomials)**.** For $0 < q < 1$ and nonzero a, b, c, d, define

$$A(z) = \frac{(1-az)(1-bz)(1-cz)(1-dz)}{(1-z^2)(1-qz^2)}$$

and, on $f(z) = f(z^{-1})$,

$$(\mathcal{D}_{AW} f)(z) = A(z)\big(f(qz) - f(z)\big) + A(z^{-1})\big(f(q^{-1}z) - f(z)\big).$$

For $x = (z + z^{-1})/2$, the *Askey–Wilson polynomial* is

$$\begin{aligned} p_n(x; a, b, c, d \mid q) = a^{-n} (ab, ac, ad; q)_n \\ \times {}_4\phi_3\left(\begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix}; q, q\right). \end{aligned}$$

Its weight on the unit circle is

$$w_{AW}(z) = \frac{(z^2, z^{-2}; q)_\infty}{(az, a/z, bz, b/z, cz, c/z, dz, d/z; q)_\infty}.$$

Exercise 14.5 (Rank-one spherical operator)**.** Verify that the displayed reflection–difference operators preserve $\mathcal{L}$, that $T_0, T_1, T_0^\vee, T_1^\vee$ satisfy the four quadratic Hecke relations and the product relation, and that they give the rank-one DAHA polynomial representation with the stated parameter map. Prove

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}],$$

and prove that the spherical action contains multiplication by $z + z^{-1}$ and $\mathcal{D}_{AW}$.

Exercise 14.6 (Askey–Wilson spectral theorem)**.** Prove

$$\mathcal{D}_{AW} p_n = (q^{-n} - 1)(1 - abcdq^{n-1}) p_n.$$

For $a, b, c, d \in (-1, 1)$, prove that $\mathcal{D}_{AW}$ is symmetric for

$$\langle f, g \rangle_{AW} = \int_{|z|=1} f(z) \overline{g(z)} w_{AW}(z) \frac{dz}{2\pi i z},$$

and deduce the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

Definition 14.9 (Wilson polynomials)**.** For positive a, b, c, d, define

$$W_n(x^2; a, b, c, d) = (a+b)_n(a+c)_n(a+d)_n \\ \times {}_4F_3\left(\begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1\right).$$

Exercise 14.7 (Wilson degeneration)**.** Set $q = e^{-\varepsilon}$ and scale the Askey–Wilson parameters and spectral variable by

$$(a, b, c, d, z) = (q^A, q^B, q^C, q^D, q^{ix}).$$

After the scalar normalization by $(1-q)^{-3n}$, prove that the Askey–Wilson polynomial tends to $W_n(x^2; A, B, C, D)$ as $\varepsilon \downarrow 0$. Derive the Wilson difference equation, weight, orthogonality, and three-term recurrence as limits of the corresponding Askey–Wilson identities.

Definition 14.10 (Type-A polynomial representation)**.** Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \ldots, x_r^{\pm 1}],$$

let X_j act by multiplication by x_j, and let s_i exchange x_i and x_{i+1}. Define

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \qquad 1 \le i < r,$$

and

$$(\pi f)(x_1, \ldots, x_r) = f(x_2, \ldots, x_r, q x_1).$$

For generic independent parameters q, t, call the operator algebra generated by $X_j^{\pm 1}$, T_i, and $\pi^{\pm 1}$ the type-A polynomial operator model.

Exercise 14.8 (DAHA polynomial quotient)**.** Prove that the displayed operators preserve $\mathcal{P}_r$ and satisfy the braid and Hecke relations. Prove

$$\pi X_i \pi^{-1} = X_{i+1} \quad (1 \le i < r), \qquad \pi X_r \pi^{-1} = q X_1,$$

and identify the two affine-Hecke subalgebras generated respectively by T_i, X_j and by T_i, Y_j. Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to q. Conclude that the polynomial operator model is a realization of $\mathsf{H}_{q,t}(GL_r)$.

Definition 14.11 (Macdonald operators and polynomials)**.** Let

$$(T_{q,x_i} f)(x_1, \ldots, x_r) = f(x_1, \ldots, q x_i, \ldots, x_r).$$

The first type-A *Macdonald operator* is

$$D_{q,t} = \sum_{i=1}^{r} \left(\prod_{j \ne i} \frac{t x_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

For generic q, t, the *Macdonald polynomial* $P_\lambda(x; q, t)$ is the symmetric polynomial triangular in the monomial symmetric functions and satisfying

$$D_{q,t} P_\lambda = \left(\sum_{i=1}^{r} q^{\lambda_i} t^{r-i} \right) P_\lambda, \qquad \lambda_1 \geq \cdots \geq \lambda_r \geq 0.$$

For $0 < q, t < 1$, define

$$\Delta_{q,t}(x) = \prod_{i \neq j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty},$$

and

$$\langle f, g \rangle_{q,t} = \frac{1}{r!} \int_{(S^1)^r} f(x) \overline{g(x)} \Delta_{q,t}(x) \prod_{j=1}^{r} \frac{dx_j}{2\pi i x_j}.$$

Exercise 14.9 (Cherednik operators)**.** Define

$$Y_i = T_i \cdots T_{r-1} \pi T_1^{-1} \cdots T_{i-1}^{-1}.$$

Use the displayed polynomial representation and the affine braid relations to prove that the Y_i commute. Prove that their symmetric Laurent polynomials preserve $\mathcal{P}_r^{S_r}$ and identify $t^{-(r-1)/2}(Y_1 + \cdots + Y_r)$ with $D_{q,t}$.

Definition 14.12 (DAHA Fourier assignment)**.** Define the DAHA Fourier assignment on the torus, winding, and Hecke generators by

$$\mathcal{F}(X_j) = Y_j^{-1}, \qquad \mathcal{F}(Y_j) = X_j, \qquad \mathcal{F}(T_i) = T_i.$$

For $\rho = (r-1, r-2, \ldots, 0)$ and a partition λ, put

$$\xi_\lambda = (q^{\lambda_1} t^{r-1}, q^{\lambda_2} t^{r-2}, \ldots, q^{\lambda_r}).$$

Exercise 14.10 (Macdonald Fourier duality)**.** Verify the DAHA relations after applying $\mathcal{F}$, prove that it extends to an automorphism, and construct its intertwining transform on the polynomial representation. Normalize P_λ by its value at t^ρ and prove the evaluation symmetry

$$\frac{P_\lambda(\xi_\mu; q, t)}{P_\lambda(t^\rho; q, t)} = \frac{P_\mu(\xi_\lambda; q, t)}{P_\mu(t^\rho; q, t)}.$$

Derive the duality between multiplication recurrences in λ and the commuting Macdonald difference equations in the spectral variable.

Exercise 14.11 (Macdonald spectral theory)**.** Order monomial symmetric functions by dominance and prove that $D_{q,t}$ is triangular with the displayed diagonal entries. For generic q, t, deduce existence and uniqueness of P_λ. Prove directly from the torus integral that $D_{q,t}$ is symmetric when $0 < q, t < 1$, and deduce orthogonality and joint diagonalization for symmetric Laurent polynomials in the commuting Y_i.

Definition 14.13 (Symmetric-function Macdonald pairing)**.** Let $\Lambda_{\text{sym}} = \mathbb{C}[p_1, p_2, \ldots]$ be graded by $\deg p_r = r$. For a partition μ, let $m_r(\mu)$ be the number of its parts equal to r, and define

$$z_\mu = \prod_{r \geq 1} r^{m_r(\mu)} m_r(\mu)!, \qquad p_\mu = \prod_j p_{\mu_j}, \qquad p_r(x) = \sum_i x_i^r.$$

Define the symmetric-function Macdonald pairing by

$$\langle p_\mu, p_\nu \rangle_{q,t}^{\mathrm{sym}} = \delta_{\mu\nu} z_\mu \prod_j \frac{1-q^{\mu_j}}{1-t^{\mu_j}}.$$

For a cell $s=(i,j)$ in the diagram of λ, define its arm and leg by

$$a_\lambda(s) = \lambda_i - j, \qquad \ell_\lambda(s) = \lambda_j' - i,$$

and put

$$b_\lambda(q,t) = \prod_{s\in\lambda} \frac{1-q^{a_\lambda(s)}t^{\ell_\lambda(s)+1}}{1-q^{a_\lambda(s)+1}t^{\ell_\lambda(s)}}.$$

Exercise 14.12 (Stable Macdonald pairing and Cauchy identity)**.** Prove that the finite-variable Macdonald polynomials are stable under adjoining a zero variable and hence determine elements P_λ of Λ_{sym}. Put $Q_\lambda = b_\lambda(q,t)P_\lambda$. Use the adjointness of the DAHA difference operators and triangularity to prove

$$\langle P_\lambda, P_\mu \rangle_{q,t}^{\mathrm{sym}} = \delta_{\lambda\mu} b_\lambda(q,t)^{-1}.$$

Deduce that P_λ and Q_λ are dual. Expand the reproducing kernel in the power sums and use the q-binomial identity to prove

$$\sum_\lambda P_\lambda(a;q,t)Q_\lambda(b;q,t) = \prod_{i,j} \frac{(ta_ib_j;q)_\infty}{(a_ib_j;q)_\infty}$$

for finite alphabets with $|a_ib_j|<1$.

Definition 14.14 (Finite Macdonald measures)**.** Let $0 \le q,t < 1$, let $a=(a_1,\dots,a_N)$ and $b=(b_1,\dots,b_M)$ be nonnegative with $a_ib_j<1$, and let P_λ, Q_λ be the preceding Macdonald functions. Define

$$\Pi(a;b) = \sum_\lambda P_\lambda(a;q,t)Q_\lambda(b;q,t), \qquad \mathbb{P}_{q,t}(\lambda) = \frac{P_\lambda(a;q,t)Q_\lambda(b;q,t)}{\Pi(a;b)}.$$

Exercise 14.13 (One Macdonald observable)**.** Apply the first Macdonald difference operator in the a variables to the Cauchy identity. For distinct a_i, prove

$$\mathbb{E}_{q,t}\left[\sum_{i=1}^N q^{\lambda_i}t^{N-i}\right] = \sum_{i=1}^N \prod_{j\neq i} \frac{ta_i - a_j}{a_i - a_j} \prod_{k=1}^M \frac{1-a_ib_k}{1-ta_ib_k},$$

and extend the identity to coincident parameters by continuity. Verify the Schur specialization $q=t$ and the q-Whittaker specialization $t=0$. For samples from the finite measure, identify the displayed expectation as a directly estimable statistic and state the positivity, truncation, and sampling assumptions required for comparison with the formula.

Definition 14.15 (Multiplicative theta function)**.** For $|p|<1$, define

$$\theta(z;p) = (z;p)_\infty (p/z;p)_\infty.$$

Definition 14.16 (Elliptic Ruijsenaars operator)**.** For type A_{r-1}, define

$$D_{p,q,t}^{\mathrm{ell}} = \sum_{i=1}^r \left(\prod_{j\neq i} \frac{\theta(tx_i/x_j;p)}{\theta(x_i/x_j;p)} \right) T_{q,x_i}.$$

Exercise 14.14 (Ruijsenaars degeneration)**.** Prove directly from $\theta(z;0) = 1 - z$ that

$$D^{\mathrm{ell}}_{p,q,t} \longrightarrow D_{q,t} \qquad (p \to 0).$$

Set $q = e^{-\varepsilon}$ and $t = e^{-g\varepsilon}$ and derive the Heckman–Opdam differential operator from the second-order term as $\varepsilon \to 0$. Take the rational scaling limit of its hyperbolic coefficients and obtain the rational Calogero–Moser operator.

Exercise 14.15 (The Calogero–Sutherland limit)**.** Write $x_i = e^{z_i}$, set $q = e^{-\epsilon}$ and $t = e^{-g\epsilon}$, and expand $D_{q,t}$ through order ϵ^2. After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i<j} \coth\left(\frac{z_i - z_j}{2}\right)(\partial_{z_i} - \partial_{z_j}).$$

On the circle, put $z_i = i\theta_i$ and conjugate by

$$\Psi_0(\theta) = \prod_{i<j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g .$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = -\sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i<j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for $r = 2$, degree two. Identify their Sutherland eigenvalues.

Exercise 14.16 (Hypergeometric degeneration)**.** Set $q = e^{-\varepsilon}$. Prove

$$\lim_{\varepsilon \to 0^+} \frac{(q^a;q)_m}{(1-q)^m} = (a)_m, \qquad \lim_{q \to 1^-} (1-q)^{1-a} \frac{(q;q)_\infty}{(q^a;q)_\infty} = \Gamma(a).$$

Deduce

$$\lim_{q \to 1^-} {}_r\phi_{r-1}\left(\begin{matrix} q^{a_1}, \dots, q^{a_r} \\ q^{b_1}, \dots, q^{b_{r-1}} \end{matrix}; q, z \right) = {}_rF_{r-1}\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix}; z \right).$$

Derive Gauss's summation from the q-Gauss summation. Prove the corresponding degeneration of the basic hypergeometric q-difference equation to the Gauss equation.

Definition 14.17 (BC_r root-system weight)**.** Write $f(z^{\pm m}) = f(z^m)f(z^{-m})$ and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon, \delta = \pm 1} f(z_i^\epsilon z_j^\delta).$$

For $|q|, |t|, |a_s| < 1$, define

$$\begin{aligned} \Delta_{BC_r}(z) = &\prod_{i=1}^r \frac{(z_i^{\pm 2};q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1};q)_\infty} \\ &\times \prod_{1 \le i < j \le r} \frac{(z_i^{\pm 1} z_j^{\pm 1};q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1};q)_\infty}. \end{aligned}$$

Use the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

Definition 14.18 (Koornwinder operator)**.** For $z = (z_1, \dots, z_r)$ put

$$A_i(z) = \frac{\prod_{s=1}^4 (1 - a_s z_i)}{(1 - z_i^2)(1 - q z_i^2)} \prod_{j \neq i} \frac{(1 - t z_i z_j)(1 - t z_i / z_j)}{(1 - z_i z_j)(1 - z_i / z_j)}.$$

On Laurent polynomials invariant under permutations and inversions define

$$\mathcal{D}_K = \sum_{i=1}^r \left[A_i(z)(T_{q,z_i} - 1) + A_i(z^{-1})(T_{q,z_i}^{-1} - 1) \right].$$

The *Koornwinder polynomial* K_λ is the Weyl-invariant Laurent polynomial triangular in orbit sums and satisfying

$$\mathcal{D}_K K_\lambda = \left(\sum_{i=1}^r \left[a_1 a_2 a_3 a_4 q^{-1} t^{2r-i-1} (q^{\lambda_i} - 1) + t^{i-1} (q^{-\lambda_i} - 1) \right] \right) K_\lambda,$$

with the scalar eigenvalue fixed by the displayed convention.

Exercise 14.17 (BC_r weight)**.** Prove that Δ_{BC_r} is invariant under permutations and independent inversions of the z_i. Identify its factors with the roots $\pm e_i$, $\pm 2e_i$, and $\pm e_i \pm e_j$. For $r = 1$, recover the Askey–Wilson weight and its order-two Weyl normalization, and identify $\mathcal{D}_K$ with $\mathcal{D}_{AW}$.

Exercise 14.18 (Koornwinder spectral theory)**.** Prove that $\mathcal{D}_K$ is symmetric for the Δ_{BC_r} inner product and that its triangular eigenfunctions are mutually orthogonal when the displayed eigenvalues are distinct.

Exercise 14.19 (BC_r limit)**.** Set $t = q^g$ with $g \in \mathbb{N}$, reduce the quotients of q-products to finite products, and compute the limit $q \to 1^-$ with $z_i = e^{i\theta_i}$.

Exercise 14.20 (Askey–Wilson spectrum)**.** Fix $M \geq 4$ and parameters $a, b, c, d \in (-1, 1)$ for which the Askey–Wilson weight is positive and the first $M + 1$ eigenvalues are distinct. Let $\mathcal{H}_M$ be the span of the normalized Askey–Wilson polynomials $p_0, \dots, p_M$ with their positive Askey–Wilson inner product. Model a finite programmable q-difference simulator by the self-adjoint Hamiltonian $\widehat{H}_M = \varepsilon \mathcal{D}_{AW}|_{\mathcal{H}_M}$, where $\varepsilon > 0$ is a calibrated energy scale. Prove that its spectral values are

$$E_n = \varepsilon (q^{-n} - 1)(1 - abcd q^{n-1}).$$

Write the three-term recurrence in the form

$$(z + z^{-1}) p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

and compute the corresponding normalized matrix elements. Deduce that a periodic perturbation proportional to the compression of $z + z^{-1}$ to $\mathcal{H}_M$ produces only neighboring transitions, with resonance frequencies $|E_{n+1} - E_n|/\hbar$ and intensities determined by A_n and C_{n+1}. Assuming ε is known and at least three nondegenerate lines are resolved, derive the equations satisfied by q and the product $abcd$. Compute the Jacobian of two independent gap ratios and state the nonvanishing and parameter-range hypotheses under which these two quantities are locally identifiable.

Form the data map from (a, b, c, d) to four selected squared normalized neighboring matrix elements. Prove that it is invariant under permutation of the parameters. At a chosen parameter point, compute its differential in the four elementary symmetric functions and state the full-rank hypothesis under which the measured strengths locally determine the unordered set $\{a, b, c, d\}$. When this rank condition fails, identify the remaining fiber instead of asserting identifiability; state also that squared strengths cannot distinguish any additional discrete ambiguity in that fiber. State that these are predictions for the engineered simulator.

Exercise 14.21 (Macdonald–Ruijsenaars spectrum)**.** Let $0 < q, t < 1$, fix $M \geq 0$, and define

$$\mathcal{H}_{q,t}^{(M)} = \operatorname{span}\{P_\lambda(x; q, t) : |\lambda| \leq M\}$$

with the positive Macdonald inner product. Put

$$d_0 = \sum_{i=1}^{r} t^{r-i}, \qquad \widehat{H}_{q,t}^{(M)} = \varepsilon(d_0 - D_{q,t})|_{\mathcal{H}_{q,t}^{(M)}}, \qquad \varepsilon > 0.$$

Identify $D_{q,t}$ with the first symmetric function of the commuting DAHA winding operators and derive its compatibility with the Hecke exchange relations. Prove that $\mathcal{H}_{q,t}^{(M)}$ is invariant, that $\widehat{H}_{q,t}^{(M)}$ is self-adjoint, and that the normalized Macdonald states have energies

$$E_\lambda = \varepsilon \sum_{i=1}^{r} t^{r-i}(1 - q^{\lambda_i}).$$

Exercise 14.22 (Macdonald transition spectrum)**.** Model a finite programmable q-difference simulator by this operator. Use the Macdonald Pieri identity for $\rho_1 = x_1 + \cdots + x_r$ and its adjoint to determine every nonzero matrix element of the compression of $\rho_1 + \rho_1^*$ to $\mathcal{H}_{q,t}^{(M)}$. For a periodic drive proportional to this compressed observable, derive the resonance frequencies

$$\Omega_{\mu\lambda} = |E_\mu - E_\lambda|/\hbar$$

and the relative intensities as squared normalized Macdonald Pieri coefficients. Explain that recovery of q, t requires known ε and r, resolved partition labels, sufficiently many nondegenerate transitions, and a fixed convention for any discrete parameter ambiguity, and that the prediction concerns the engineered simulator.

Exercise 14.23 (Macdonald–Jack degeneration)**.** Set $q = e^{-\delta}$ and $t = e^{-g\delta}$; after the scalar and center-of-mass normalizations in the Calogero–Sutherland limit exercise, prove that the rescaled spectral gaps and transition coefficients converge to the Sutherland energies and Jack Pieri coefficients. State how measured line locations and strengths test this degeneration. Explain that recovery of the deformation parameter requires resolved partition labels and sufficiently many nondegenerate transitions.

Definition 14.19 (Elliptic shifted factorials and gamma function)**.** For $|p|, |q| < 1$, define

$$\theta(z_1, \ldots, z_s; p) = \prod_{j=1}^{s} \theta(z_j; p),$$

$$(a; q, p)_m = \prod_{j=0}^{m-1} \theta(aq^j; p),$$

$$(a_1, \ldots, a_s; q, p)_m = \prod_{i=1}^{s} (a_i; q, p)_m,$$

and

$$\Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1}q^{k+1}/z}{1 - p^j q^k z}.$$

Definition 14.20 (Deformation hierarchy)**.** For a series $\sum_m c_m$, set $h_m = c_{m+1}/c_m$. It is *ordinary hypergeometric* when h_m is rational in m, and *basic hypergeometric* when h_m is rational in q^m. It is *elliptic hypergeometric* when, after writing $x = q^m$, its term ratio has the form

$$h(x) = z\frac{\prod_{i=1}^{s}\theta(a_i x; p)}{\prod_{i=1}^{s}\theta(b_i x; p)}.$$

Its parameters are *balanced* when

$$\prod_{i=1}^{s} a_i = \prod_{i=1}^{s} b_i.$$

By theta quasi-periodicity this is equivalent to

$$h(px) = h(x).$$

Exercise 14.24 (Elliptic degenerations)**.** Prove

$$\theta(pz; p) = -z^{-1}\theta(z; p), \qquad \Gamma(qz; p, q) = \theta(z; p)\Gamma(z; p, q),$$

$$\Gamma(pz; p, q) = \theta(z; q)\Gamma(z; p, q), \qquad \Gamma(z; p, q)\Gamma(pq/z; p, q) = 1.$$

Prove

$$\theta(z; 0) = 1 - z, \qquad (a; q, 0)_m = (a; q)_m, \qquad \Gamma(z; 0, q) = \frac{1}{(z; q)_\infty}.$$

Definition 14.21 (Frenkel–Turaev sum)**.** For $a^2 q^{n+1} = bcde$, define

$$\begin{aligned} {}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) = \sum_{m=0}^{n} &\frac{\theta(aq^{2m}; p)}{\theta(a; p)} \\ &\times \frac{(a, b, c, d, e, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq^{n+1}; q, p)_m} q^m. \end{aligned}$$

Exercise 14.25 (Theta addition and Frenkel–Turaev summation)**.** Prove the multiplicative theta addition formula

$$\begin{aligned} &\theta(xz, x/z, yw, y/w; p) - \theta(xw, x/w, yz, y/z; p) \\ &= \frac{y}{z}\theta(xy, x/y, zw, z/w; p) \end{aligned}$$

by comparing quasi-periodicity and zeros. Use it to telescope the difference between the nth and $(n-1)$st terminating sums and prove

$${}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q, p)_n}{(aq/b, aq/c, aq/d, aq/(bcd); q, p)_n}.$$

Take $p \to 0$ and identify the terminating very-well-poised basic hypergeometric summation.

Definition 14.22 (Elliptic face weights and dynamical exchange)**.** Fix logarithms of p, q with $|p|, |q| < 1$ and put

$$[x]_{p,q} = q^{(1-x)/2}\frac{\theta(q^x; p)}{\theta(q; p)}.$$

For integer heights i, j, k, l satisfying

$$|i - j| = |j - k| = |k - l| = |l - i| = 1,$$

define $W(i, j, k, l \mid x)$ by

$$\begin{aligned} W(l\pm1, l\pm2, l\pm1, l \mid x) &= [x+1]_{p,q}, \\ W(l\mp1, l, l\pm1, l \mid x) &= \frac{[l+1\pm1]_{p,q}[x]_{p,q}}{[l+1]_{p,q}}, \\ W(l\pm1, l, l\pm1, l \mid x) &= \frac{[l+1\mp x]_{p,q}}{[l+1]_{p,q}}, \end{aligned}$$

whenever the denominators are nonzero, and set all other weights equal to zero. On $V = \mathbb{C}e_+ \oplus \mathbb{C}e_-$ let $he_\pm = \pm e_\pm$ and define $R(x, \lambda) \in \operatorname{End}(V \otimes V)$ at $\lambda = l+1$ by

$$\langle e_{j-k} \otimes e_{k-l}, R(x, l+1)(e_{i-l} \otimes e_{j-i})\rangle = W(i, j, k, l \mid x),$$

using the orthonormal distinguished weight lines and extending meromorphically in λ.

Definition 14.23 (Vertex–face intertwiners)**.** Let $R^{\mathrm{vert}}(u) \in \operatorname{End}(V \otimes V)$ be a spectral exchange operator. For each admissible oriented height edge $|a-b| = 1$, let $\psi(a, b \mid u) \in V$ be meromorphic in u. The pair $(R^{\mathrm{vert}}, \psi)$ satisfies the vertex–face relation for W when

$$\begin{aligned} &R^{\mathrm{vert}}(u-v)(\psi(a, b \mid u) \otimes \psi(b, c \mid v)) \\ &= \sum_{\substack{b' \\ |a-b'|=|b'-c|=1}} W(a, b, c, b' \mid u-v)\psi(a, b' \mid v) \otimes \psi(b', c \mid u). \end{aligned}$$

The intertwiners are nondegenerate when the maps from admissible height-path spaces induced by their tensor products are injective for generic spectral parameters.

Exercise 14.26 (Elliptic face Yang–Baxter relation)**.** Prove $[-x]_{p,q} = -[x]_{p,q}$ and rewrite the theta addition identity as the corresponding four-term identity for the brackets $[x]_{p,q}$. Verify that $R(x, \lambda)$ has weight zero and prove the dynamical Yang–Baxter equation

$$\begin{aligned} &R_{12}(x, \lambda + h^{(3)})R_{13}(x+y, \lambda)R_{23}(y, \lambda + h^{(1)}) \\ &\quad = R_{23}(y, \lambda)R_{13}(x+y, \lambda + h^{(2)})R_{12}(x, \lambda) \end{aligned}$$

by reducing every nontrivial weight component to that bracket identity. Translate the two sides into the two sums over the admissible internal height of a three-face patch. Show directly that the shifts by $h^{(1)}, h^{(2)}, h^{(3)}$ are the changes of the adjacent face height and that λ, unlike x and y, is therefore the dynamical parameter.

For periodic admissible height paths, form the row-to-row transfer maps by multiplying the displayed face weights and summing over the intermediate path. Use the face Yang–Baxter relation to prove commutation of transfer maps with distinct spectral parameters. For fixed boundary heights of a three-face patch with two admissible internal heights, compute their normalized probabilities from the two products of W and prove that a star–triangle move preserves the boundary partition function. State the real parameter and gauge conditions under which the normalized weights are positive statistical-mechanical probabilities.

Exercise 14.27 (Vertex–face transfer correspondence)**.** Assume nondegenerate vertex–face intertwiners. Apply the ordinary spectral Yang–Baxter equation for R^{vert} to

$$\psi(a, b \mid u) \otimes \psi(b, c \mid v) \otimes \psi(c, d \mid w)$$

and use the vertex–face relation in the two orders. Prove that equality of the resulting coefficients is the face Yang–Baxter relation and hence the dynamical Yang–Baxter equation of the preceding exercise. Conversely, use nondegeneracy to prove the ordinary Yang–Baxter equation on the image of the height-path map from the face relation.

For inhomogeneities $z_1, \ldots, z_N$, tensor the edge intertwiners around a periodic row to obtain the map Φ_u from periodic height paths to the corresponding vertex sector. Prove the row-transfer intertwining identity

$$\mathsf{T}_N^{\mathrm{vert}}(u)\Phi_u = \Phi_u \mathsf{T}_N^{\mathrm{face}}(u)$$

with the boundary twist dictated by the net height change. Deduce equality of the matched transfer spectra and periodic-strip partition functions. In the trigonometric degeneration $p \to 0$, identify the vertex transfer operator with the six-vertex family and the face transfer operator with the trigonometric solid-on-solid model. State the boundary-sector, normalization, injectivity, and face-gauge conditions required when their finite-size spectra are compared.

Definition 14.24 (Terminating elliptic $6j$ series)**.** For $bcdefg = a^3q^{n+2}$, define

$$\begin{aligned} &{}_{12}V_{11}(a; b, c, d, e, f, g, q^{-n}; q, p) \\ &\quad = \sum_{m=0}^{n} \frac{\theta(aq^{2m}; p)}{\theta(a; p)} \frac{(a, b, c, d, e, f, g, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq/f, aq/g, aq^{n+1}; q, p)_m} q^m. \end{aligned}$$

Exercise 14.28 (Frenkel–Turaev elliptic face recoupling)**.** Let

$$\lambda = \frac{a^2q}{bcd}, \qquad g = \frac{\lambda a q^{n+1}}{ef},$$

and assume no displayed denominator vanishes. Expand the series on the right, interchange the resulting finite sums, and apply the Frenkel–Turaev ${}_{10}V_9$ summation to prove

$$\begin{aligned} &{}_{12}V_{11}(a; b, c, d, e, f, g, q^{-n}; q, p) \\ &\quad = \frac{(aq, aq/(ef), \lambda q/e, \lambda q/f; q, p)_n}{(aq/e, aq/f, \lambda q/(ef), \lambda q; q, p)_n} \\ &\qquad \times {}_{12}V_{11}\left(\lambda; \frac{\lambda b}{a}, \frac{\lambda c}{a}, \frac{\lambda d}{a}, e, f, g, q^{-n}; q, p\right). \end{aligned}$$

Fuse n consecutive elementary face weights by summing over admissible internal height paths. Factor the endpoint theta products and show that the remaining path sum has the displayed ${}_{12}V_{11}$ term ratio. Identify the two sides of the transformation with the two fusion orders and hence with the two recouplings of the four external height paths. Deduce the tetrahedral change of channel for the resulting elliptic $6j$ coefficients and recover the face Yang–Baxter relation for fused weights.

Choose a numerator parameter so that the transformed series reduces to its zeroth term and recover the preceding ${}_{10}V_9$ summation. Take $p \to 0$ to obtain Bailey's terminating very-well-poised ${}_{10}\phi_9$ transformation, and then take the spectral degeneration to the Q-Racah quantum $6j$ coefficients developed earlier. For a finite fused face strip, express the relative Boltzmann probabilities of two resolved intermediate heights as ratios of normalized fused face weights. Separately, in a unitary recoupling specialization, express channel probabilities as squared normalized elliptic $6j$ coefficients. State the positivity, face-gauge, boundary-height, unitarity, and spectral-resolution assumptions required for the two interpretations.

Definition 14.25 (Elliptic beta kernel)**.** Write

$$\Gamma(tz^{\pm 1};p,q)=\Gamma(tz;p,q)\Gamma(t/z;p,q),\qquad \kappa_{p,q}=\frac{(p;p)_\infty(q;q)_\infty}{2}.$$

For $|t_j|<1$ with $\prod_{j=1}^{6}t_j=pq$, define

$$\mathcal{B}(t_1,\ldots,t_6)=\kappa_{p,q}\int_{\mathbb{T}}\frac{\prod_{j=1}^{6}\Gamma(t_jz^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)}\,\frac{dz}{2\pi iz}.$$

Exercise 14.29 (Elliptic beta recurrence)**.** Use the elliptic-gamma difference equations to derive the parameter-shift recurrences of $\mathcal{B}$.

Exercise 14.30 (Elliptic beta integral)**.** Evaluate a residue degeneration for which two parameters multiply to pq and use analytic continuation to prove

$$\mathcal{B}(t_1,\ldots,t_6)=\prod_{1\le i<j\le 6}\Gamma(t_it_j;p,q).$$

Take $p\to 0$ and identify the resulting basic hypergeometric beta integral.

Definition 14.26 (Elliptic integral operators)**.** For $|t|<1$ define

$$(\mathcal{M}_tf)(w)=\frac{\kappa_{p,q}}{\Gamma(t^2;p,q)}\int_{\mathbb{T}}\frac{\Gamma(tw^{\pm 1}z^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)}f(z)\,\frac{dz}{2\pi iz},$$

and

$$\mathcal{D}_s(y,w)=\Gamma\Big((pq)^{1/2}s^{-1}y^{\pm 1}w^{\pm 1};p,q\Big).$$

Exercise 14.31 (Elliptic star–triangle relation)**.** Apply the elliptic beta integral to the internal integration variable and prove the operator identity

$$\mathcal{M}_s\mathcal{D}_{st}(y,\cdot)\mathcal{M}_t=\mathcal{D}_t(y,\cdot)\mathcal{M}_{st}\mathcal{D}_s(y,\cdot)$$

whenever the contours separate reciprocal pole sequences and the parameters satisfy the corresponding modulus bounds. Interpret the kernels as edge weights and $\mathcal{D}$ as a site weight of a continuous spin model. Prove that the two local triangular partition functions are equal and that the row transfer operators obtained by the two factorizations commute. For a finite periodic strip, deduce equality of the transfer spectra and partition functions under a star–triangle move; state the positivity regime and contour assumptions under which these are statistical-mechanical observables.